



AOS-CX 10.17.xxxx Monitoring Guide For 6300, 6400 Switch Series

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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

Latest version available online

Command syntax notation conventions

About the examples

Identifying switch ports and interfaces

Identifying modular switch components

Applicable products

This document applies to the following products:

- HPE Aruba Networking 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A, R8S89A, R8S90A, R8S91A, R8S92A, S0E91A, S0X44A, S3L75A, S3L76A, S3L77A, S4P41A, S4P42A, S4P43A, S4P44A, S4P45A, S4P46A, S4P47A, S4P48A)
- HPE Aruba Networking 6400 Switch Series (R0X31A, R0X38B, R0X38C, R0X39B, R0X39C, R0X40B, R0X40C, R0X41A, R0X41C, R0X42A, R0X42C, R0X43A, R0X43C, R0X44A, R0X44C, R0X45A, R0X45C, R0X26A, R0X27A, JL741A, S0E48A, S0E48A #0D1, S1T83A, S1T83A #0D1)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention

Usage

example - text	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example - text	In code and screen examples, indicates text entered by a user.
Any of the following: • <example - text> • <example - text> • example - text • example - text	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or	Ellipsis:
...	• In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch (CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch (config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch (config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch (config-vlan-<VLAN-ID>) #
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 6300 Switch Series

- member: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 10.
The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.

- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on member 1.

On the HPE Aruba Networking 6400 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/1 and 1/2.
 - Line modules are on the front of the switch starting in slot 1/3.
- port: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: member/power supply:
 - member: 1.
 - power supply: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: member/tray/fan:
 - member: 1.
 - tray: 1 to 4.
 - fan: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: member/module:
 - member: 1.
 - member: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Monitoring hardware through visual observation

View the LEDs on the switch to monitor your hardware.

Subtopics

Confirming normal operation of the switch by reading LEDs

Detecting if the switch is not ready for a failover event

Finding faulted components using the switch LEDs

Confirming normal operation of the switch by reading LEDs

This task describes using the switch LEDs to confirm that the switch is operating normally.



NOTE

For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Quick check: Verify that the chassis has power and there are no fault conditions.
On the front of the switch, verify that the states of the following LEDs are On Green:
2.
 - Power
 - HealthVerify that the Health LEDs of all installed line modules are On Green.
3. Verify that the Health LEDs of all installed management modules are On Green.
4. Verify that the network ports are operating normally.
 - a. On the active management module, check the Status Front section. Verify that each LED that indicates a line module is in one of the following states:
 - On Green (normal operation)
 - Off (no line module installed)
 - b. On each line module, verify that each port LED is in one of the following states:
 - On Green, Half-Bright Green, or Flickering Green (normal operation)

- Off (no cable connected or port off by default in config)
5. Verify that the power supplies are operating normally.
 - a. On the active management module, check the Status Front section. Verify that each LED that indicates a power supply is in one of the following states:
 - On Green (normal operation)
 - Off (no power supply installed)
 - b. On each power supply, verify that LEDs are in the following states:
 - Power LED: On Green
 - Fault LED: Off
 6. Verify that the rear components are operating normally by checking the Status Rear section of the active management module:
 - a. Verify that the LEDs for the fabric modules are in one of the following states:
 - On Green (normal operation)
 - Off (component not installed)
 - b. Verify that the LEDs for the fan trays and fans are On Green.
 7. Verify that the standby management module is ready to take over as the active management module. On the standby management module, verify the states of the following LEDs:
 - Health LED is On Green.
 - Management state standby (Stby) LED is On Green.

Detecting if the switch is not ready for a failover event

This task describes using the switch LEDs to detect if the switch is not ready for the loss of a fabric module or for a failover from the active management module to the standby management module.



NOTE

Although you can detect power supply failures by viewing the LEDs, you must use software commands to determine if the power supply redundancy is sufficient to power the chassis if a power supply fails. For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Detect if the standby management module is shut down.

If the standby management module is shut down, the LED states are as follows:

- The standby management module health LED is Off.
- The standby management state active (Actv) LED is Off.
- The standby management state standby (Stby) LED is Off.
- On the active management module in the Status Front Management Modules section, the LED for the standby management module is Off. For example, if the active management module is Management Module LED 5, Management Modules LED 6 is Off.

2. Detect if the standby management module is in a transient state. If the standby management module is booting, updating, or in another transient state, the LED states are as follows:

- The standby management module health LED is Slow Flash Green when the service operating system is running or during an operating system update.
- The standby management module Booting LED is Slow Flash Green when the AOS-CX operating system is booting.
- The standby management state active (Actv) LED is Off.
- The standby management state standby (Stby) LED is Off.
- On the active management module in the Status Front Management Modules section, the LED for the standby management module is Slow Flash Green.

3. Detect if a fabric module is shut down or not present. If a fabric module is shut down or not present, the LED states are as follows:

- On the active management module, in the Status Rear section, the LED for the fabric module is Off.
- On the rear display module, the LED for the fabric module is Off.
- On the fabric module, the health LED is Off. However, the fabric module is behind fan 1 and is not directly visible.

Finding faulted components using the switch LEDs

This task describes using the switch LEDs to find components that are in a fault condition.



NOTE

All green LEDs—except for chassis power LEDs and the Usr1 LED—are off when the LED mode is set to Light Faults (The Usr1 LED of the LED Mode section of the active management module is On Green and the default behavior for the Usr1 LED is being used.). For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Find the switch that has the fault condition, which is indicated by a chassis health LED in the state of Slow Flash Orange.

The chassis health LED is located on the front of the switch and on the rear panel of the switch.

2. If you are at the back of the switch, on the rear panel, look for LEDs that are in the Slow Flash Orange state:

The Status Rear area has LEDs for power supplies, fabric modules, fan trays, and fans. The number on the LED represents the unit number of the component.

If the only LED in a state of Slow Flash Orange is the Chassis health LED, go to the front of the switch.

3. At the front of the switch, on the active management module, look for LEDs that are in the Slow Flash Orange state:

- The Status Front area has LEDs for power supplies, line and fabric modules, and management modules. The number on the LED indicates the slot number of the component.
- The Status Rear area has LEDs for fabric modules and fan trays, with a single LED for all the fans in the fan tray. The number on the LED represents the slot or bay number of the component.

4. Use the number indicated by the LED that is flashing to locate the slot that contains the faulted component.

The fabric modules are located behind the fan trays, and the fabric module number corresponds to the fan tray number.

5. At the front of the switch, on line modules, look for LEDs that are in the Slow Flash Orange state: Module LEDs and Port LEDs indicate faults if their states are Slow Flash Orange.

IP Flow Information Export

IP Flow Information Export (IPFIX) is an embedded network flow analysis tool that compiles characteristic and measured properties of flows and sends flow reports to internal or external flow collectors. IPFIX is

configurable via the command-line or REST interfaces. With IPFIX, customers configure flow records with match (key) fields and collection (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collection fields are the set of fields that identify information to collect for a flow, such as packet and byte counters.

A flow exporter defines where and how to export flow reports. Flow exporters are created as standalone entities in the **config** context to provide flow monitors the ability to export flow reports.

Compatibility with Traffic Insight

The AOS-CX traffic insight feature allows monitoring of large amount of data that it collects from various flow exporters like IPFIX, and provides the ability to filter, aggregate, and sort the data based on user flow monitor requests. Traffic insight tracks different monitor requests simultaneously and provides monitor reports per request. For more information on configuring the Traffic Insight features, refer to the AOS-CX Security Guide.

Compatibility with Application Recognition

If the **application recognition** feature is enabled, then the application data and the flow properties collected by AR and IPFIX are exported to external or internal IPFIX collectors. For more information on configuring the Application Recognition and Traffic Insight features, refer to the AOS-CX Security Guide.

Information Elements

The IPFIX Information Elements (IE) are entities that are defined and maintained by the Internet Assigned Numbers Authority (IANA). They are characterized by a unique piece of information they can provided about a flow. Information Elements may be either private or public. Private Information Elements are exported with a Private Enterprise Number (PEN).

AOS-CX can act as an intermediate collecting process for flow reports from hardware to append certain additional IPFIX information elements to the flow reports. When configured, the software will act as an intermediate exporting process to export the augmented flow reports to any configured flow exporters.

AOS-CX supports the standard and private information elements shown in the tables below.



NOTE

Not all switches support all information elements. View a list of information elements supported by your switch using the command **show flow monitor <monitor-name> information-elements**. All Standard Information Element registration information can be found on the IANA website, at <https://www.iana.org/assignments/ipfix/>.

Standard Information Elements

octetDeltaCount

packetDeltaCount

Private Information Elements

tlsClientVersion (ID 1029, 4823
Aruba)

Standard Information Elements

protocolIdentifier

sourceTransportPort

sourceIPv4Address

ingressInterface

destinationTransportPort

destinationIPv4Address

sourceIPv6Address

destinationIPv6Address

sourceIPv6Address

destinationIPv6Address

classId

sourceMacAddress

vlanId

ipVersion

destinationMacAddress

applicationDescription

applicationId

applicationName

flowEndReason

flowStartMicroseconds

flowEndMicroseconds

dnsResponseCode

ingressPhysicalInterface

egressPhysicalInterface

Private Information Elements

tlsServerVersion (ID 1030, 4823 Aruba)

ja3 (ID 1031, 4823 Aruba)

ja3s (ID 1032, 4823 Aruba)

supportedNextProtocol (ID 1033, 4823 Aruba)

CertificateIssuer (ID 1034, 4823 Aruba)

CertificateIssueDate (ID 1035, 4823 Aruba)

CertificateExpiryDate (ID 1036, 4823 Aruba)

applicationType (ID 1100, 4823 Aruba)

tcp3WayHsServerRespTime (ID 1101, 4823 Aruba)

tcp3WayHsClientRespTime (ID 1102, 4823 Aruba)

dropGroup1 (ID 1200, 14823 Aruba)

dropReason1 (ID 1201, 14823 Aruba)

dropStartMilliseconds1 (ID 1202, 14823 Aruba)

dropEndMilliseconds1 (ID 1203, 14823 Aruba)

dropGroup2 (ID 1210, 14823 Aruba)

dropReason2 (ID 1211, 14823 Aruba)

Standard Information Elements

applicationCategoryName

Private Information Elements

dropStartMilliseconds2 (ID 1212, 14823 Aruba)

dropEndMilliseconds2 (ID 1213, 14823 Aruba)

dropGroup3 (ID 1220, 14823 Aruba)

dropReason3 (ID 1221, 14823 Aruba)

dropStartMilliseconds3 (ID 1222, 14823 Aruba)

dropEndMilliseconds3 (ID 1223, 14823 Aruba)

dropGroup4 (ID 1230, 14823 Aruba)

dropReason4 (ID 1231, 14823 Aruba)

dropStartMilliseconds4 (ID 1232, 14823 Aruba)

dropEndMilliseconds4 (ID 1233, 14823 Aruba)

cngQueueThreshold (ID 490, 14823 Aruba)

cngQueueObservedTimeStamp (ID 491, 14823 Aruba)

About individual Information Elements

The following IEs are used for both IPv4 and IPv6 flows:

- cngSourceIP
- cngDestinationIP

They are always 16 bytes wide. The IP version of the flow is indicated in the **ipVersion** IE. For IPv4 addresses, the first 12 octets are zero and the IPv4 address is encoded in the last 4 octets. For IPv6 addresses, the full 16 bytes encode the address.

The following IEs reflect the hardware interface ID used by the associated flow:

- ingressPhysicalInterface
- egressPhysicalInterface

The hardware interface ID does not match the front-panel port number of the interface. The hardware interface ID can be mapped to the front-panel port number by using the REST API to query for **hw_intf_info** for all interfaces and saving the **switch_intf_id** value together with the associated front-panel name to form a mapping table.

Example using curl and jq:

```
bash
$ curl -X GET
  "{SWITCH}/rest/{VERSION}/system/interfaces?
attributes=hw_intf,info,name&depth=2"
-H "x-csrf-token: {CSRFTOKEN}"
-b {AUTH COOKIE FILE}
| jq 'to_entries
      | map({port: .key, switch_intf_id:
(.value.hw_intf_info.switch_intf_id | tonumber)})
      | sort_by(.switch_intf_id)
      | map(({.port): .switch_intf_id})
      | add'
{
  "1/1/6": 1,
  "1/1/2": 2,
  "1/1/1": 3,
  ...
}
```

The **IPFIX Drop Exceptions** feature is a network debugging and triaging tool supported on 6300 Switch series. It is configured using the **collect drop ingress-exceptions** command in the **config-flow-record** context. The private information elements 1200-1233 contain possible drop reason information for why dropped packets are seen for a flow. Up to four drop reasons are supported for each flow. Data in consecutive values (for example, 1200, 1201, 1202, 1203) indicate the data in those fields are associated with each other.

For each drop reason, AOS-CX reports the drop group name, drop reason name, and drop start and end timestamps in milliseconds.

Table 1. Drop reasons for the IPFIX Drop Exceptions feature

Drop reason	Example Cause	Supported Platforms
Unknown VLAN	The packet has an unknown VLAN	5420, 6200, 6300, 6400, 8100, 8325, 8325P, 8325H, 8360, 10000 Switch series
MTU Failure	The packet size is larger than the IP Maximum Transmission Unit (MTU) configured in the interface.	5420, 6200, 6300, 6400, 8100, 8325, 8325P, 8325H, 8360 Switch series
Unroutable Unicast	The unicast packet can't be routed as destination address is not resolved.	5420, 6200, 6300, 6400, 8100, 8360 Switch series.
Unicast TTL expired	For IPv4: The unicast packet has a Time - to - Live (TTL) value of 0 or 1. For IPv6: The unicast packet has a hop limit value of 0 or 1.	5420, 6200, 6300, 6400, 8100, 8360 Switch series.
Security Ingress IP Lockdown	Packet's source IP address, VLAN, MAC or Interface does not match the IP binding entry.	5420, 6200, 6300, 6400, 8100, 8360 Switch series.
Security Ingress MAC Lockout	The packet is sourced from a locked - out MAC address.	5420, 6200, 6300, 6400, 8100, 8360 Switch series.
IPTCAM invalid IP address	Packet has invalid source or destination IP address. (For example, an invalid IPv4 address of 0.0.0.0 or 127.0.0.0 , or 240.0.0.0 , or an invalid IPv6 address of :: or ::1 .)	5420, 6200, 6300, 6400, 8100, 8360 Switch series.
Security learn	Packet is received on a security - enabled interface and is learned for the first time.	5420, 6200, 6300, 6400, 8100, 8360 Switch series.
Security drop	Packet is received on a security - enabled interface but is not allowed, as it is not authorized on the VLAN.	5420, 6200, 6300, 6400, 8100, 8360 Switch series.
Meter drop	The packet is dropped because of exceeding the rate limit configured	5420, 6200, 6300, 6400, 8100, 8360 Switch series.

Drop reason	Example Cause	Supported Platforms
	on port or via a port - access policy.	
ASIC drop	The packet is dropped by ASIC because of parity error, parser error or invalid lookup entry.	5420, 6200, 6300, 6400, 8100, 8360 Switch series

For 6300 Switch series:

If there is no data in the **drop group name** or **drop reason name** fields, there have been less than four drop reasons for the flow. Every time the drop reason changes, the next set of drop reason IEs will be populated. Every time the cache inactive timeout value has passed between drops for the same drop reason, the next set of drop reason IEs will be populated. If more than four drop reasons are seen for a given flow, data for the oldest drop reason will be purged, and data for the new drop reason will be added in its place. Analyzing the timestamps for all drop reasons will indicate which reason is the newest.

For Switch Series:

If the **drop end** timestamp has the same value as the **drop start** timestamp, that means only one packet of the flow was dropped. If the drop end timestamp is different from the drop start timestamp, that means more than one packet was dropped within 30 seconds.

Subtopics

[Flow monitors](#)

[Flow Records](#)

[Flow Exporters](#)

[Destinations](#)

[Configuring IP Flow Information Export on 6300 and 6400 Switches](#)

[Role-based IPFIX](#)

[FAQs and Troubleshooting](#)

[Flow monitoring commands](#)

Flow monitors

A flow monitor is applied to an interface to perform network traffic monitoring. A flow monitor consists of a flow record, a flow cache, and optional flow exporters. A flow record must be created and assigned to the flow monitor for the monitoring process to function. Flow data is compiled from the network traffic on the interface and stored in the flow cache based on the match (key) and collect (non-key) fields in the flow record. Data from the flow cache is exported by the flow exporters assigned to the flow monitor. 6300 and 6400 series support a maximum of sixteen flow monitors with a limit of two flow exporters that can be applied to a single flow monitor.

Flow Records

A flow record defines match (key) fields and collection (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collection fields are the set of fields that identify information to collect for a flow, such as packet and byte counters. On the 5420, 6200, 6300, 6400, 8100, and 8360 switch series a maximum of sixteen flow records can be created.

There are six mandatory match fields, of which the IP match fields must be of the same type (IPv4 or IPv6).



NOTE

A flow record is invalid if it does not contain one of the supported sets of match fields.

The supported sets of match fields are:

- IPv4 version
- IPV4 source address
- IPv4 destination address
- IPv4 protocol
- Transport destination port
- Transport source port
- IPv6 version
- IPv6 source address
- IPv6 destination address
- IPv6 protocol
- Transport destination port
- Transport source port

Flow Exporters

A flow exporter defines where and how to export flow reports. Flow exporters are created as standalone entities in the **config** context to provide flow monitors the ability to export flow reports. 6300 and 6400 Switch series support a maximum of sixteen flow monitors with a limit of two flow exporters that can be applied to a single flow monitor.

Destinations

About this task

The destination specifies where flow reports are sent. There is one possible destination for a flow exporter within the 6300 and 6400 switch series:

Procedure

Hostname or IP address of a device with an optional VRF .

Results

A flow exporter can only send flow reports to one destination. The destination type specifies which destination to use. If no destination type is specified, the default destination type is the hostname or IP address of a device . If a VRF is not specified, the default VRF will be used. A destination of each type can be configured, but only the one corresponding to the destination type is used. If there is no destination corresponding to the destination type, then the flow exporter configuration is incomplete. If a new destination of a particular type is configured, it will replace the destination of that type that was previously configured.

- Traffic Insight instance

Configuring IP Flow Information Export on 6300 and 6400 Switches

The following list describes the steps required to configure a IP flow information export (IPFIX) solution:

- Step one: Create flow records
- Step two: Configure flow exporter(s)
- Step three: Configure monitor(s)
- Step four: Apply a flow monitors to interface(s)



NOTE

IPv6 related commands are only applicable to switches that support IPv6 protocol.

Step one: Create Flow Records

Flow Records are used to define the data that will be added to the IPFIX template. This example configures one record for IPv4 and one for IPv6.

```
switch(config)# flow record flowRecordv4
switch(config-flow-record)# match ip protocol
```

```

switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip version
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# match transport source port
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect application name
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
switch(config-flow-record)# collect application https url
switch(config)# flow record flowRecordv6
switch(config-flow-record)# match ipv6 protocol
switch(config-flow-record)# match ipv6 source address
switch(config-flow-record)# match ipv6 destination address
switch(config-flow-record)# match ipv6 version
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# match transport source port
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect application name
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
switch(config-flow-record)# collect application https url
switch(config-flow-record)# collect application dns response-code
switch(config-flow-record)# collect application tls-attributes
switch(config-flow-record)# collect application https url"
switch(config-flow-record)# collect application type
switch(config-flow-record)# collect egress interface
switch(config-flow-record)# collect egress queue
switch(config-flow-record)# collect egress vlan
switch(config-flow-record)# collect forwarding-status

```

Next, use the **show flow record** command to verify the configuration.

Step two: Configure flow exporter(s)

In this step, you can define an exporter to send to an external destination by hostname or IP address, or to an internal destination such as Traffic Insight. The example below configures IPFIX to export data to an external address/hostname:

```

switch(config)# flow exporter flowExternal
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 11.1.1.1
switch(config-flow-exporter)# show flow exporter
-----

```

```

-----
Flow exporter 'flowExternal'
-----
-----
Status                : Accepted
Export Protocol        : ipfix
Destination Type       : Hostname or IP address
Destination            : 11.1.1.1
Transport Configuration
Protocol               : udp
Port                   : 4739

```

To configure IPFIX to export to Traffic Insight, first configure Traffic Insight.

```

switch(config)# traffic-insight TI
switch(config-ti-TI)# source ipfix
switch(config-ti-TI)# monitor topN type topN-flows
switch(config-ti-TI)# monitor appFlow type application-flows
switch(config-ti-TI)# monitor dns type application-flows
switch(config-ti-TI)# monitor raw-mon type raw-flows
switch(config-ti-TI)# enable

```

Next, configure the flow exporter for Traffic Insight

```

switch(config)# flow exporter flowExpTI
switch(config-flow-exporter)# export-protocol ipfix
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight TI

```

You can use the **show flow exporter** command to verify the flow exporter configuration for Traffic Insight

```

switch(config)# show flow exporter flowExpTI
-----
-----
Flow exporter 'flowExpTI'
-----
-----
Status                : Accepted
Export Protocol        : ipfix
Destination Type       : Traffic Insight
Destination            : TI
Transport Configuration
Protocol               : udp
Port                   : 4739

```

Finally, use the **show run traffic-insight** command to verify the Traffic Insight configuration:

```

switch(config)# show running-config traffic-insight
traffic-insight TI
enable

```

```

source ipfix
!
monitor topN type topN-flows entries 5
monitor appFlow type application-flows
monitor raw type raw-flows

```

Step three: Configure the monitor(s)

First, configure an IPv4 flow monitor.

```

switch(config)# flow monitor flowMonv4
switch(config-flow-monitor)# record flowRecordv4
Switch (config-flow-monitor)# exporter flowExternal
switch(config-flow-monitor)# exit

```

Next, configure an IPv6 flow monitor.

```

switch(config)# flow monitor flowMonv6
switch(config-flow-monitor)# record flowRecordv6
switch(config-flow-monitor)# exporter flowExternal
switch(config-flow-monitor)# exit

```

Once both flow monitors are created, use the **show flow monitor** command to verify the flow monitor configurations.

```

switch(config-flow-monitor)# show flow monitor

```

```

-----
----
Flow monitor 'flowMonv4'
-----

```

```

----
Status                : Accepted
Flow Record            : flowRecordv4
Flow Exporter(s)       : flowExternal
Cache Configuration
Inactive Timeout       : 120
Active Timeout         : 1800

```

```

-----
----
Flow monitor 'flowMonv6'
-----

```

```

----
Status                : Accepted
Flow Record            : flowRecordv6
Flow Exporter(s)       : flowExternal
Cache Configuration
Inactive Timeout       : 120
Active Timeout         : 1800

```

Step four: (Optional) Enable Application Recognition and apply a flow monitor to interfaces



NOTE

Enable Application Recognition only if you are using IPFIX to send an application ID. You do not need to enable Application Recognition for IPFIX to be able to report information to an external collector or for internal analytics reports

If you want to use IPFIX to send an application ID to the Application Recognition feature, you must first enable Application Recognition.

```
switch(config)# no ip source-lockdown resource-extended
switch(config)# app-recognition
switch(config-app-recognition)# enable
switch(config-app-recognition)# exit
```

Next, apply flow monitor to IPv4 and IPv6 interfaces

```
switch(config)# int 1/1/1-1/1/28
switch(config-if)# app-recognition enable
switch(config-if)# ip flow monitor flowMonv4 in
switch(config-if)# ipv6 flow monitor flowMonv6 in
switch(config-if)# exit
```

Finally, use the **show run interface** command to verify that the flow monitor was applied to interface.

```
switch(config-if)# show run int 1/1/1
interface 1/1/1
no shutdown
no routing
vlan access 1
app-recognition enable
ip flow monitor flowMonv4 in
ipv6 flow monitor flowMonv6 in
exit
```

A collecting process is not configured with IPFIX.

Role-based IPFIX

If IPFIX monitoring is configured on a port, a network administrator who wants to monitor traffic from clients must preconfigure IPFIX monitor before onboarding the clients. Also, if a client moves from one port to another, the monitor must be reconfigured accordingly. To reduce all these complexities, 5420, 6200, 6300, and 6400 switch series allow you to configure an IPFIX monitor on a port access role.

An IPFIX deployment with monitoring configured on a port uses TCAM to get flow statistics. Since TCAM is a high-demand resource which is shared across multiple protocols including the security protocols, security policy applications could fail if IPFIX consumes most of the TCAM resources. Port access clients can use a

large amount of of TCAM resources for policy applications, making it difficult support port access clients with TCAM based IPFIX.

With role-based IPFIX, when a user plugs a device into to a specific port, their user role is applied to the port. Whenever the user plugs that device into to a different port again, the role is applied to that specific port. This gives mobility and accessibility to the user and provides permissions specific to the user's role when the user's device moves across ports. Role-based IPFIX enables colorless port behavior.

There are some mutually exclusive features that will disable role-based IPFIX when they are enabled on the 6200, 6300, and 6400 platforms. The mutually exclusive features are:

- UBT
- MAC Lockout
- Extended Router MAC
- Source Port Filter
- IP Lockdown
- L3VNI

The following behaviors are observed in a deployment with role-based IPFIX:

- If a user applies the IPFIX monitor configuration on the port, it uses the existing TCAM infrastructure.
- If a port has both Role based IPFIX and Port based IPFIX, the Port based IPFIX implementation takes the priority.
- If IPFIX monitor configuration on a port moves from role based to TCAM based or vice-versa, existing IPFIX flows learned on that port are deleted.
- When the user role has an IPFIX configuration and fails to apply it, port access logs it, but it does not prevent the client from getting authorized.
- If port access is in the client mode and if there is more than one client with a conflicting IPFIX monitor configuration, traffic from all the clients will be exported through the IPFIX monitor configured for the first client.
- Role-based IPFIX and TCAM port-based IPFIX implementations can co-exist on a switch.
- Flows from unauthorized clients are not monitored.

These are some limitations to role-based IPFIX:

- Role based IPFIX can be enabled only through port access role assignment.
- Starting with 10.17, role-based IPFIX supports monitoring in both directions. If only ingress is enabled, it monitors traffic in the ingress direction. Port-based IPFIX continues to support only ingress flow monitoring.

FAQs and Troubleshooting

- When IPFIX is used with Application Recognition, these features do not support LAGs or MLAGs (VSX LAGs).
- The following messages are displayed to indicate an illegal argument:
 - % The flow exporter <EXPORTER-NAME> does not exist.
 - % The flow record <RECORD-NAME> does not exist.
 - % The flow monitor <MONITOR-NAME> does not exist.
 - Invalid destination IP address or hostname entered.
 - Unable to create the flow exporter. The maximum allowed number of flow exporters (<max>) has been reached.
 - Unable to create the flow record. The maximum allowed number of flow records (<max>) has been reached.
 - Unable to create the flow monitor. The maximum allowed number of flow monitors (<max>) has been reached.
 - Flow monitor cannot be applied while interface is part of LAG <LAG-NAME>.
 - Flow monitor could not be applied.
 - Flow monitor could not be unapplied.

Flow monitoring commands

Select a command from the list in the left navigation menu.

Subtopics

alert-threshold aggregate active-flows percentage
alert-threshold endpoint active-flows
alert-threshold endpoint new-connections accepted
alert-threshold endpoint new-connections initiated
diag-dump ipfix basic
description
exporter
flow exporter
flow monitor
flow record

- [flow statistics](#)
- [flow-tracking](#)
- [ip|ipv6 flow monitor \(interface\)](#)
- [show flow exporter](#)
- [show flow monitor](#)
- [show flow record](#)
- [show flow-tracking](#)
- [show tech ipfix](#)
- [show ip flow-table-utilization](#)
- [show IPv4 flow endpoint statistics](#)
- [show IPv6 flow session statistics](#)

alert-threshold aggregate active-flows percentage

Syntax

```
[no] alert-threshold aggregate active-flows percentage <1-100>
```

Description

Sets the alert threshold as a percentage of maximum IP flow table entries per linecard or stack member. An alert is triggered when the number of active flows reaches or exceeds this percentage. The **show ip flow-table-utilization** command displays the IP flow table size on each LC or stack member.

Parameter	Description
<1-100>	Percentage of the maximum allowed aggregate active flows for alert.

Examples

Configuring the aggregate active flows alert threshold per linecard or stack member:

```
switch(config-flow-tracking)# alert-threshold aggregate active-flows percentage 50
```

Removing the aggregate active flows alert threshold:

```
switch(config-flow-tracking)# no alert-threshold aggregate active-flows percentage
```


Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	config-flow-tracking	Administrators or local user group members with execution rights for this command.

alert-threshold endpoint active-flows

Syntax

```
[no] alert-threshold endpoint active-flows <50-1000>
```

Description

Sets the alert threshold for the number of active flows per endpoint. An alert is triggered when an endpoint's active flow count reaches or exceeds this threshold.

Parameter	Description
<50-1000>	Endpoint session count alert threshold.

Examples

Configuring the endpoint active flows alert threshold:

```
switch(config-flow-tracking)# alert-threshold endpoint active-flows 50
```

Removing the endpoint active flows alert threshold:

```
switch(config-flow-tracking)#no alert-threshold endpoint active-flows
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	config-flow-tracking	Administrators or local user group members with execution rights for this command.

alert-threshold endpoint new-connections accepted

Syntax

```
[no] alert-threshold endpoint new-connections accepted <10-500>
```

Description

Sets the alert threshold for the number of new connections accepted by an endpoint within a 30-second monitoring window. An alert is triggered when an endpoint's accepted connection count reaches or exceeds this threshold.

Parameter	Description
<10-500>	Endpoint new connections accepted alert threshold.

Examples

Configuring the endpoint new connections accepted alert threshold:

```
switch(config-flow-tracking)# alert-threshold endpoint new-connections  
accepted 50
```

Removing the endpoint new connections accepted alert threshold:

```
switch(config-flow-tracking)#no alert-threshold endpoint new-connections  
accepted
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	config-flow-tracking	Administrators or local user group members with execution rights for this command.

alert-threshold endpoint new-connections initiated

Syntax

```
[no] alert-threshold endpoint new-connections initiated <10-500>
```

Description

Sets the alert threshold for the number of new connections initiated by an endpoint within a 30-second monitoring window. An alert is triggered when an endpoint's initiated connection count reaches or exceeds this threshold.

Parameter	Description
<10-500>	Endpoint new connections initiated alert threshold.

Examples

Configuring the endpoint new connections initiated alert threshold:

```
switch(config-flow-tracking)# alert-threshold endpoint new-connections  
initiated 50
```

Removing the endpoint new connections initiated alert threshold:

```
switch(config-flow-tracking)#no alert-threshold endpoint new-connections  
initiated
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	config-flow-tracking	Administrators or local user group members with execution rights for this command.

diag-dump ipfix basic

Syntax

```
diag-dump ipfix basic
```

Description

Displays diagnostic information for IPFIX.

Examples

```
diag-dump ipfix basic  
=====
```

[Start] Feature ipfix Time : Tue Apr 11 02:23:03 2023

```
=====
```

[Start] Daemon ipfixd

```

-----
- IPFIX Record Cache dump -
- IPFIX Record ipfix -
....
:- IPFIX Monitor v6ti completed -
- End of IPFIX Monitor Cache dump -
-----

[End] Daemon ipfixd
-----
-----

[Start] Daemon ops-switchd
-----

Key format: <traffic_type>_<coalescence_id>_<agent_id>_<asic_port>
Key                                TCAM Entry ID    Count
-----
1_1532781829_3_20                 0xffff7c7e7a00   1
1_3217499901_1_12                 0xffff91187580   1
1_3217499901_1_13                 0xffff91183d80   1
1_3217499901_1_14                 0xffff91186e80   1
....
-----

[End] Daemon ops-switchd
-----
=====

[End] Feature ipfix
=====

Diagnostic-dump captured for feature ipfix

```

Command History

Release	Modification
10.11	Command introduced on 6300 and 6400 Switch series.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	Manager (#)	Administrators or local user group members with execution rights for this command.

description

Syntax

```
description <DESCRIPTION>
no description <DESCRIPTION>
```

Description

Configures the description for the following telemetry options:

- flow monitor in the **config-flow-monitor** context
- flow congestion monitor in the **config-flow-congestion-monitor** context
- flow exporter in the **config-flow-exporter** context
- flow record in the **config-flow-record** context

The no form deletes a flow congestion monitor.

Parameter	Description
<DESCRIPTION>	Displays a string of 256 characters, maximum, including spaces.

Examples

Adding or modifying the description of flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# description Used for analyzing basic ipv4
traffic
```

Adding or modifying the description of flow record, **flow-record-1**:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# description Used for basic traffic analysis
```

Adding or modifying the description of flow exporter, **flow-exporter-1**:

```
switch(config)# flow exporter flow-exporter-1
switch(config-flow-exporter)# description Exports flows to 10.2.3.45:2055
```

Removing the description of flow monitor, **flow-monitor-1**:

```
switch(config)# flow exporter flow-monitor-1
switch(config-flow-exporter)# no description
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	config-flow-record config-flow-monitor config-flow-congestion-monitor config-flow-exporter	Administrators or local user group members with execution rights for this command.

exporter

Syntax

```
exporter <EXPORTER-NAME>
no exporter <EXPORTER-NAME>
```

Description

Assigns a flow exporter to **flow monitor** in the **config-flow-monitor** context and **flow congestion monitor** in the **config-flow-congestion-monitor** context.

The **no** form of this command removes the flow exporter from the selected monitor.

Parameter	Description
<code><EXPORTER-NAME></code>	Specifies the name of the flow exporter assigned to the monitor.



NOTE

Examples

Assigning flow exporter, **flow-exporter-1**, to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# exporter flow-exporter-1
```

Removing a flow exporter assigned to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# no exporter flow-exporter-1
```

Attempting to assign non-existent flow exporter, **flow-exporter-5**, to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# exporter flow-exporter-5
Flow exporter 'flow-exporter-5' does not exist.
switch(config-flow-monitor)#
```

Assigning flow exporter, **flow-exporter-1**, to flow congestion monitor, **congestion-monitor-1**:

```
switch(config)# flow congestion-monitor congestion-monitor-1
switch(config-flow-congestion-monitor)# exporter flow-exporter-1
```

Removing a flow exporter assigned to flow congestion monitor:

```
switch(config)# flow congestion-monitor congestion-monitor-1
switch(config-flow-congestion-monitor)# no exporter flow-exporter-1
```

Attempting to assign non-existent flow exporter, **flow-exporter-5**, to flow congestion monitor, **congestion-monitor-1**:

```
switch(config)# flow congestion-monitor congestion-monitor-1
switch(config-flow-congestion-monitor)# exporter flow-exporter-5
Flow exporter 'flow-exporter-5' does not exist.
switch(config-flow-congestion-monitor)#
```

Assigning more than one flow exporter to flow monitor, **flow-monitor-2**:

```
switch(config)# flow monitor flow-monitor-2
switch(config-flow-monitor)# exporter flow-exporter-2
```



```
switch(config-flow-monitor)# exporter flow-exporter-3
```

Attempting to assign flow exporter, **flow-exporter-6**, to monitor, **flow-monitor-3**, when two exporters are assigned:

```
switch(config)# flow monitor flow-monitor-3
switch(config-flow-monitor)# exporter flow-exporter 4
switch(config-flow-monitor)# exporter flow-exporter 5
switch(config-flow-monitor)# exporter flow-exporter 6
```

Cannot assign more than two flow exporters to a flow monitor.

```
switch(config-flow-monitor)#
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	config-flow-monitor config-flow-congestion-monitor	Administrators or local user group members with execution rights for this command.

flow exporter

Syntax

```
flow exporter <name>
  export-protocol ipfix
  description <description>
  destination
<hostname>
[vrf <vrfname>]
```

```

<ipaddr>
[vrf <vrfname>]
<ip6addr>
[vrf <vrfname>]
type
hostname-or-ip-addr
traffic-insight}
traffic-insight <instance-name>
    no ...
    template data timeout <timeout>
    transport udp <port>

```

Description

A flow exporter is the part of the IP Flow Information Export (IPFIX) feature that defines how a flow monitor exports flow reports. You can assign the same flow exporter configuration to more than one flow monitor. Each flow exporter includes a destination setting that identifies the device to which the flow reports are sent.

Parameter	Description
<name>	Name of the flow exporter, up to 64 characters.
export-protocol ipfix	Define an export protocol for the flow exporter. The default ipfix protocol is the only protocol currently available.
description <description>	A description of the flow exporter, up to 256 characters and spaces.
destination	Configure the export destination
<hostname>	The exporter sends flow records to the specified hostname destination. The hostname can be a string of up to 64 characters.
<IPAddr>	The exporter sends flow records to this IPv4 address destination.
<ip6addr>	The exporter sends flow records to this IPv6 address destination.
vrf <vrfname>	You can optionally include the name of the destination VRF in the destination definition on 5420, 6200, 6300, 6400, 8100, 8360, 9300S Switch series.
type	Configure the type of the destination.
hostname-or-ip-addr	Define the destination type as a hostname or IP address.

Parameter	Description
<code>traffic-insight <name></code>	Define the destination type as a traffic insight instance.
<code>no ...</code>	Negate any configured parameter.
<code>traffic-insight <INSTANCE-NAME></code>	Specify the a Traffic Insight instance to be used as the destination.
<code>template data timeout <timeout></code>	A flow exporter template describes the format of exported flow reports. Therefore, flow reports cannot be decoded properly without the corresponding templates. This setting defines how often the flow exporter will resend templates to the flow monitor. The supported range is 1 - 86400 seconds, and the default is 600 seconds.
<code>transport udp <port></code>	Transport protocol and port for sending flow record reports. The default port is port 4739.

Usage

The following table shows the maximum supported of flow monitors and flow exporters for each switch model.

Switch	Maximum Flow Monitors	Maximum Flow Exporters
6300	16	Two flow exporters can be applied to a single flow monitor.
6400		

Examples

The following example creates a flow exporter configuration named **exporter-1**:

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# dscp 34
switch(config-flow-exporter)# destination 192.0.2.1 vrf VRF1
switch(config-flow-exporter)# template data timeout 1200
switch(config-flow-exporter)# description Exports flows to 192.0.2.1
```

The following example sets a Traffic Insight instance as the destination for a flow exporter:

```
switch(config)# flow exporter exporter-3
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight instance-1
```

The following example adds a destination of each possible type and set **hostname-or-ip-addr** as the type to use:

```
switch(config)# flow exporter exporter-4  
switch(config-flow-exporter)# destination collector-1  
switch(config-flow-exporter)# destination traffic-insight instance-1  
switch(config-flow-exporter)# destination type hostname-or-ip-addr
```

The following example sets an IPv4 address as the destination for a flow exporter:

```
switch(config)# flow exporter exporter-1  
switch(config-flow-exporter)# destination type hostname-or-ip-addr  
switch(config-flow-exporter)# destination 192.168.0.1
```

The following example sets a hostname as the destination for a flow exporter:

```
switch(config)# flow exporter exporter-1  
switch(config-flow-exporter)# destination type hostname-or-ip-addr  
switch(config-flow-exporter)# destination collector1
```

The following example sets an IPv6 address as the destination for a flow exporter:

```
switch(config)# flow exporter exporter-1  
switch(config-flow-exporter)# destination type hostname-or-ip-addr  
switch(config-flow-exporter)# destination 2001:db87::8a2e:370a:7334
```

The following example sets an IPv4 address as the destination for a flow exporter with the VRF to which the IPv4 address belongs:

```
switch(config)# flow exporter exporter-2  
switch(config-flow-exporter)# destination type hostname-or-ip-addr  
switch(config-flow-exporter)# destination 192.0.2.1 vrf VRF1
```

The following example sets a Traffic Insight instance as the destination for a flow exporter:

```
switch(config)# flow exporter exporter-3  
switch(config-flow-exporter)# destination type traffic-insight  
switch(config-flow-exporter)# destination traffic-insight instance-1
```

The following example adds a destination of each possible type and set **hostname-or-ip-addr** as the type to use:

```
switch(config)# flow exporter exporter-4  
switch(config-flow-exporter)# destination collector-1  
switch(config-flow-exporter)# destination traffic-insight instance-1  
switch(config-flow-exporter)# destination type hostname-or-ip-addr
```

The following example removes the destination type of **traffic-insight** from a flow exporter:

```
switch(config)# flow exporter exporter-3  
switch(config-flow-exporter)# no destination traffic-insight
```

The following example removes the destination type of **hostname-or-ip-addr** from a flow exporter:

```
switch(config)# flow exporter exporter-1  
switch(config-flow-exporter)# no destination
```

Command History

Release	Modification
10.11	Command introduced on 6300 and 6400 Switch series.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	config config-flow-exporter	Administrators or local user group members with execution rights for this command.

flow monitor

Syntax

```
flow monitor <name>  
    exporter <name>  
cache timeout {inactive <timeout>|{active <timeout>  
    description <description>  
    record <name>
```

Description

On HPE Aruba Networking 5420, 6200, 6300, 6400, 8325, 8325H, 8325P, 8100, 8360, 9300, 9300S, and 10040 Switch series, a flow monitor is the part of the IP Flow Information Export (IPFIX) feature that performs network monitoring for the selected interface. A flow monitor configuration consists of a flow record, a flow cache, and one or more associated flow exporters. A flow monitor compiles data from the network traffic on the interface and stores it in the flow cache in a format defined by the flow record. The flow exporters associated with the monitor then export data from the flow cache to the flow exporter destination.



NOTE

HPE Aruba Networking 5420, 6200, 6300, 6400, 8100, 8360 Switch series support a maximum of sixteen flow monitors with a limit of two flow exporters that can be applied to a single flow monitor. If no software augmentation of flows is required, there is no need to configure a flow collector or flow monitor.

Parameter	Description
<code><name></code>	Name of the flow monitor, up to 64 characters.
<pre>cache timeout active <timeout></pre>	Use the cache timeout parameter to define an active or inactive timeout for the flow monitor. A flow monitor closes a flow session that is active for longer than the active timeout or inactive for longer than the inactive timeout. The active timeout range is 30 - 604800. The default active timeout value is 1800 and inactive timeout value is 30.
<pre>cache timeout inactive <timeout></pre>	Use the cache timeout parameter to define an inactive timeout for the flow monitor. A flow monitor closes a flow session that is active for longer than the active timeout or inactive for longer than the inactive timeout. For 5420, 6200, 6300, 6400, 8100, 8360, 9300S, 10040 Switch Series, the inactive timeout range is 30 - 604800. The default active timeout value is 1800 and inactive timeout value is 30.
<code>description</code>	A description up to 256 characters long, including spaces.
<code>exporter <name></code>	Assign a flow exporter to a flow monitor.
<code>record <name></code>	(For HPE Aruba Networking 5420, 6200, 6300, 6400, 8100, 8325, 8325P, 8325H, 8360, 9300, 9300S, 10040 Switch series) Assigns a flow record to a flow monitor.

Examples

The following example creates a flow monitor configuration named **monitor-1**.

```
switch(config)# flow monitor monitor-1
switch(config-flow-monitor)# description Monitor for analyzing basic ipv4
traffic
switch(config-flow-monitor)# exporter flow-exporter-1
switch(config-flow-monitor)# exporter flow-exporter-2
switch(config-flow-monitor)# record flow-record-1
switch(config-flow-monitor)# cache timeout inactive 120
switch(config-flow-monitor)# cache timeout active 1500
```

The following workflow changes the flow record assigned to a flow monitor.

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# record flow-record-2
```

Create more than the maximum number of allowed flow monitors

```
switch(config)# flow monitor monitor-1
switch(config)# flow monitor monitor-2
<--OUTPUT OMITTED FOR BREVITY-->
switch(config)# flow monitor monitor-16
switch(config)# flow monitor monitor-17
No more than 16 flow monitors can be configured. Another flow monitor
must be removed first.
```

Assign more than one flow exporter to flow monitor **flow-monitor-2**

```
switch(config)# flow monitor flow-monitor-2
switch(config-flow-monitor)# exporter flow-exporter-2
switch(config-flow-monitor)# exporter flow-exporter-3
```

Add or modify the description of flow record **flow-record-1**

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# description Used for basic traffic analysis
```

Add or modify the description of flow exporter **flow-exporter-1**

```
switch(config)# flow exporter flow-exporter-1
switch(config-flow-exporter)# description Exports flows to 10.2.3.45:2055
switch(config-flow-exporter)# description Exports flows to Traffic Insight
```

Command History

Release	Modification
10.11	Command introduced on 6300, 6400 Switch series.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	config config-flow-monitor	Administrators or local user group members with execution rights for this command.

flow record

Syntax

```
flow record <record-name>
  match
ip {source|destination} address
ip protocol|version
ipv6 {source|destination} address
ipv6 protocol|version
transport {source | destination} port
  collect
application name
application https url
application dns response-code
application tls-attributes
forwarding-status
drop ingress-exceptions
egress vlan
egress interface
egress queue
counter {packets|bytes}
tcp establishment-time
transport {source | destination} port
timestamp absolute first
timestamp absolute last
  descriptioN <description>
```

Description

Creates or modifies a flow record and switches to the **config-flow-record** context for the flow record. Define data to be included in a flow record by configuring flow record match and collect fields.

A flow record defines match (key) fields and collection (non-key) fields. Customers configure flow records with **match** (key) fields and **collect** (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collect fields are the set of fields that identify information to collect for a flow, such as packet and byte counters.

Traffic with matching attributes (for example, traffic coming from the same interface, sent to the same destination with the same protocol) are classified as a single flow. Information for some or all of the matched

settings can be collected and exported to a destination defined by the flow exporter assigned to the flow monitor.



NOTE

Traffic must match a match rule definition before it can be collected and sent. You cannot collect and send data that is not matched.



NOTE

For 6200, 6300, 6400, 8100, and 8360 Switch series, a maximum of 16 flow records can be created.



NOTE

It is advised to configure collect egress interface and queue also when threshold is being configured.

Parameter	Description
<record-name>	Name of the flow monitor, up to 64 characters.
match	match traffic according to one or more of the following key attributes: • ip source: Match traffic from the same IPv4 source. • ip destination: Match traffic to the same IPv4 destination. • ip protocol: Match traffic using the same IP version • ip version: Match traffic using the same IP protocol • ipv6 source: Match traffic from an IPv6 source. • ipv6 destination: Match traffic to an IPv6 destination. • ipv6 protocol: Match traffic using the same IPv6 version • ipv6 version: Match traffic using the same IPv6 protocol • transport {source destination} port: Match traffic by source or destination transport port
description	A description for the flow record up to 256 characters long, including spaces.
collect	Configures data fields to be included a flow record. • application name: Specify the application name as a non - key field in a flow record.

Parameter	Description
	• application https url: Specify the HTTP/HTTPS application URL as a non - key field in a flow record. • application dns response - code: Specify the DNS parameters and DNS response code as a non - key field in the flow record. • application tls - attributes: Specifies TLS Attributes as a non - key field in a flow record • application tcp establishment - time: Specifies TCP connection establishment time as a non - key field in a flow record • drop ingress - exceptions: On 5420, 6200, 6300, 6400, 8100, 8325, 8325P, 8325H, 9300 and 10000 Switch series, this enables the IPFIX drop - reason feature, which specifies drop ingress - exceptions as a non - key field in a flow record. • counter bytes: Collect counter data for bytes in the flow. Byte count represents the number of incoming bytes since the previous report. • counter packets: Collect counter data for packets in the flow. Packet count represents the number of incoming packets since the previous report. • egress interface: Specifies an egress interface as a non - key field in a flow record. • egress queue: Specifies an egress queue as a non - key field in a flow record. • egress vlan: Specifies an egress VLAN ID as a non - key field in a flow record. • forwarding status: Specifies forwarding status as a non - key field in a flow record • timestamp absolute first: Collect absolute timestamp of the first packet observed. • timestamp absolute last: Collect absolute timestamp of the last packet observed. • tcp establishment - time: Specifies TCP connection establishment time as a non - key field in a flow record.

Examples

Adding IPv4 and transport match fields to **flow-record-1**:

```

switch(config)# flow record flow-record-1
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip version
switch(config-flow-record)# match transport source port
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# description Record used for basic ipv4 traffic
analysis

```

The following example adds egress packet metadata collect fields to **flow record flow-record-1**. Egress interface details will not be available until the MAC entry is not programmed after ARP resolution. As a result, the egress interface information is exported only after the next active/inactive timeout after the MAC entry is programmed.

```

switch(config)# flow record flow-record-1
switch(config-flow-record)# collect egress interface
switch(config-flow-record)# collect egress queue

```

The following example adds a TCP connection establishment time to flow record **flow-record-1** as a collect field. During this process, ARC inspects the first few packets of each flow and only those packets are copied to the switch. Time measurement for established TCP flows is possible during the TCP connection establishment phase, and all three packets are received in the order.

```

switch(config-flow-record)# collect application tcp establishment-time

```

Adding counter and timestamp collect fields to **flow-record-1**:

```

switch(config)# flow record flow-record-1

switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last

```

Creating more than the maximum number of allowed flow records:

```

switch(config)# flow record record-1
switch(config)# flow record record-1
switch(config)# flow record record-2
<--OUTPUT OMITTED FOR BREVITY-->
switch(config)# flow record record-16
switch(config)# flow record record-17
Unable to add another flow record; the limit is 16.

```

Adding IPv4 match fields to flow record **flow-record-1** using the **ip** keyword:

```

switch(config)# flow record flow-record-1
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip protocol

```

```
switch(config-flow-record)# match ip version
```

Adding IPv6 match fields to flow record **flow-record-2**:

```
switch(config)# flow record flow-record-2
switch(config-flow-record)# match ipv6 source address
switch(config-flow-record)# match ipv6 destination address
switch(config-flow-record)# match ipv6 protocol
switch(config-flow-record)# match ipv6 version
```

Adding IPv6 collect fields to flow record **flow-record-2**:

```
switch(config)# flow record flow-record-2
switch(config-flow-record)# collect application tcp establishment-time
switch(config-flow-record)# collect egress-vlan
switch(config-flow-record)# collect egress interface
switch(config-flow-record)# collect forwarding-status
switch(config-flow-record)# collect egress queue
```

Removing the IPv4 destination address match field from flow record **flow-record-1** using the **ip** keyword:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match ip destination address
```

Removing the IPv4 destination address match field from flow record **flow-record-1** using the **ipv4** keyword:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match ipv4 destination address
```

Removing the transport destination port match field from flow record **flow-record-1**:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match transport destination port
```

Adding queue congestion threshold to **flow-record-1** as a collect field:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# collect egress queue threshold
```



NOTE

To export queue threshold attribute, capture-flows threshold profile should be configured on the egress interfaces. If threshold profile is not configured, congested flows can't be detected and the IEs values are exported as 0. If threshold profile is configured, but collect field is not configured, these IEs are not exported. So in either case, congested flows are not known. For more information on features that use this command, refer to the Monitoring Guide for your switch model.

Command History

Release	Modification
10.16	The drop ingress - exceptions parameter was added for the 6300 Switch series.
10.15.1000	The collect tcp establishment - time parameter is introduced on the 6300 and 6400 Switch series.
10.15	The egress vlan, egress interface and egress queue parameters were added.
10.14	The ipv4 parameter is deprecated and replaced with ip .
10.13	Added application https url and dns response - code parameters.
10.11	Command introduced on the 6300 and 6400 Switch Series.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	config config-flow-record	Administrators or local user group members with execution rights for this command.

flow statistics

Syntax

```
flow statistics [enable] | ingress-only  
no flow statistics [enable] | ingress-only
```

Description

Configures collection of statistics for flows that are tracked.

Parameter	Description
enable	Enables collection of statistics for flows that are tracked.

Parameter	Description
<code>ingress-only</code>	Enables collection of statistics for flows that are tracked on ingress interface - only.



CAUTION

If any of the mutually exclusive configurations are present on the switch, flow statistics will not be enabled and an error message is displayed.



NOTE

Flow statistics gets operational only after you run the flow-tracking enable command; systems with flow-tracking already enabled must disable it before configuring the statistics and then re-enable, or toggle it afterwards.

Examples

Enable flow statistics collection:

```
switch(config-flow-tracking)# flow statistics enable
```

Disable flow statistics collection:

```
switch(config-flow-tracking)#no flow-statistics enable
```

Enabling flow collection on ingress interface only:

```
switch(config-flow-tracking)# flow statistics ingress-only
```

Disabling flow collection on ingress interface only:

```
switch(config-flow-tracking)#no flow statistics ingress-only
```

Command History

Release	Modification
10.14	Command introduced on 6300 and 6400 Switch Series

Command Information

Platforms	Command context	Authority
6300	<code>config</code>	Administrators or local user group members with execution rights for this command.
6400 (v2 profile only)		

flow-tracking

Syntax

```
flow-tracking
  enable
  icmp-ageout
  interface-flow-limit
  flow
statistics
enable
ingress-only
  tcp-ageout
  track
icmp
  udp-ageout
no ...
```

Description

Configures flow tracking for TCP and UDP flows, and optionally, ICMP flows. The **no** form of this command deletes the flow tracking configuration context.

In order to optimize the flow removal process, flows that have aged-out are flushed in batches. A flow that has aged out is flushed only when the next batch processes. This can cause some flows to stay inactive for a slightly longer time than the value configured here.

Parameter	Description
enable	Enables flow tracking.
icmp-ageout	Configures an age - out time for ICMP flows, in seconds. Range: 10 to 86400. Default: 15.
interface-flow-limit	Configures global concurrent flow limit for flow tracking enabled interfaces. Range:For 6300 and 6400 Switch series, 64 to 25 000. . Default: none.
flow	Configures flow parameters.
statistics	Configures flow statistics.
enable	Enables flow statistics collection.

Parameter	Description
<code>ingress-only</code>	Restricts flow tracking to the ingress direction.
 NOTE The application visibility feature requires traffic flows in both the ingress and egress directions. Configuring this option prevents features that depend on bi-directional flows from functioning.	
<code>tcp-ageout</code>	Configures age-out time for established TCP flows in seconds. Range: 120 to 86400. Default: 600.
<code>track icmp</code>	Enables tracking of ICMP flows, in addition to the TCP/UDP flows tracked by default.
<code>udp-ageout</code>	Configures age-out time for established UDP flows in seconds. Range: 30 to 86400. Default: 30.

Examples

Configuring flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)#
```

Deleting flow tracking:

```
switch(config)# no flow-tracking
switch(config)#
```

Enabling flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# enable
```

Disabling flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no enable
```

Configuring an established ICMP flow age-out to 600 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# icmp-ageout 600
```

Removing an established ICMP flow age-out of 600 seconds:


```
switch(config) # flow-tracking  
switch(config-flow-tracking) # no icmp-ageout 600
```

Configuring an established TCP flow age-out to 1000 seconds:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # tcp-ageout 1000
```

Removing an established TCP flow age-out of 1000 seconds:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # no tcp-ageout 1000
```

Configuring an established UDP flow age-out to 1000 seconds:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # udp-ageout 1000
```

Removing an established UDP flow age-out of 1000 seconds:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # no udp-ageout 1000
```

Configuring global level interface flow limit to 256 interfaces:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # interface-flow-limit 256
```

Removing global level interface flow limit to 256 interfaces:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # no interface-flow-limit 256
```

Related Commands

Command	Description
IP source lockdown resource extended	no ip source - lockdown resource - extended must be disabled to enable flow - tracking

Command History

Release	Modification
10.17	The flow statistics ingress - only parameter was introduced.
10.14	The flow statistics enable parameter is introduced.
10.14	The track icmp parameter is introduced.
10.13	Command introduced. on 6300 and 6400 Switch Series

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	config	Administrators or local user group members with execution rights for this command.

ip|ipv6 flow monitor (interface)

Syntax

```
[no] ip|ipv6 flow monitor (interface)
```

Description

Enable flow monitoring on inbound and outbound interfaces by assigning a flow monitor to that interface. Only physical interfaces and LAG interfaces can be monitored. A flow monitor cannot be applied to an interface that is part of a LAG. If an unsupported application is attempted, an error message will be displayed. If the flow monitor is associated with a flow record that contains application fields as collect fields, then Application Recognition should be enabled on the same interface.

The **[no]** form of command disables the flow monitoring.

Examples

Enable a flow monitor configuration named **flow-monitor-1** for IPv4 traffic on a physical interface.

```
switch(config)# interface 1/1/1  
switch(config-if)# ip flow monitor flow-monitor-1 in
```

Associate a flow monitor configuration named **flow-monitor-2** for IPv4 traffic on a LAG interface.

```
switch(config)# interface lag 1
switch(config-if)# ip flow monitor flow-monitor-2 in
```

Associate a flow monitor configuration named **flow-monitor-3** for IPv6 traffic on a physical interface.

```
switch(config)# interface 1/1/1
switch(config-if)# ip flow monitor flow-monitor-3 in
```

Associate a flow monitor configuration named **flow-monitor-4** for IPv6 traffic on a physical interface.

```
switch(config)# interface lag 1
switch(config-if)# ipv6 flow monitor flow-monitor-1 in
```

Command History

Release	Modification
10.11	Command introduced on the 6300 and 6400 Switch Series

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	config config-flow-monitor	Administrators or local user group members with execution rights for this command.

show flow exporter

Syntax

```
show flow exporter [<name>]
[statistics]

[vsx-peer]
```

Description

Displays flow exporter statistics, configuration and status. When no exporter name is specified, the output of this command displays information for all flow exporters.

The output of this command can indicate the following status types:

- Accepted
- Rejected (Internal error: exporter does not exist)
- Rejected (Internal error: destination type does not exist)
- Rejected (Destination type is hostname or IP address, but no destination is specified)
- Rejected (Destination type is hostname or IP address, but the specified hostname or IP address is invalid)
- Rejected (Destination type is Traffic Insight, but no destination is specified)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance does not exist)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance is not enabled)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance source is not IPFIX)
- Rejected (Internal error: destination type is Traffic Insight, but the specified Traffic Insight instance is invalid)
- Rejected (Internal error: the specified destination VRF is invalid)
- Rejected (Internal error: the specified destination VRF is not ready)

Parameter	Description
<name>	Name of the flow exporter.
statistics	Adds statistical information about the flow exporter to the output.
vsx-peer	Displays flow collector configuration for the VSX peer.

Examples

Display the configuration of a flow exporter named **exporter-1**.

```
switch# show flow exporter exporter-1
-----
----
Flow exporter 'exporter-1'
-----
----
Description           : Exports to the first collector
Status                : Accepted
Export Protocol       : ipfix
```

```

Destination Type      : Hostname or IP address
Destination           : 192.168.0.1
Transport Configuration
  Protocol             : UDP
  Port                 : 9995

```

Display statistics information for all flow exporters

```
switch# show flow exporter statistics
```

```

-----
----
Flow exporter 'exporter-1'
-----
----
Reports sent           : 14961
-----
----
Flow exporter 'exporter-2'
-----
----
Reports sent           : 5

```

Display information with no flow exporters configured

```

switch# show flow exporter
No flow exporters configured.
switch# show flow exporter statistics
No flow exporters configured.

```

Display a flow exporter information with TI as a destination

```

switch# show flow exporter exporter-5
-----
----
Exporter Name          : exporter-5
-----
----
Description            : Exporter configured with TI as the destination
Status                : Rejected (Destination type is Traffic Insight,
but the specified Traffic Insight instance does not exist)
Export Protocol        : ipfix
Destination Type       : Traffic Insight
Destination            : instance-1
Transport Configuration
  Protocol             : UDP
  Port                 : 2055

```

Display information for a flow exporter that has multiple destinations of different types configured and **hostname-or-ip-addr** is specified as the destination type to use.

```

switch# show flow exporter exporter-6
-----
Flow exporter 'exporter-6'
-----
Description           : TI and hostname configured, but type is hostname-
or-ip-addr
Status                : Accepted
Export Protocol        : ipfix
Destination Type       : Hostname or IP address
Destination            : collector-1
Destination VRF        : mgmt
Transport Configuration
Protocol               : UDP
Port                   : 4821

```

Command History

Release	Modification
10.11	Command introduced on the 6300 and 6400 Switch series.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	Manager (#)	Administrators or local user group members with execution rights for this command.

show flow monitor

Syntax

```

show flow monitor
[statistics | information-elements]

show flow monitor <MONITOR-NAME>

```

```
[statistics]
```

Description

Displays flow monitor configuration and status. When no monitor name is specified, the output of this command displays information for all flow monitors.

Parameter	Description
<name>	Name of the flow monitor.
statistics	Includes the statistics parameter to display additional flow and cache statistics.
information-elements	Displays the configured information elements for the flow monitor.

Usage

The output of this command can indicate the following status types:

- Possible status types for all switches
 - Accepted
 - Rejected (Internal error: monitor does not exist)
 - Rejected (The state of one or more of the assigned flow exporters is rejected)
- Possible Status types for 4100i, 5420 ,6000, 6100, 6200, 6300, 6400, 8100, 8325 ,8325P, 8325H, 8360, 9300, 9300S, and 10040 Switch series:
 - Rejected (A record must be assigned to the monitor, but no record is assigned)
 - Rejected (The state of the assigned record is rejected)
 - Rejected (Internal error: failure in processing the record configuration)

The possible statistics for a flow monitor are:

Statistics name	Meaning	Switches that support this statistic
Current Entries	Current number of flows in the flow cache for this flow monitor	6300 6400 (v2 profile only)

Statistics name	Meaning	Switches that support this statistic
Flows Added	Total number of flows added to the flow cache for this flow monitor since it was created	6300 6400 (v2 profile only)
Total Flows Terminated	Total number of flows removed from the flow cache for this flow monitor since it was created due to any flow end reason	6300 6400
Flows Aged	Number of flows removed from the flow cache for this flow monitor since it was created due to active or inactive timeout	6300 6400
Active Timeout	Number of flows removed from the flow cache for this flow monitor since it was created due to active cache timeout.	
Inactive Timeout	Number of flows removed from the flow cache for this flow monitor since it was created due to inactive cache timeout.	6300 6400
End of Flow Detected	Number of flows removed from the flow cache for this flow monitor since it was created due to the detection of signals indicating the end of the flow. For example, a TCP FIN/RST flag.	
Forced End	Number of flows removed from the flow cache for this flow monitor since it was created due to some external event. For example, the shutdown of a flow monitor.	6300 6400
Current Dropped Entries	Current number of dropped flows in the flow cache for this flow monitor	
Current Forwarded Entries	Current number of forwarded flows in the flow cache for this flow monitor	

Examples

Display information for a flow monitor on 6200, 6300, 6400, 8100 or 8360 Switch series:

```
switch# show flow monitor 'monitor-1'
```

```
-----  
----
```

```
Flow monitor 'monitor-1'
```



```

-----
----
Description          : Used for IPv4 traffic analysis
Status               : Accepted
Flow Record          : record-1
Flow Exporter(s)     : exporter-1, exporter-2
Cache Configuration
Inactive Timeout     : 1800
Active Timeout       : 300
-----
----

```

Display information and statistics for a flow monitor on 5420, 6200, 6300, 6400, 8100 or 8360 Switch series:

```
show flow monitor statistics
```

```

-----
----
Flow monitor 'monitor-1'
-----
----
Current Entries      : 2
Flows Added          : 6
Total Flows Terminated : 4
  Flows Aged         : 2
    Active Timeout   : 1
    Inactive Timeout : 1
  End of Flow Detected : 2
  Forced End         : 0
  Flows Aged         : 4

```



NOTE

The flow monitor statistics counters will be reset to zero after a VSF ISSU switchover.

Command History

Release	Modification
10.11	Command introduced on the 6300 and 6400 Switch series.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	config config-flow-monitor	Administrators or local user group members with execution rights for this command.

show flow record

Syntax

```
show flow record [<name>]
```

Description

Display flow record configuration and status. When no record name is specified, the output of this command displays information for all flow records.

The output of this command can indicate the following status types:

- Accepted
- Rejected (Internal error: failed to process record)
- Rejected (Mix of IPv4 and IPv6 match fields is not allowed. Specify match fields of the same IP version (IPv4 or IPv6))

Parameter	Description
<name>	Name of the flow record.

Examples



NOTE

IPv6 related commands are only applicable to switches that support IPv6 protocol.

Display the configuration of a flow record named **flow-record-1**.

```
switch# show flow record record-1
```

```

-----
Flow record  'record-1'
-----
-----
Description          : Used for IPv4 traffic analysis
Status               : Accepted
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
    transport destination port
    transport source port
Collect Fields
    application name
counter bytes
    counter packets
    application https URL

    drop ingress-exceptions

```

Display the information of a specific flow record.

```

switch# show flow record record-1
-----
-----
Flow record  'record-1'
-----
-----
Description          : Used for IPv4 traffic analysis
Status               : Accepted
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source address

    transport destination port
    transport source port
Collect Fields
    application name

    application https url
    counter bytes
    counter packets

```

Display information for all flow records

```
switch# show flow record
-----
----
Flow record  'record-1'
-----
----
Description           : Used for IPv4 traffic analysis
Status                 : Accepted
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
    transport destination port
    transport source port
Collect Fields
    application name
    counter bytes
    counter packets
-----
----
Flow record  'record-2'
-----
----
Description           : Used for IPv6 traffic analysis
Status                 : Accepted
Match Fields
    ipv6 destination address
    ipv6 protocol
    ipv6 source address
    ipv6 version
    transport destination port
    transport source port
Collect Fields
    application name
    counter bytes
    counter packets
...
```

Display information for a specific flow record

```
switch# show flow record record-3
-----
----
Flow record  'record-3'
```

```

-----
----
Description          : Used for IPv4 traffic analysis
Status               : Rejected (Incomplete match fields. The mandatory
match fields are: version, source address,
destination address, protocol, transport destination port, and transport
source port.)
Match Fields
ipv4 destination address
ipv4 protocol
ipv4 source address
ipv4 version
Collect Fields
application name
counter bytes
counter packets

```

Display information with no flow records configured

```

switch# show flow record
No flow records configured

```

Command History

Release	Modification
10.17	The output of this command includes drop ingress - exceptions information for the 6300 Switch series.
10.11	Command introduced on the 6300 and 6400 Switch series.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	Manager (#)	Administrators or local user group members with execution rights for this command.

show flow-tracking

Syntax

```
show flow-tracking
```

Description

Displays flow-tracking and statistics collection configurations and status.

Examples

```
switch (config)# show flow-tracking
Flow Tracking Global Configuration
Configuration status          : Enabled
Operational status           : Enabled
Failure Reason                : NA
UDP Ageout                    : 60   (Seconds)
TCP Ageout                    : 600  (Seconds)
ICMP Ageout                   : 15   (Seconds)
Interface Flow limit          : None
Tracked Protocols             : TCP, UDP
Alert Thresholds:
    Aggregate active flows percentage : 50
    Endpoint active flows              : 250
    Endpoint new connection initiated : 50
    Endpoint new connection accepted  : 60
Statistics Collection
Configuration Status          : Enabled
Operational Status            : Enabled
Ingress-only                  : Enabled
Failure Reason                : NA
Flow Tracking Port Configuration
Interface      App Recognition    IPFIX      Operation Status
-----
1/1/1          Disabled           Disabled   Disabled
1/1/2          Disabled           Disabled   Disabled
1/1/3          Disabled           Disabled   Disabled
1/1/4          Disabled           Disabled   Disabled
1/1/5          Disabled           Disabled   Disabled
1/1/6          Disabled           Disabled   Disabled
1/1/7          Disabled           Disabled   Disabled
1/1/8          Disabled           Disabled   Disabled
1/1/9          Disabled           Disabled   Disabled
1/1/10         Disabled           Disabled   Disabled
1/1/11         Disabled           Disabled   Disabled
1/1/12         Disabled           Disabled   Disabled
1/1/13         Disabled           Disabled   Disabled
```

1/1/14	Disabled	Disabled	Disabled
1/1/15	Enabled	Disabled	Enabled
1/1/16	Disabled	Disabled	Disabled
1/1/17	Disabled	Disabled	Disabled
1/1/18	Disabled	Disabled	Disabled
1/1/19	Disabled	Disabled	Disabled
1/1/20	Disabled	Disabled	Disabled
1/1/21	Disabled	Disabled	Disabled
1/1/22	Disabled	Disabled	Disabled
1/1/23	Disabled	Disabled	Disabled
1/1/24	Disabled	Disabled	Disabled
1/1/25	Disabled	Disabled	Disabled
1/1/26	Disabled	Disabled	Disabled
1/1/27	Disabled	Disabled	Disabled
1/1/28	Disabled	Disabled	Disabled

Display the configuration of role based flow tracking on a 6300 or 6400 Switch series.

```
switch(config)# show flow-tracking
Flow Tracking Global Configuration
Configuration status           : Enabled
Operational status            : Enabled
Failure Reason                 : NA
UDP Ageout                     : 30   (Seconds)
TCP Ageout                     : 600  (Seconds)
ICMP Ageout                    : 15   (Seconds)
Interface Flow limit           : None
Tracked Protocols              : TCP, UDP
Statistics Collection
  Configuration Status         : Enabled
  Operational Status           : Enabled
  Ingress-only                 : Enabled
  Failure Reason               : NA
Flow Tracking Port Configuration
Interface      App Recognition  Reflexive ACL      IPFIX
Operation Status
-----
1/1/1          Enabled          Disabled           Enabled
Enabled
1/1/2          Enabled          Disabled           Disabled
Enabled
1/1/3          Enabled          Disabled           Disabled
Enabled
1/1/4          Enabled          Disabled           Disabled
```

Enabled			
1/1/5	Enabled	Disabled	Disabled
Enabled			
1/1/6	Enabled	Disabled	Disabled
Enabled			
1/1/7	Enabled	Disabled	Enabled
Enabled			
1/1/8	Enabled	Disabled	Disabled
Enabled			
1/1/9	Enabled	Disabled	Disabled
Enabled			
1/1/10	Disabled	Disabled	Disabled
Disabled			
1/1/13	Enabled	Disabled	Enabled
Enabled			
1/1/14	Enabled	Disabled	Disabled
Enabled			
1/1/15	Enabled	Disabled	Disabled
Enabled			
1/1/16	Enabled	Disabled	Disabled
Enabled			
1/1/17	Enabled	Disabled	Disabled
Enabled			
1/1/18	Enabled	Disabled	Disabled
Enabled			
1/1/19	Enabled	Disabled	Disabled
Enabled			
1/1/20	Enabled	Disabled	Disabled
Enabled			
1/1/21	Enabled	Disabled	Disabled
Enabled			
1/1/23	Enabled	Disabled	Disabled
Enabled			
1/1/24	Enabled	Disabled	Enabled
Enabled			
1/1/28	Disabled	Disabled	Disabled
Disabled			

Related Commands

Command	Description
IP source lockdown resource extended	IP source lockdown must be disabled with the no ip source - lockdown resource - extended command before enabling flow - tracking

Command History

Release	Modification
10.17	Added Ingress - only information for Statistics Collection.
10.14	Added role based IPFIX and ICMP ageout information.
10.13	Command introduced on 6300 and 6400 Switch Series.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	Manager (#)	Administrators or local user group members with execution rights for this command.

show tech ipfix

Syntax

```
show tech ipfix
```

Description

Shows the IPFIX configuration settings.

If applicable source IP address or source interface is configured for the IPFIX protocol, that configuration is used.

For 6200, 6300, 6400, 8360, 8100 and 9300S Switch Series, If a valid source is configured, the exporter sends flows to an external collector using the effective configured source IP address as the source IP address

of the flow packets. In the context of this application, a valid source IP address is any IP address configured in the exporter's VRF namespace.

Examples

The example shows the IPFIX configuration settings.

```
switch#show tech ipfix
=====
Show Tech executed on Tue Apr 11 02:43:06 2023
=====
[Begin] Feature ipfix
=====
*****
Command : show flow exporter
*****

-----
Flow exporter 'ipfix'
-----
-----
Status                : Accepted
Export Protocol        : ipfix
Destination Type       : Traffic Insight
Destination            : t1
Transport Configuration
Protocol               : udp
Port                   : 4739
-----
Flow exporter 'V6E1'
-----
....
=====
[End] Feature ipfix
=====
```

Command History

Release	Modification
10.11	Command introduced on 6300, 6400, 8100, and 8360 Switch series.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	Manager (#)	Administrators or local user group members with execution rights for this command.

show ip flow-table-utilization

Syntax

```
show ip flow-table-utilization
```

Description

Displays the current IP flow table utilization per linecard (LC), including both the maximum capacity and the current number of active flows.

Examples

Configuring the endpoint new connections accepted alert threshold:

```
switch# show ip flow-table-utilization
Line Modules
=====
Slot name | flow-table capacity | flow-count
-----|-----|-----
1/1       | 28672               | 15000
1/3       | 28672               | 5000
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	Manager (#)	Administrators or local user group members with execution rights for this command.

show IPv4 flow endpoint statistics

Syntax

```
show ip flows endpoint <A.B.C.D> [vrf <vrf_name>]
```

Description

Displays the number of active flows of VRF with name **vrf_name**, the flows observed in the last 30 seconds, and details about new connections initiated and accepted by this endpoint during the same period. If the **vrf_name** is not specified, default VRF active flows are displayed.

Parameter	Description
IPv4 address	Show endpoint statistics for the IPv4 address. Required
vrf	Display the routes in the VRF. Optional
vrf_name	Specify the VRF name.

Parameter	Description
	Required



NOTE

The output displays the following information:

- **VRF:** The specified vrf name.
- **Endpoint address:** The specified IPv4 address.
- **Active flows:** Number of active flows for the endpoint.
- **Flows seen last 30sec:** Number of flows observed in the last 30 seconds.
- **No of new connections (last-30sec):**
 - **Initiated:** Number of new connections initiated by the endpoint in the last 30 seconds.
 - **Accepted:** Number of new connections accepted for the endpoint in the last 30 seconds.

Examples

This example shows the number of active flows of the default VRF:

```
switch# show ip flows endpoint 192.168.1.1
VRF                                     : default
Endpoint address                       : 192.168.1.1
Active flows                           : 10
Flows seen last 30sec                  : 20
Number of new connections (last-30sec):
    Initiated                          : 2
    Accepted                           : 0
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	Manager (#)	Administrators or local user group members with execution rights for this command.

show IPv6 flow session statistics

Syntax

```
show ipv6 flows endpoint <A.B.C.D> [vrf <vrf_name>]
```

Description

Displays the number of active flows of VRF with name **vrf_name**, the flows observed in the last 30 seconds, and details about new connections initiated and accepted by this endpoint during the same period. If the **vrf_name** is not specified, default VRF active flows are displayed.

Parameter	Description
IPv6 address	Show endpoint statistics for the IPv6 address. Required
vrf	Display the routes in the VRF. Optional
vrf_name	Specify the VRF name.

Parameter	Description
	Required



NOTE

The output displays the following information:

- **VRF:** The specified vrf name.
- **Endpoint address:** The specified IPv6 address.
- **Active flows:** Number of active flows for the endpoint.
- **Flows seen last 30sec:** Number of flows observed in the last 30 seconds.
- **No of new connections (last-30sec):**
 - **Initiated:** Number of new connections initiated by the endpoint in the last 30 seconds.
 - **Accepted:** Number of new connections accepted for the endpoint in the last 30 seconds.

Examples

This example shows the number of active flows of the specified VRF:

```
switch# show ipv6 flows endpoint 2001::1 vrf red
VRF                                     : red
Endpoint address                       : 2001::1
Active flows                           : 10
Flows seen last 30sec                  : 20
Number of new connections (last-30sec):
    Initiated                          : 2
    Accepted                           : 0
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400 (v2 profile only)	Manager (#)	Administrators or local user group members with execution rights for this command.

Queue Monitoring

AOS-CX provides multiple features that assist with monitoring queuing behavior such as Queue Monitoring.

The Queue Monitoring feature allows the switch to collect queue statistics at specific time intervals, including queue depth, and display this information as a queue statistics history. This data provides a history of activity for each monitored queue and can be viewed directly on the switch. These aggregate statistics can be used to monitor network health, troubleshoot network issues, and identify normal network behavior patterns and performance anomalies. The feature is disabled by default, but can be enabled on a maximum of 52 interfaces per switch. For more information on enabling this feature, see [queue-monitor](#).

Queue statistics history data can be displayed in the command-line interface as a list or a histogram reflecting the last eight hours of queue statistics history, or as a table showing the previous five minutes of queue statistics history.

As performance anomalies are identified, the interface queue experiencing the anomaly can further analyzed to identify the cause of the behavior. For example, traffic patterns for flows competing for network bandwidth can be examined to determine if corrective action is needed to address potential problems.

If no queue depth is specified, the output of the **show interface queue-monitor** command displays statistics for queues with a minimum depth of one kbyte, which includes all queues which have reached a depth greater than zero.

Subtopics

[Queue statistics history](#)

[Queue depth](#)

[Data retention limits](#)

[Using collected data](#)

[Congestion Event Detection](#)

[Congestion event detection commands](#)

Queue statistics history

Queue monitoring stores collected queue depth data in a time-series database within the switch. This data, accessible on the switch for reviewing the queue statistics history, helps with identifying network issues or behavior patterns.

Queue depth

Queue depth is the memory size required to buffer packets in a queue during congestion. The value changes depending on the arrival rate of the packets destined to egress a given queue, and the rate at which that queue is able to transmit packets.

Data retention limits

Data retention limits are enforced within the switch by reducing time precision of the stored data within specific windows of time. The retention duration for queue statistics data is derived from the frequency of the polling interval.

Table 1. Queue monitoring data retention schedule

Polling interval	Maximum statistic age
1 seconds	8 hours
5 seconds	8 hours
10 seconds	8 hours
30 seconds	8 hours
60 seconds	8 hours



NOTE

The switch has limited storage for retaining information. For longer monitoring durations, use an external monitor with the queue monitoring feature to collect and store data.

Using collected data

Queue monitoring provides the ability to look into switch congestion-related performance issues and identify when problems have occurred. The methods available for reviewing performance issues are:

- Congestion events
- Per-interface congestion histograms
- Per-interface congestion-detailed view

As anomalies are identified, the interface queue that experienced the anomaly is the starting point for identifying the cause of the behavior. Examine traffic patterns for flows competing for network bandwidth to determine if corrective action is needed to address potential problems.

Congestion Event Detection

The Congestion Event Detection feature is an extension of Queue Monitoring that allows users to configure alert thresholds for queue depth, queue rate, and queue drops. Alerts can be viewed on-switch and can also be sent to the switch event log on a selective basis.

Congestion Event Detection is supported on 8325, 8325H, 8325P and 10000 switch series. This feature enables users to configure queue statistics thresholds which are actively checked by the switch on interfaces where monitoring and thresholds are applied. This mechanism leverages queue monitoring or a Flow Congestion-Monitor as the source of periodic queue statistics samples for comparison against the configured thresholds. If a statistic has crossed a threshold, information about the occurrence is saved within the switch and an optional action can also be taken at the time of the occurrence. Accumulated occurrences (events) can be viewed on the switch through the CLI or requested through the REST API interface.

Congestion event detection commands

Select a command from the list in the left navigation menu.

Subtopics

[apply congestion-event profile](#)
[congestion-event profile](#)

apply congestion-event profile

Syntax

```
apply congestion-event profile [<NAME>]  
no apply congestion-event profile [<NAME>]
```

Description

Applies a congestion event profile to an interface.

The **no** form removes the congestion event profile from the interface.

Parameter	Description
<NAME>	Specifies the name of a congestion event profile. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Usage

In order for the congestion event profile entry associated with an applied congestion event profile to be effective, the interface needs to be configured for **queue-monitor**. Once the queue statistics are being monitored, apply the congestion event profile to the interface to begin the process of actively checking the monitored queue statistics for congestion events based on the thresholds configured in the applied congestion event profile. User-specified congestion event profiles and profile entries may be modified and deleted while applied on an interface. If a congestion event profile or entry is modified while applied on an interface, any ongoing event occurrences will end immediately.

Examples

Applying congestion event profile, **prof1**, to an interface where the queues on the interface have a monitoring source:

```
switch(config-if) # apply congestion-event profile prof1
```

Applying non-existent congestion event profile, **prof22**, to an interface:

```
switch(config-if) # apply congestion-event profile prof22  
Profile prof22 does not exist.
```

Applying congestion event profile, **prof1**, to an interface where the queues on the interface do not have a queue data monitoring source. Without a source for queue statistics, no congestion event threshold checking will be performed.

```
switch(config-if) # apply congestion-event profile prof1  
Warning: Queue monitoring must also be enabled using the 'queue-monitor'  
command for congestion events to be detected.
```

Removing any congestion event profile from an interface.

```
switch(config-if) # no apply congestion-event profile
```

Removing congestion event profile, **prof1**, from an interface:

```
switch(config-if) # no apply congestion-event profile prof1
```

Removing congestion event profile, **prof2**, from an interface where congestion event profile, **prof1**, is currently applied.

```
switch(config-if) # no apply congestion-event profile prof2
```

The profile to remove does not match the currently configured profile.

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
	config-if	Administrators or local user group members with execution rights for this command.

congestion-event profile

Syntax

```
congestion-event profile <NAME>
no congestion-event profile <NAME>
```

Description

Creates a new congestion event profile and switches to the **config-cng-prof** context for the profile.

The no form deletes the profile and removes it from all applied interfaces.



NOTE

If the specified profile already exists, this command switches to the **config-cng-prof** context for the named profile.

Parameter	Description
<name>	Name of the congestion event profile, up to 64 characters.

Usage

A congestion event profile contains zero or more congestion event entries that specify the thresholds to use for identifying conditions of interest that should be saved as an event occurrence. Use **show congestion-event profile [NAME]** to view the status of all congestion event profiles or their settings.

Examples

Creating a congestion-event profile named **prof1**:

```
switch(config)# congestion-event profile prof1
```

Deleting a congestion event profile named **prof1**:

```
switch(config)# no congestion-event profile prof1
```

Creating a congestion event profile with an invalid name:

```
switch(config)# congestion-event profile prof##
Invalid name. Please enter a string of up to 64 alphanumeric,
underscore, hyphen, and period characters.
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
	config	Administrators or local user group members with execution rights for this command.

IP Flow Path Trace

IP Flow Path Trace (flowtraced) feature helps to trace the complete path taken by an Application Flow while traversing from Source device to Destination device. There is a new dedicated daemon (flowtraced) created to support the feature. A new proprietary UDP protocol is designed to handle the feature requirements. These CPU generated packets traverse through the network and collect all the necessary telemetry information from each of the HPE ANW OS-CX devices traversed where flow path trace is supported. All the information collected from each of the device traversed is stored in

the payload as TLVs.



NOTE

IP Flow Path Trace is supported on 5420, 6200, 6300, 6400, 8360 and 8100 Switch Series.

Subtopics

Limitations, Conflicts or Exclusions

IP Flow Path Trace commands

Limitations, Conflicts or Exclusions

- Platform should support Forwarding Info infra to fetch the nexthop and interface details.
- Traffic Insight feature needs to be running on all devices in the path to match on the Policy/ACL drop matches. If Traffic Insight is not supported on the devices, then the Flow drop will not be detected.
- If Traffic Insight feature is not supported or enabled along with Application recognition, Flow Tracking, IPFix, TI with AppFlow monitor on the first-hop device, then the flow trace query should have the complete 5 tuple information. Partial query having only the Client IP and destination (AppName, DomainName) will result in lookup failure.
- Client-Insight ARPtoGW needs to be enabled on the first Access switch if acting as an L2 device to make sure that the MAC-IP association of the GW are available.
- Client-IP-Tracker needs to be enabled on the first Access switch if acting as an L2 device to make sure that the MAC-IP association of the Client/Source are available.
- If the Clients have static IPs, the **gateway-ip** needs to be provided in the flow trace query.
- The feature sends out proprietary packets using UDP Source L4 Port set to 55000 and Destination L4 Port set to 55001, which are private ports as per IANA.
- In case of Flow Trace capability unaware devices present on the path, the pathtrace post that device could result in an incorrect path.
- If the last-hop device IP is not provided along with the 5-tuple query, then packet would get leaked to the destination device if the last hop switch is not Flow Trace capable.
- The MAC and Routing tables may be constantly changing. The egress port received is valid for that snapshot in time. It may not remain the same in the future.
- This feature is not applicable for broadcast, multicast or unknown unicast packets.

- IPv6 support is not provided.
- Below drops are not captured in current release:
 - GBP Policy drops
 - QoS drops
 - Per-Hop latency
 - Path Latency
 - GBP relay issues
 - PBR drops
- Support for UBT, IPSEC or GRE tunnels is not present.
- Periodic probes for the 5-tuple query is not supported.
- Number of non-flowtrace capable devices in the path (applicable for L3 only).
- Automatic Reverse Flow Trace is not supported.
- If Application provided is not active in TI for the provided source IP, then the flow trace query will fail.
- If destination details are not provided, then the flow trace query will fail.
- Uses IP exception CoPP class which can be updated while executing the **flowtrace** query.
- If the ARP table in case of L3 or the CIPT/CIARPGW table in case of L2 does not have the MAC-IP binding on the first-hop device, then the query is not initiated and failed.
- LLDP neighbor information is used to figure out if the adjacent device is a Flow Trace capable or not. So all the devices in the network need to have LLDP enabled to make sure that the device classification is correct.
- If the TLV packet payload increases above the MTU of the egress interface, the flow trace will be stopped on that device and a partial trace information will be sent back to the initiating device. Each node will add multiple TLV information depending on the attributes getting collected.
- It is advisable to have the MTU in the network to be jumbo capable to make sure the flow trace is complete especially when there are more than 7-8 nodes between the source and destination device.

IP Flow Path Trace commands

Select a command from the list in the left navigation menu.

Subtopics

IP Flow Path Trace

IP Flow Path Trace

Syntax

```
flowtrace {source-ip-address <IP-ADDR>} {destination-type <ipv4|domainname|  
appname> destination <string>}  
[source-l4-port <L4-PORT> destination-l4-port <L4-PORT>] [transport-  
protocol <PROTOCOL>]  
[vrf <VRF-NAME>] [last-hop-device-ip <IP-ADDR>] [gateway-ip <IP-ADDR>]
```

Description

This command is used to get the complete flow path trace of the 5 tuple query which is initiated from the CLI. The Source IP and the Destination details are mandatory for the command to initiate the query. The other fields are optional parameters. If **last-hop-device-ip** is not provided, then the proprietary packets will traverse until the last hop, HPE ANW OS-CX, where flow path trace is supported, or until the destination device, in case the device before destination does not support flow path trace. The **gateway-ip** is also optional. This is useful especially when the first-hop device is an L2 device to get the next-hop MAC details needed for forwarding info lookup.



NOTE

The flow path trace command does not validate the transport protocol of the flows and provides output using the user-specified protocol number instead of verifying the actual transport protocol.

Parameter	Description
IP-ADDR	Specify the IPv4 address (A.B.C.D). Required
L4-PORT	Specify the L4 Port (1 - 65535). Optional
PROTOCOL	Specify the transport protocol number (1 - 255) Optional
VRF-NAME	Specify the VRF. Optional

Parameter	Description
DOMAIN-NAME	Specify the Domain Name. Optional
APP-NAME	Specify the Application Name. Optional

Examples

This example shows the complete Flow Path trace successful until the last-hop-device:

```
switch# flowtrace source-ip-address 50.0.0.5 destination-type ipv4
destination 80.0.0.2 transport-protocol 6 source-l4-port 23456
destination-l4-port 34567 vrf default last-hop-device-ip 60.0.0.1
IP Flow Trace Summary:
-----
src_ip      : 50.0.0.5          dst_ip      : 80.0.0.2
src_port    : 23456            dst_port    : 34567
protocol    : 6
-----
----
PathNode 1 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXXXAL
System MAC       : 90:aa:aa:aa:aa:80      Egress Subnet  : NA
Ingress Interface : 1/1/43                Egress Interface : 1/1/45
Ingress VLAN     : 10                    Egress VLAN     : 10
Flow Status      : Forwarded
PathNode 2 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXXXBL
System MAC       : f8:bb:bb:bb:bb:00      Egress Subnet  : ECMP
70.0.0.0/24
Ingress Interface : 1/3/47                Egress Interface : 1/5/47
Ingress VLAN     : 10                    Egress VLAN     : 20
Flow Status      : Forwarded
PathNode 3 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXXXCL
System MAC       : f8:cc:cc:cc:cc:00      Egress Subnet  :
60.0.0.0/24
Ingress Interface : 1/6/3                Egress Interface : lag20-
>1/3/2
```

Ingress VLAN	: 20	Egress VLAN	: 30
Flow Status	: Forwarded		
PathNode 4 :			
Device Type	: Last_Hop_Device	Device ID	:
SGXXXXXXDL			
System MAC	: 90:dd:dd:dd:dd:00	Egress Subnet	:
80.0.0.0/24			
Ingress Interface	: 1/2/2	Egress Interface	: 1/1/1
Ingress VLAN	: 30	Egress VLAN	: 40
Flow Status	: Forwarded		

In this example the flow Path trace failed because of a Policy drop in an intermediate device:

```
switch# flowtrace source-ip-address 50.0.0.5 destination-type ipv4
destination 80.0.0.2 transport-protocol 6 source-l4-port 23456
destination-l4-port 34567 vrf default last-hop-device-ip 60.0.0.1
IP Flow Trace Summary:
-----
src_ip      : 50.0.0.5          dst_ip      : 80.0.0.2
src_port    : 23456            dst_port    : 34567
protocol    : 6
-----
-----
PathNode 1 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXAL
System MAC       : 90:aa:aa:aa:aa:80      Egress Subnet  : NA
Ingress Interface : 1/1/43                Egress Interface : 1/1/45
Ingress VLAN     : 10                    Egress VLAN     : 10
Flow Status      : Forwarded
PathNode 2 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXBBL
System MAC       : f8:bb:bb:bb:bb:00      Egress Subnet  : ECMP
70.0.0.0/24
Ingress Interface : 1/3/47                Egress Interface : 1/5/47
Ingress VLAN     : 10                    Egress VLAN     : 20
Flow Status      : Forwarded
PathNode 3 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXCL
System MAC       : f8:cc:cc:cc:cc:00      Egress Subnet  : ECMP
70.0.0.0/24
Ingress Interface : 1/3/47                Egress Interface : 1/5/47
Ingress VLAN     : 10                    Egress VLAN     : 20
```

```
Flow Status      : Dropped
Drop Reason:
Policy : testPolicy1
Class  : testClass1
RuleID : 30
```

This example shows Flow Path trace via an intermediate Non-Flow trace capable device.

```
switch# flowtrace source-ip-address 50.0.0.5 destination-type ipv4
destination 80.0.0.2 transport-protocol 6 source-l4-port 23456
destination-l4-port 34567 vrf default last-hop-device-ip 60.0.0.1
IP Flow Trace Summary:
-----
src_ip      : 50.0.0.5                dst_ip      : 80.0.0.2
src_port    : 23456                  dst_port    : 34567
protocol    : 6
-----
----
PathNode  1 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXXXAL
System MAC       : 90:aa:aa:aa:aa:80      Egress Subnet  : NA
Ingress Interface : 1/1/43                Egress Interface : 1/1/45
Ingress VLAN     : 10                    Egress VLAN     : 10
Flow Status      : Forwarded
PathNode  2 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXXXBBL
System MAC       : f8:bb:bb:bb:bb:00      Egress Subnet  : ECMP
70.0.0.0/24
Ingress Interface : 1/3/47                Egress Interface : 1/5/47
Ingress VLAN     : 10                    Egress VLAN     : 20
Flow Status      : Forwarded
PathNode  3 :
Device Type      : NonFlowTrace_Capable   Device ID      :
SGXXXXXXXXCL
System MAC       : f8:cc:cc:cc:cc:00
Flow Status      : Forwarded
PathNode  4 :
Device Type      : Last_Hop_Device        Device ID      :
SGXXXXXXXXDL
System MAC       : 90:dd:dd:dd:dd:00      Egress Subnet  :
80.0.0.0/24
Ingress Interface : 1/2/2                Egress Interface : 1/1/1
```

Ingress VLAN	: 30	Egress VLAN	: 40
Flow Status	: Forwarded		

This example shows a successful complete Flow Path trace for an Application Based query:

```
switch# flowtrace source-ip-address 10.0.0.10 destination-type app_name
destination ssh
IP Flow Trace Summary:
-----
src_ip      : 10.0.0.10          dst_ip      : 60.0.0.2
src_port    : 23456              dst_port    : 22
protocol    : 6                  app_name     : ssh
-----
-----
PathNode 1 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXXXAL
System MAC       : 90:ea:ea:ee:aa:50      Egress Subnet  : NA
Ingress Interface : 1/1/1                  Egress Interface : 1/1/48
Ingress VLAN     : 10                      Egress VLAN     : 10
Flow Status      : Forwarded
PathNode 2 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXXXBBL
System MAC       : f8:bb:bb:bb:bb:00      Egress Subnet  :
70.0.0.0/24
Ingress Interface : 1/1/47                  Egress Interface : 1/5/47
Ingress VLAN     : 10                      Egress VLAN     : 20
Flow Status      : Forwarded
PathNode 3 :
Device Type      : Last_Hop_Device        Device ID      :
SGXXXXXXXXCL
System MAC       : f8:cc:cc:cc:cc:00      Egress Subnet  :
60.0.0.0/24
Ingress Interface : 1/6/3                    Egress Interface : 1/6/6
Ingress VLAN     : 20                      Egress VLAN     : 30
Flow Status      : Forwarded
```

This example shows a successful complete Flow Path trace for a VxLAN based topology:

```
switch# flowtrace source-ip-address 10.0.0.10 destination-type app_name
destination ssh
IP Flow Trace summary:
-----
src_ip      : 10.0.0.10          dst_ip      : 60.0.0.2
src_port    : 23456              dst_port    : 22
protocol    : 6
```

```

-----
----
PathNode 1 :
Device Type      : FlowTrace_Capable      Device ID      :
SGXXXXXXXXAL
System MAC       : 90:aa:aa:aa:aa:80      Egress Subnet  : NA
Ingress Interface : 1/1/43                Egress Interface : 1/1/45
Ingress VLAN     : 10                    Egress VLAN     : 10
Flow Status      : Forwarded
PathNode 2:
Device Type      : Tunnel_Entry_Device    Device ID      : SGXXXXXXXXBL
System MAC       : BB:A4:7D:29:C4:50      Egress IP      :
30.0.0.1
Ingress Interface : 1/1/10                Egress Interface : 1/1/12
Egress VLAN      : 30                    Flow Status     :
Forwarded
PathNode 3:
Device Type      : Flow_Trace Capable     Device ID      :
SGXXXXXXXXCL
System MAC       : CC:A4:7D:29:C4:50      Egress IP      :
40.0.0.1
Ingress Interface : 1/1/13                Egress Interface : 1/1/15
Egress VLAN      : 40                    Flow Status     :
Forwarded
PathNode 4:
Device Type      : Tunnel_End_Device      Device ID      :
SGXXXXXXXXDL
System MAC       : DD:A4:7D:29:C4:50      Egress IP      :
50.0.0.1
Ingress Interface : 1/1/4                Egress Interface : 1/1/5
Egress VLAN      : 30                    Flow Status     :
Forwarded
PathNode 5:
Device Type      : Last_Hop_Device        Device ID      :
SGXXXXXXXXEL
System MAC       : EE:A4:7D:29:C4:60      Egress IP      :
60.0.0.1
Ingress Interface: 1/1/20                Egress Interface : 1/1/21
Egress VLAN: 50                          Flow Status     :
Forwarded

```

This example shows an error due to a missing Source IP:

```

switch# flowtrace destination-type ipv4 destination 50.0.0.50
transport-protocol 6 source-l4-port 23456 destination-l4-port 34567

```

```
vrf default last-hop-device-ip 50.0.0.1
```

```
Missing Required Parameter Source IP
```

This example shows an error since there is no active Application:

```
switch# flowtrace source-ip-address 100.0.0.10 destination-type appname  
destination ssh
```

```
No active session to the 'ssh' application from source 100.0.0.10
```

This example shows an error since Forwarding Info infra is not successful

```
switch# flowtrace source-ip-address 10.0.0.10 destination-type ipv4  
destination 50.0.0.50 transport-protocol 6 source-l4-port 23456  
destination-l4-port 34567 vrf default last-hop-device-ip 50.0.0.1  
Forwarding-info lookup was not successful
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

gRPC network management interface

gRPC is a RPC framework developed by Google to create distributed systems. This protocol allows a client running on one system to call the services defined in a remote server as a local server.

The gRPC Network Management Interface (gNMI) feature provides secure, high-performance, standards-based network management and gRPC-based access to network device configuration and operational data. Using industry-standard OpenConfig models, gNMI enables the building of modern network automation and monitoring solutions.

gNMI is implemented as an HTTP/2 server supporting encrypted communication and role-based access control. It allows network management systems to interact with the device for configuration and telemetry using gRPC, supporting VRF-aware access and role-based control.

gNMI supports access over SVIs, loopback, and data ports.

gNMI allows for the following:

- Streaming of real-time telemetry data from switches
- Monitoring of interface statistics, system performance, and hardware health
- Integration with modern network management platforms
- Implementation of event-driven network automation

Refer to [gNMI commands](#) for information on enabling and configuring gNMI.

gNMI commands

Select a command from the list in the left navigation menu.

crypto pki application gnmi certificate

Syntax

```
crypto pki application gnmi certificate <CERT-NAME>
```

Description

Configures a certificate for the gNMI server. **local-cert** is used by default. For more details, refer to the [Public Key Infrastructure guide] (./Functionality_Guide_PKI.md)

Parameter	Description
<CERT-NAME>	Specifies the certificate name.

Examples

Configure sign-cert for the gNMI server:

```
switch(config) # crypto pki application gnmi certificate sign-cert
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

gnmi vrf

Syntax

```
gnmi vrf <VRF NAME>  
no gnmi vrf <VRF NAME>
```

Description

Enables the gNMI server on a given VRF. gNMI can be enabled on multiple VRFs simultaneously.



NOTE

There is a maximum of 30 gNMI subscriptions per request and a maximum of 9 concurrent streams per switch. These are system-wide limits shared among all authenticated users regardless of connection method (local or remote).

The **no** form of this command removes the configuration.

Disabling the gNMI server on a VRF immediately closes any active streams.

Parameter	Description
<VRF NAME>	Specifies the VRF name.

Examples

Enable the gNMI server on VRF mgmt, this allows access to the gNMI server from the OOBM port in the "management VRF":

```
switch(config) # gnmi vrf mgmt
```

Removing the configuration of gNMI server on mgmt:

```
switch(config) # no gnmi vrf mgmt
```


Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

show gnmi

Syntax

```
show gnmi
```

Description

Displays the current gNMI configuration.

Examples

Display the current gNMI configuration:

```
switch(config)# show gnmi
gNMI Configuration
-----
VRF                : mgmt, default
Access mode        : read-only
Global stream limit : 9
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

diag-dump

Syntax

```
diag-dump gnmi basic
```

Description

Provides detailed service status information for gNMI and nginx-configurator daemon.

Examples

Showing feature information:

```
=====
[Start] Feature gnmi Time : Thu Jun 26 14:30:13 2025
=====
-----
[Start] Daemon nginx-configurator
-----
gNMI nginx VRF service status dump:
-----
-----
| VRF                                | LoadState   | ActiveState |
SubState          |
-----
-----
| VRF_2                                | loaded       | active      |
running          |
-----
-----
-----
[End] Daemon nginx-configurator
-----
=====
[End] Feature gnmi
```

=====

D diagnostic-dump captured for feature gnmi

Command History

Release	Modification
10.17.1000	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

debug nginxconfigurator

Syntax

```
debug nginxconfigurator <all|gnmi> all severity <level>
```

Description

Enables nginx configurator and YANG resolver debug logs.

Examples

Enabling nginx configurator and YANG resolver debug logs:

```
switch# debug nginxconfigurator all severity debug
switch# debug yang all severity debug
```

Reviewing debug logs:

```
switch(config)# show debug buffer module nginxconfigurator
switch(config)# show debug buffer module yang-resolver
```

Command History

Release	Modification
10.17.1000	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

show tech

Syntax

```
show tech gnmi
```

Description

Provides configuration, logs, and status information relevant to the gNMI HTTP/2 server.

Examples

Showing gNMI configuration information:

```
switch# show tech gnmi
=====
Show Tech executed on Thu Jun 26 14:30:33
2025=====
=====
Begin] Feature gnmi
=====
*****
Command : show gnmi
*****
gNMI Configuration
-----
VRF                : mgmt, default
Access mode        : read-only
Max streams globally : 9

*****
Command : diag-dump gnmi basic
*****

=====
[Start] Feature gnmi Time : Thu Jun 26 14:30:33 2025
=====
-----
[Start] Daemon nginx-configurator
-----
```

```
gNMI nginx VRF service status dump:
```

```
-----  
-----  
| VRF                               | LoadState   | ActiveState |  
SubState      |
```

```
-----  
-----  
| VRF_2                               | loaded      | active      |  
running        |
```

```
-----  
[End] Daemon nginx-configurator  
-----
```

```
=====
```

```
[End] Feature gnmi  
=====
```

Command History

Release	Modification
10.17.1000	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

Boot commands

Subtopics

Boot commands

Boot commands

Select a command from the list in the left navigation menu.

Subtopics

- [boot fabric-module](#)
- [boot line-module](#)
- [boot management-module](#)
- [boot management-module \(recovery console\)](#)
- [boot set-default](#)
- [boot system](#)
- [show boot-history](#)

boot fabric-module

Syntax

```
boot fabric-module <SLOT-ID>
```

Description

Reboots the specified fabric module.

Parameter	Description
<SLOT-ID>	Specifies the member and slot of the module in the format member/slot . For example, to specify the module in member 1 slot 3, enter 1/3 .

Usage

The **boot fabric-module** command reboots the specified fabric module. Traffic performance is affected while the module is down.

If the specified module is the only fabric module in an up state, rebooting that module stops traffic switching between line modules and the line modules power down. The line modules power up when one fabric module returns to an up state.

This command is valid for fabric modules only.

Examples

Rebooting the fabric module in slot **1/3** when auto-confirm is not enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module. Traffic performance
may
be affected while the module is down. Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n)? y
switch#
```

Rebooting the fabric module in slot **1/1** when auto-confirm is enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module. Traffic performance
may
be affected while the module is down. Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n) y (auto-confirm)
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot line-module

Syntax

```
boot line-module <SLOT-ID>
```

Description

Reboots the specified line module.

Parameter	Description
<code><SLOT-ID></code>	Specifies the member and slot of the module in the format member/slot . For example, to specify the module in member 1 slot 3, enter 1/3 .

Usage

This command is supported on switches that have multiple line modules.

Reboots the specified line module. Any traffic for the switch passing through the affected module (SSH, TELNET, and SNMP) is interrupted. It can take up to 2 minutes to reboot the module. During that time, you can monitor progress by viewing the event log.

This command is valid for line modules only.

Examples

Reloading the module in slot 1/1:

```
switch# boot line-module 1/1
This command will reboot the specified line module and interfaces on this
module will not send or receive packets while the module is down. Any
traffic passing through the line module will be interrupted. Management
sessions connected through the line module will be affected. It might take
up to 2 minutes to complete rebooting the module. During that time, you can
monitor progress by viewing the event log.
Do you want to continue (y/n)? y
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot management-module

Syntax

```
boot management-module {active | standby | <SLOT-ID>}
```

Description

Reboots the specified management module. Choose the management module to reboot by role (active or standby) or by slot number.

Parameter	Description
<i>active</i>	Selects the active management module.
<i>standby</i>	Selects the standby management module.
<i><SLOT-ID></i>	Specifies the member and slot of the management module in the format member/slot . For example, to specify the module in member 1 slot 5, enter 1/5 .

Usage

This command is supported on switches that have multiple management modules.

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the **show images** command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the **boot** command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot** command is aborted.



NOTE

Hewlett Packard Enterprise recommends that you use the **boot management-module** command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the **boot management-module** command enables you to save the configuration, cancel the reboot, or both.

Examples

Rebooting the active management module when the standby management module is available:

```
switch# boot management-module active
The management-module in slot 1/5 is going down for reboot now.
```

Rebooting the active management module when the standby management module is not available:

```
switch# boot management-module 1/5
The management module in slot 1/5 is currently active and no
standby management module was found.
This will reboot the entire switch.
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot management-module (recovery console)

Syntax

```
boot management-module {local|remote}
```

Description

Reboots the specified management module by specified location (local or remote).

Parameter	Description
<code><local></code>	Reboots the local management module.
<code><remote></code>	Reboots the remote management module.

Usage

This command is supported on switches that have multiple management modules.

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the **show images** command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the **boot** command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot** command is aborted.



NOTE

Hewlett Packard Enterprise recommends that you use the **boot management-module** command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the **boot management-module** command enables you to save the configuration, cancel the reboot, or both.

Examples

Booting a remote management module:

```
switch# boot management-module remote
There is no other management module installed.
Aborting.
switch#
```

Command History

Release	Modification
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

boot set-default

Syntax

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
primary	Selects the primary network operating system image.
secondary	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```



NOTE

For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

Syntax

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
primary	Selects the primary operating system image for this reboot and sets the configured default operating system image to primary for future reboots.
secondary	Selects the secondary operating system image for this reboot and sets the configured default operating system image to secondary for future reboots.
serviceos	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovers OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.
You can use the **boot set-default** command to change the configured default operating system image.
- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

Syntax

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

For switches that support line modules (such as 6400 switch series) including the **all** parameter displays information for the active management module and all available line modules.



NOTE

To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
all	Optional. Shows boot information for the active management module. For switches that support line modules, including this parameter displays information for and all available line modules.
vsf member <1-10>	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
Index	The position of the boot in the history file. Range: 0 to 3.
Boot ID	A unique ID for the boot . A system - generated 128 - bit string.
Current Boot, up for <time>	For the current boot, the show boot - history command shows the number of seconds the module has been running on the current software.
<Timestamp>: boot reason	For previous boot operations, the show boot - history command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: • <DAEMON - NAME> crash: The daemon identified by <DAEMON - NAME> caused the module to boot. • Kernel crash: The operating system software associated with the module caused the module to boot. • Uncontrolled reboot: The reason for the reboot is not known. • Reboot requested through database: The reboot occurred because of a request made through the CLI or other API. For details, see

Table 1. Description of reboots handled through the database

Boot History String	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset button
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.
Configuration version migration	A configuration version migration occurred which required a reboot.
Console error	The console failed to start.

Boot History String	Description
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the allotted time.
Critical service fault (error)	A daemon critical to switch operation has stopped functioning. An extra error string may be present to describe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	A user requested a renumber of a VSF member.
VSF switchover requested	A user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis low critical temperature	Chassis temperature below the minimum operating threshold.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceeded.
ISSU SMM update	Standby management module reboot triggered by an In - Service Software Upgrade (ISSU).
ISSU switchover	Redundancy switchover triggered by an In - Service Software Upgrade.
ISSU aborted	Standby management module reset triggered by failure during an In - Service Software Upgrade.
Rollback timer expired	Reset triggered by the ISSU rollback timer expiring.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====
Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs
Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version:
FL.10.14.0000-1619-ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2
Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1
Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 : Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 : Switch boot count is 3
25 Aug 23 10:29:02 : Primary overtemperature fault detected with 0x2 in
PSU 1/1
```

(For 6400 Switch series) Showing the boot history of the active management module and all line modules:

```
switch#
Management module
=====
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
```

```

=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
Management module
=====
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...

```

In the event of a reset triggered by a power supply unit (PSU), or a PSU input fault, the output of this command also displays information about why the PSU initiated a reboot. The following example displays the boot history of a switch with a reboot initiated by a PSU.

```

switch# show boot-history
Management module
=====
Index : 2
Boot ID : a61ad00d10864c748bc7893a5d4af2e4
15 Dec 23 19:02:02 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:02:02 : Switch boot count is 0
15 Dec 23 19:02:17 : PSU 1/1: Fault detected
Index : 1
Boot ID : 30d831bbfdfa425baf50a629ee01b185
15 Dec 23 19:01:58 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:01:58 : Switch boot count is 0

```

The following example displays the boot history for the VSF member **2**.

```

switch# show boot-history vsf member 2
Member-2
=====
Index : 0

```

```
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs
Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2
```

Command History

Release	Modification
10.13.1000	The output of this command is enhanced to display additional information about the reason for the reboot, if available.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

External storage

The switch has limited capacity to store data, collected by switch features and protocols. You can provide virtually unlimited storage capacity by adding user-supplied external storage volumes. Supported volume types and storage protocols include: NFSv3, NFSv4, and SCP (sshfs).

One application of external storage is the saving and restoring of DHCP lease files over SCP or NFS network attached storage systems. SCP file system protocol uses a user mode process to emulate a network file system. The key advantage is packet level encryption and simple configuration. The key disadvantage is slow performance.

You can set up external storage volume credentials and then enable it. A storage management process acts on your requests by enabling the storage volume using the requested storage protocol. You can disable the external storage volume or set it up but leave it disable.

The feature maintains storage volume state. The states are: **disabled** (down), **connecting** (establishing connection), **operational** (up), and **unaccessible** (unavailable).

If a storage volume is unavailable, the system attempts to reconnect periodically. Multiple volumes could connect concurrently. If one connection times out the others can connect immediately.

The system supports server connection through data and management ports.

Data port support requires server IP address on a default VRF.

Once a storage volume is enabled, applications can use the volume to store retrieve and delete files and directories.

Subtopics

External storage commands

External storage commands

Select a command from the list in the left navigation menu.

Subtopics

address

directory

disable

enable

external-storage

password (external-storage)

show external-storage

show running-config external-storage

type

username

vrf

address

Syntax

```
address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}  
no address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
```

Description

Specifies the NAS IP address or hostname.

The **no** form of this command deletes an IP address or hostname.

Parameter	Description
<code><IPV4-ADDR></code>	Specifies the NAS server IPv4 address, Global.
<code><IPV6-ADDR></code>	Specifies the IPv6 address of the NAS server.
<code><HOSTNAME></code>	Specifies the hostname of the NAS server. String.

Examples

Creating the logfiles storage volume with IP address 10.1.1.1:

```
switch(config) # external-storage logfiles
switch(config-external-storage-logfiles) # address 10.1.1.1
```

Deleting an external storage volume named logfiles:

```
switch(config) # external-storage logfiles
switch(config-external-storage-logfiles) # no address 10.1.1.1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-external-storage- <code><VOLUME-NAME></code>	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority

directory

Syntax

```
directory <DIRECTORY-NAME>
no directory <DIRECTORY-NAME>
```

Description

Selects an existing directory on the external storage volume.

The **no** form of this command clears a directory of an external storage volume.

Parameter	Description
<DIRECTORY-NAME>	Specifies the external storage directory for mapping the volume .

Examples

Creating a volume named logfiles that is mapped under /home on the server:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# directory /home
```

Clearing the directory /home:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no directory /home
```


Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config- external- storage- <VOLUME-NAME>	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

disable

Syntax

```
disable  
no disable
```

Description

Disables the external storage volume.

The **no** form of this command enables the external storage volume. This is identical to the `enable` command.

Examples

Disabling a volume named logfiles:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config- external- storage- <VOLUME-NAME>	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

enable

Syntax

```
enable  
no enable
```

Description

Enables the external storage volume.

The **no** form of this command disables the external storage volume. This is identical to the `disable` command.

Examples

Creating and then enabling a volume named logfiles:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# enable
```

Disables the external storage volume:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config- external- storage- <VOLUME-NAME>	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

external-storage

Syntax

```
external-storage <VOLUME-NAME>  
no external-storage <VOLUME-NAME>
```

Description

Creates or updates an external storage volume.

The **no** form of this command deletes an external storage volume.

Examples

Creating the logfiles storage volume:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)#
```

Deleting the logfiles storage volume:

```
switch(config)# no external-storage logfiles
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

password (external-storage)

Syntax

```
password [{plaintext | ciphertext} <PASSWORD>]  
no password {plaintext | ciphertext} <PASSWORD>
```

Description

Sets the password for network attached storage server login.

The **no** form of this command clears the password for network attached storage server login.

Parameter	Description
{ciphertext plaintext}	Selects the password format.
<PASSWORD>	Specifies the password. NOTE When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Creating a volume named logfiles with password Xj#9:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# password plaintext Xj#9
```

Creating a volume named bak1 with a prompted plaintext password:

```
switch(config)# external-storage bak1
switch(config-external-storage-bak1)# password
Enter the NAS server password: *****
Re-Enter the NAS server password: *****
```

Clearing the password for volume logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no password plaintext Xj#9
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config- external- storage- <VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

show external-storage

Syntax

```
show external-storage [<VOLUME-NAME>]
```

Description

Shows external storage configuration and state for all volumes or for a specified volume.

Parameter

Description

<VOLUME-NAME>

Specifies the external storage volume name that the show command will use.

Examples

```
switch# show external-storage
-----
-----
                Address      VRF      Username      Type      Directory
State
-----
-----
nfsvol      10.1.1.1      nas      ---      NFSv3      /home
operational
nfsfiles    20.1.1.1      nas      netstorage    NFSv4      /netstor
disabled
scpdev      nasserver      nas      scpstor      SCP      /scp
unaccessible
```

Command History

Release

Modification

10.07 or earlier - -

Command Information

Platforms

Command context

Authority

6300	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.
6400		

show running-config external-storage

Syntax

```
show running-config external-storage
```

Description

Shows the running configuration of the external storage.

Examples

```
switch# show running-config external-storage
external-storage nfsvol
    address    10.1.1.1
    vrf        nas
    type       nfsv4
    directoty  /home
    enable
external-storage scpdev
    address    30.1.1.1
    vrf        nas
    username   switchuser
    password   ciphertext xxx
    type       scp
    directoty  /home
    enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

type

Syntax

```
type {nfsv3 | nfsv4 | scp}
no type {nfsv3 | nfsv4 | scp}
```

Description

Sets the network attached storage access type for reaching the external storage volume.

The **no** form of this command deletes an external storage volume.

Parameter	Description
nfsv3	Specifies the NFSv3 network access protocol.
nfsv4	Specifies the NFSv4 network access protocol.
scp	Specifies the SCP network access protocol.

Examples

Creating the logfiles volume using NFSV4:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# type nfsv4
```

Clearing the external storage access type:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no type nfsv4
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config- external- storage- <VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

username

Syntax

```
username <USER-NAME>  
no username <USER-NAME>
```

Description

Sets the username for logging in to a network attached storage server.

The **no** form of this command clears a username.

Parameter	Description
<USER-NAME>	Specifies the username.

Examples

Creating a volume named logfiles with the user name nassuser:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# username nasuser
```

Clearing the user name nasuser from accessing the logfiles volume:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no username nasuser
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
6300 6400	config- external- storage- <VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

vrf

Syntax

```
vrf <VRF-NAME>  
no vrf <VRF-NAME>
```

Description

Setting a VRF to reach network attached storage.

The **no** form of this command clears access of a VRF to network attached storage.

Parameter	Description
<VRF-NAME>	Specifies the VRF name.

Examples

Creating the logfiles volume and setting a VRF named nas to access the network attached storage:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# vrf nas
```

Clearing access of a VRF named nas to the network attached storage:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no vrf nas
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config- external- storage- <VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

IP-SLA

The IP Service Level Agreement (IP-SLA) is a feature that enables the measuring of network performance between two nodes in a network for different service level agreement parameters such as round-trip time (RTT), one-way delay, jitter, reachability, packet loss, and voice quality scores. These two nodes can span across area in access, distribution or core inside a LAN as well as across WAN between core to core or core to Data Centre switches. This feature helps you measure the SLA for different protocols or applications such as UDP echo, UDP jitter (for voice and video), TCP connect, HTTP, and ICMP echo. This guide provides details for managing and monitoring different types of IP-SLAs.

Subtopics

IP-SLA guidelines

IP-SLA guidelines

- AOS-CX supports only SLA configuration through CLI and thresholds can be configured using NAE agents using WebUI/REST.
- AOS-CX supports only forever tests. On-demand tests are not supported.
- Maximum sessions: IP-SLA source 500, IP-SLA responder 500.
- NAE can effectively monitor a maximum of 300 parameters, reducing the maximum supported session by 300.
- NAE supports only syslog.
- NAE agents must be triggered for each IP-SLA test on every switch.
- If multiple IP addresses are received for a DNS query, DNS works with the first resolved IP.
- When the DNS server IP is not explicitly configured, the system automatically uses the first DNS server available in its default configuration.
- The source interface/IP option is not applicable for SLAs configured on 'mgmt' VRF, as it has only one interface.
- A system time change because of NTP or a manual change causes an incorrect calculation.
- There is no interoperability of UDP echo SLA between AOS-CX and FlexFabric switches.
- Source IP and source port combination must be unique across SLA sessions in a same switch.
- Do not use the same source port across the source and responder sessions in a switch.
- The configuration of history results is limited to a maximum of 8 IP-SLA sessions. This means that you can enable and store historical performance data, such as response times and availability, for up to 8 individual IP SLA sessions at any given time.
- NTP synchronization is a must for SLA types involving one-way delay such as UDP jitter VoIP.
- It is mandatory to set default CoPP to the maximum value when UDP jitter SLA is enabled. Otherwise, 100% packet loss can be seen and UDP jitter SLA probes will result in failure:

```
copp-policy default
  class hypertext priority 6 rate 50000 burst 64
```

```
default-class priority 6 rate 99999 burst 9999
```

Limitations with VoIP SLAs

- A maximum of 80 concurrent VoIP SLAs can be scheduled in a 20 second slot.
- A single VoIP probe takes 20 seconds to complete.
- The default and minimum probe interval for VoIP SLA is 120 seconds.
- SLAs scheduled in the same slot, periodically sends 1000 probe packets for 120 seconds in 20 second intervals.
- Default 120 second probe interval is divided in to 6 slots of 20 seconds to avoid synchronization of all configured VoIP SLAs sending probes at the same time.
- SLAs started at the same time exceeding the concurrent limit of 80 must wait for the next 20 second VoIP slot to open before moving to 'running' state.
- The maximum number of VoIP SLAs supported is $80 \times 6 \text{ slots} = 480 \text{ SLAs}$.
- SLAs exceeding 480 will continue to remain in the 'waiting for VoIP slot' until any slot is freed by stopping the running SLA.
- To avoid high RTT, a single switch with more than 20 SLAs should not have single responder SLA.
- When IP is received dynamically (e.g. using DHCP) for interfaces other than management interface, IPSLA source or responder has to be configured only using interface name.

IP-SLA commands

Select a command from the list in the left navigation menu.

Subtopics

[http](#)

[https](#)

[icmp-echo](#)

[ip-sla](#)

[ip-sla responder](#)

[show ip-sla all](#)

[show ip-sla responder](#)

show ip-sla
start-test
stop-test
tcp-connect
udp-echo
udp-jitter-voip
vrf

http

Syntax

```
http {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache  
disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}]  
[probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|  
<IPV6-ADDR>|IFNAME}}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-  
NUMBER>] [http-raw-request <RAW-PAYLOAD>]  
no http {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache  
disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}]  
[probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|  
<IPV6-ADDR>|IFNAME}}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-  
NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

Description

Configures HTTP as the IP-SLA test mechanism. Requires destination URL and type of HTTP request (raw/get).

The **no** version of this command disables HTTP as the IP-SLA test mechanism.

Parameter	Description
{get raw}	Selects HTTP request type as get or raw where the system will generate or provide HTTP payload.
<URL>	Specifies HTTP URL address of syntax http://<HOST NAME/IP - ADDRESS>:<PORT>/<PATH>.
history-interval <HISTORY-INTERVAL>	Configures the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 60 to 7200.
cache disable	Selects cache option for the HTTP server. By default the option is enabled.

Parameter	Description
<code>name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}</code>	Specifies the DNS server for destination hostname resolution.
<code>probe-interval <PROBE-INTERVAL></code>	Specifies the probe interval in seconds. Range: 30 to 604800.
<code>proxy <PROXY-URL></code>	Specifies the probe interval in seconds. Range: 30 to 604800.
<code>source {<IPV4-ADDR> <IPV6-ADDR> <IFNAME>}</code>	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
<code>source-port <SOURCE-PORT-NUM></code>	Specifies the value of the source port for the IP - SLA probes.
<code>version <VERSION-NUMBER></code>	Specifies the source interface to use for sending IP - SLA probes.
<code>http-raw-request <RAW-PAYLOAD></code>	Specifies the HTTP raw request. String.

Examples

Configuring HTTP get with parameters, including history interval:

```
switch(config)# ip-sla 1
switch(config-ipsla-1)# http get http://device.arubanetworks.com/root/home.html history-interval 120 cache disable name-server 10.10.10.2 probe-interval 30
```

Configuring HTTP raw with parameters:

```
switch(config-ipsla-1)# http raw http://2.2.2.2 http-raw-request "GET /en/US/hmpgs/index.html HTTP/1.0\r\n\r\n"
```

Disabling HTTP get with parameters:

```
switch(config-ipsla-1)# no http get http://device.example.com/root/home.html name-server 10.10.10.2 history-interval 120
```

Disabling HTTP raw with parameters:

```
switch(config-ipsla-1)# no http raw http://device.example.com/root/
home.html http-raw-request "GET /en/US/hmpgs/index.html HTTP/1.0\r\n\r\n"
```

Command History

Release	Modification
10.16.1000	Added new parameter history - interval . Also, IPv6 addresses can be used.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

https

Syntax

```
https {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache
disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}]
[probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|
<IPV6-ADDR>|IFNAME}}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-
NUMBER>] [http-raw-request <RAW-PAYLOAD>]
no https {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache
disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}]
[probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|
<IPV6-ADDR>|IFNAME}}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-
NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

Description

Configures HTTPS as the IP-SLA test mechanism. Requires destination URL and type of HTTPS request (get/raw).

The **no** form of this command removes the configuration.



NOTE

For HTTPS IP-SLA sessions, it is not required to install a certificate on the switch.

Parameter	Description
{get raw}	Selects HTTPS request type as get or raw where the system will generate or provide HTTPS payload.
<URL>	Specifies HTTPS URL address of syntax. https://<HOST NAME/IP - ADDRESS>:<PORT>/<PATH>.
history-interval <HISTORY-INTERVAL>	Configures the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 60 to 7200.
cache disable	Selects cache option for the HTTPS server. By default the option is enabled.
name-server {<IPv4-ADDR-DNS-SERVER> <IPv6-ADDR-DNS-SERVER>}	Specifies the IPv4 address of DNS server.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 30 to 604800.
proxy <PROXY-URL>	Specifies the probe interval in seconds. Range: 30 to 604800.
source {<IPv4-ADDR> <IPv6-ADDR> IFNAME}	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
source-port <SOURCE-PORT-NUM>	Specifies the value of the source port for the IP - SLA probes.
version <VERSION-NUMBER>	Specifies the source interface to use for sending IP - SLA probes.
https-raw-request <RAW-PAYLOAD>	Specifies the HTTPS raw request. String.

Examples

Configuring HTTPS get with parameters:

```
switch(config-ipsla-1)# https get https://device.arubanetworks.com/root/home.html
```

Configuring HTTPS raw with parameters:

```
switch(config-ipsla-1)# https raw https://device.arubanetworks.com/root/home.html raw-request "GET /en/US/hmpgs/index.html"
```

Removing the HTTPS raw:

```
switch(config-ipsla-1)# no https raw https://device.arubanetworks.com/root/home.html raw-request "GET /en/US/hmpgs/index.html"
```

Command History

Release	Modification
10.16.1000	Added new parameters history - interval . Also, IPv6 addresses can be used.
10.12.1000	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

icmp-echo

Syntax

```
icmp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} [history-interval  
<HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-  
SERVER>}] [payload-size <PAYLOAD-SIZE>] [probe-interval <PROBE-INTERVAL>]  
[source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [timeout  
<TIMEOUT>] [tos <TYPE-OF-SERVICE>]  
no icmp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} [history-
```

```
interval <HISTORY-INTERVAL> [name-server {<IPv4-ADDR-DNS-SERVER>|<IPv6-ADDR-DNS-SERVER>}] [payload-size <PAYLOAD-SIZE>] [probe-interval <PROBE-INTERVAL>] [source {<SOURCE-IPv4-ADDR>|<SOURCE-IPv6-ADDR>|<IFNAME>}] [timeout <TIMEOUT>] [tos <TYPE-OF-SERVICE>]
```

Description

Configures ICMP echo as the IP-SLA test mechanism. Requires destination address for the IP-SLA test.

The **no** form of this command disables the ICMP echo as the IP-SLA test mechanism.

Parameter	Description
{<DEST-IPv4-ADDR> <DEST-IPv6-ADDR> <HOSTNAME>}	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP - SLA.
history-interval <HISTORY-INTERVAL>	Specifies the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 10 to 7200.
name-server {<IPv4-ADDR-DNS-SERVER> <IPv6-ADDR-DNS-SERVER>}	Specifies the DNS server for destination hostname resolution.
payload-size <PAYLOAD-SIZE>	Specifies the payload size of an SLA probe. Range: 0 to 1440.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 5 to 604800.
source {<SOURCE-IPv4-ADDR> <SOURCE-IPv6-ADDR> <IFNAME>}	Selects the source IPv4 or IPv6 address for SLA probes or the source interface to use for sending IP - SLA probes.
timeout <TIMEOUT>	Specifies the interval before a probe is timed out. Range: 5 to 604800.
tos <TYPE-OF-SERVICE>	Specifies the type of service value to be used in probe packets. Range: 0 to 255.

Examples

Configuring icmp-echo:

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# icmp-echo 2.2.2.2 name-server 4.4.4.4 source
```

3.3.3.3

Configuring ICMP echo with several parameters, including history interval and timeout:

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# icmp-echo 2.2.2.2 history-interval 160 name-
server 4.4.4.4 payload-size 400 probe-interval 80 source 3.3.3.3 timeout
20 tos 255
```

Command History

Release	Modification
10.16.1000	Added two new parameters history - interval and timeout . Also, IPv6 addresses can be used.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

ip-sla

Syntax

```
ip-sla <IP-SLA-NAME>
no ip-sla <IP-SLA-NAME>
```

Description

Creates an IP Service Level Agreement (SLA) profile and switches to the **config-ip-sla** context.

The **no** form of this command deletes an IP-SLA profile. By default, all profile use the default VRF (default).

Parameter	Description
<code><IP-SLA-NAME></code>	Specifies an IP - SLA profile name. Length: 1 to 64 characters.

Examples

Creating an IP-SLA:

```
switch(config) # ip-sla 1
switch(config-ip-sla-1) #
```

Deleting an IP-SLA:

```
switch(config) # no ip-sla 1
switch(config) #
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

ip-sla responder

Syntax

```
ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip)
[<PORT-NUM>] [source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [vrf
<VRF-NAME>] [ipv6]
no ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip)
```

```
[<PORT-NUM>] [source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [vrf <VRF-NAME>] [ipv6]
```

Description

Selects the IP-SLA responder. The responder can be configured for udp-echo, tcp-connect, udp-jitter-voip type. It requires the SLA name, SLA type, and port number as arguments.

The **no** form of this command removes the IP-SLA responder.

Parameter	Description
<code><SLA-NAME></code>	Specifies the IP - SLA responder name. Length: 1 to 64 characters.
<code>udp-echo</code>	Enables responder for udp - echo probes.
<code>tcp-connect</code>	Selects TCP connect as the IP - SLA test mechanism.
<code>udp-jitter-voip</code>	Selects VOIP jitter as the IP - SLA test mechanism.
<code><PORT-NUM></code>	Specifies the port number to listen for IP - SLA probes. Range: 1 to 65535.
<code>source {<SOURCE-IPV4-ADDR> <SOURCE-IPV6-ADDR> <IFNAME>}</code>	Selects the source IPv4 or IPv6 address for SLA probes or the source interface to use for sending IP - SLA probes.
<code>vrf <VRF-NAME></code>	Specifies the name of the VRF to use.
<code>ipv6</code>	Configures IPv6 responder. This keyword is required if an IPv6 address is being used by the source interface or VRF. By default, it will be considered as an IPv4 address.

Usage

The IPv6 keyword is required if an IPv6 address is being used by the source interface or VRF. Otherwise, by default, it will be considered as an IPv4 address.

Examples

Configuring IP-SLA responder for udp-echo:

```
switch(config) # ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Configuring IP-SLA responder with IPv6:

```
switch(config) #ip-sla responder SLA1 udp-echo 8000 source 1/1/1 ipv6
```

Configuring IP-SLA responder for udp-jitter-voip:

```
switch(config) #ip-sla responder SLA1 udp-jitter-voip 1025 vrf <VRF>
```

Disabling IP-SLA responder:

```
switch(config) # no ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Command History

Release	Modification
10.15	Added ipv6 parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

show ip-sla all

Syntax

```
show ip-sla all
```

Description

Shows all ip-sla source configuration and status.

Examples

Showing results for ip-sla all:

```
switch# show ip-sla all
```

```
SLA Name           : 2 (non-persistent)
Status             : running
SLA Type           : udp-echo
VRF                : default
Source IP          :
Source Interface    :
Domain Name Server  :
Payload Size       : 28
TOS                : 0
Probe Interval(seconds) : 60
History Interval(seconds) : 0
Timeout Interval(seconds) : 45
IP-SLA session status
IP-SLA Name        : 2 (non-persistent)
IP-SLA Type        : udp-echo
Destination Host Name/IP Address : 10.1.1.2
Destination Port    : 8888
Source IP Address/IFName :
Source Port        :
Status             : running
IP-SLA Session Cumulative Counters
Total Probes Transmitted : 10
Probes Timed-out        : 10
Bind Error              : 0
Destination Address Unreachable : 0
DNS Resolution Failures : 0
Reception Error         : 0
Transmission Error      : 0
Operational Status     : down
IP-SLA Latest Probe Results
Last Probe Time        :
Packets Sent           : 1
Packets Received       : 0
Packet Loss in Test    : 100%
Minimum RTT(ms)        :
Maximum RTT(ms)        :
Average RTT(ms)        :
DNS RTT(ms)            :
```

```
-----
--
SLA Name           : echo-udp-sess2 (non-persistent)
Status             : running
```



```

SLA Type           : udp-echo
VRF                 : default
Source IP           :
Source Interface    :
Domain Name Server  :
Payload Size        : 28
TOS                 : 0
Probe Interval(seconds) : 60
History Interval(seconds) : 0
Timeout Interval(seconds) : 45
IP-SLA session status
IP-SLA Name         : echo-udp-sess2 (non-persistent)
IP-SLA Type         : udp-echo
Destination Host Name/IP Address : 100.1.1.2
Destination Port     : 8888
Source IP Address/IFName :
Source Port          :
Status              : running
IP-SLA Session Cumulative Counters
Total Probes Transmitted : 10
Probes Timed-out         : 0
Bind Error               : 0
Destination Address Unreachable : 0
DNS Resolution Failures  : 4
Reception Error          : 0
Transmission Error       : 0
Operational Status      : Up
IP-SLA Latest Probe Results
Last Probe Time         :
Packets Sent            : 1
Packets Received        : 1
Packet Loss in Test     : 0.0000%
Minimum RTT(ms)         :
Maximum RTT(ms)         :
Average RTT(ms)         :
DNS RTT(ms)             :
-----
--

```

Showing results for non-configured ip-sla all:

```

switch# show ip-sla all
IPSLA source is not configured

```

Command History

Release	Modification
10.16.1000	Updated to display history interval .
10.12.1000	Updated to display https as an IP - SLA type.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show ip-sla responder

Syntax

```
show ip-sla responder <SLA-NAME> [initiator {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>}] [<SOURCE-PORT-NUM>] [results]
```

Description

Shows the given IP-SLA responder configuration and operation status.

Parameter	Description
<code><SLA-NAME></code>	Specifies the SLA name.
<code>initiator {<SOURCE-IPV4-ADDR> <SOURCE-IPV6-ADDR>}</code>	Selects the source IPv4 or IPv6 address for SLA probes to use.
<code><SOURCE-PORT-NUM></code>	Configures the source port for the IP - SLA test. Range: 1 to 65 535.

Parameter	Description
results	Displays the statistics for a given source IP and port.

Examples

Showing IP-SLA responder configuration:

```
switch# show ip-sla responder SLA3
  SLA Name       : SLA3
  IP-SLA Type    : Udp-echo
  VRF            : Default
  Responder Port : 8000
  Responder IP   : 2.2.2.3
  Responder Interface : 1/1/1
  Responder Status : Running
```

Showing IP-SLA responder with initiator and results parameters:

```
switch# show ip-sla responder SLA1 initiator 2.2.2.1 8000 results
IP-SLA Type       : Udp-echo
VRF Name          : Default
Source IP         : 2.2.2.1
Source Port       : 8000
Responder Port    : 8888
Responder IP      : 2.2.2.3
Responder Interface :
Responder Status  : Running
Packets Received  : 2
Packets Sent      : 2
```

Command History

Release	Modification
10.16.1000	Added new parameters: initiator and results .
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
6300	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show ip-sla

Syntax

```
show ip-sla <SLA-NAME> [{results | history-results}]
```

Description

Shows the given IP-SLA source configuration and status.

Parameter	Description
<code><SLA-NAME></code>	Specifies the SLA name.
<code>results</code>	Displays the statistics calculated for an SLA type.
<code>history-results</code>	Displays the history statistics calculated for the SLA ID.

Examples

Showing results for ip-sla:

```
switch# show ip-sla xyz results
IP-SLA session status
IP-SLA Name           : xyz
IP-SLA Type           : tcp-connect
Destination Host Name/IP Address: 2.2.2.1
Destination Port       : 8888
Source IP Address/IFName : 2.2.2.2
Source Port           : 5555
Status                : running
IP-SLA session cumulative counters
Total Probes Transmitted : 1
Probes Timed-out         : 0
Bind Error               : 0
Destination Address Unreachable : 0
```

```

DNS Resolution Failures      : 0
Reception Error              : 0
Transmission Error           : 0
IP-SLA Latest Probe Results
Last Probe Time              : 2018 Jul 13 02:00:35
Packets Sent                  : 1
Packets Received              : 1
Packet Loss in Test           : 0.0000%
Minimum RTT(ms)               : 12
Maximum RTT(ms)               : 12
Average RTT(ms)               : 12
DNS RTT(ms)                   : 0
TCP RTT(ms)                   : 12

```

Showing history results for ip-sla:

```
switch# sh ip-sla abcd history-results
```

```

IP-SLA Name: abcd Session Details ===== IP-SLA Type : tcp-connect Status : running
Probe Interval(seconds) : 30 History Interval(seconds) : 600 Source Port : 3000 Destination Port :
4000 Source IP Address : 10.0.0.1 Dest Host Name/IP Address : 10.0.0.2 History Probe Results
===== Packet Stats ----- Probes Transmitted : 4 Packets Sent : 4 Packets
Received : 4 Loss Percentage : 0.0000% Error Stats ----- Transmission Errors : 0 Reception Errors :
0 Bind Errors : 0 Dest. Unreachable : 0 Probes Timed-Out : 0 DNS Resolution Failures : 0 Probe RTT Stats
----- Min RTT(ms) : 0 Max RTT(ms) : 0 Avg RTT(ms) : 0 DNS RTT Stats ----- Min RTT(ms) :
Max RTT(ms) : Avg RTT(ms) :

```

Command History

Release	Modification
10.16.1000	Added new parameter history - results . Updated to display history interval .
10.12.1000	Updated to display https as an IP - SLA type.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

start-test

Syntax

```
start-test
```

Description

Starts the IP-SLA probes.

Examples

```
switch(config)# ip-sla test  
switch(config-ip-sla-test)# start-test
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

stop-test

Syntax

```
stop-test
```

Description

Stops the IP-SLA probes.

Examples

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# stop-test
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

tcp-connect

Syntax

```
tcp-connect {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM>
[history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|
<IPV6-ADDR-DNS-SERVER>} [probe-interval <PROBE-INTERVAL>] [source {<IPV4-
ADDR>|<IPV6-ADDR>|IFNAME>}] [source-port <PORT-NUM>]
no tcp-connect {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-
NUM> [history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-
SERVER>|<IPV6-ADDR-DNS-SERVER>} [probe-interval <PROBE-INTERVAL>] [source
{<IPV4-ADDR>|<IPV6-ADDR>|IFNAME>}] [source-port <PORT-NUM>]
```

Description

Configures TCP connect as the IP-SLA test mechanism. Requires destination address/hostname and destination port for the IP-SLA of tcp-connect IP-SLA type.

The **no** form of this command removes the the TCP connection.

Parameter	Description
<code>{<DEST-IPV4-ADDR> <DEST-IPV6-ADDR> <HOSTNAME>}</code>	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP - SLA.
<code><DEST-PORT-NUM></code>	Destination port for the IP - SLA. Range: 1 to 65535.
<code>history-interval <HISTORY-INTERVAL></code>	Configures the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 60 to 7200.
<code>name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}</code>	Specifies the DNS server for destination hostname resolution.
<code>probe-interval <PROBE-INTERVAL></code>	Probe interval in seconds. Range: 30 to 604800.
<code>source {<IPV4-ADDR> <IPV6-ADDR> <IFNAME>}</code>	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
<code>source-port <PORT-NUM></code>	Specifies the port for the IP - SLA test.

Examples

Configuring TCP connect echo with parameters, including history interval:

```
switch(config)# ip-sla tcp
switch(config-ip-sla-tcp)# tcp-connect https://device.example.com 8080 name-server 10.10.10.2 history-interval 180 source 1/1/1 source-port 6000
```

Disabling the TCP connect:

```
switch(config-ip-sla-tcp)# no tcp-connect 10:1::1:1 8080 source 1/1/1 source-port 6000
```

Command History

Release	Modification
10.16.1000	Added new parameters history - interval . Also, IPv6 addresses can be used.

Release	Modification
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

udp-echo

Syntax

```
udp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM>
[history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|
<IPV6-ADDR-DNS-SERVER>} [payload-size <PAYLOAD-SIZE>] [probe-interval
<PROBE-INTERVAL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port
<SOURCE-PORT-NUM>] [timeout <TIMEOUT>] [tos <TYPE-OF-SERVICE>]
no udp-echo {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-NUM>
[history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|
<IPV6-ADDR-DNS-SERVER>} [payload-size <PAYLOAD-SIZE>] [probe-interval
<PROBE-INTERVAL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port
<SOURCE-PORT-NUM>] [timeout <TIMEOUT>] [tos <TYPE-OF-SERVICE>]
```

Description

Configures UDP echo as the IP-SLA test mechanism. Requires destination address/hostname and destination port number for the IP-SLA of udp-echo SLA type.

The **no** form of this command removes the UDP echo configuration.

Parameter	Description
{<DEST-IPV4-ADDR> <DEST-IPV6-ADDR> <HOSTNAME>}	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP - SLA.
	Specifies the destination port for the IP - SLA. Range: 1 to 65535.

Parameter	Description
<code><DEST-PORT-NUM></code>	
<code>history-interval <HISTORY-INTERVAL></code>	Configures the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 10 to 7200.
<code>name-server {<IPv4-ADDR-DNS-SERVER> <IPv6-ADDR-DNS-SERVER>}</code>	Specifies the DNS server for destination hostname resolution.
<code>payload-size <PAYLOAD-SIZE></code>	Specifies the payload size of an SLA probe. Range: 28 to 1440.
<code>probe-interval <PROBE-INTERVAL></code>	Specifies the probe interval in seconds. Range: 5 to 604800.
<code>source {<IPv4-ADDR> <IPv6-ADDR> <IFNAME>}</code>	Selects the source, either an IPv4 address, an IPv6 address, or a hostname for SLA probes.
<code>source-port <SOURCE-PORT-NUM></code>	Configures the source port for the IP - SLA test. Range: 1 to 65535.
<code>timeout <TIMEOUT></code>	Specifies the interval before a probe is timed out. Range: 5 to 604800.
<code>tos <TYPE-OF-SERVICE></code>	Specifies the type of service. Range: 0 to 255.

Examples

Configuring UDP echo with parameters, including history interval and timeout:

```
switch(config-ipsla-1)# udp-echo https://device.example.com 4000 history-
interval 180 name-server 2.2.2.2 payload-size 100 probe-interval 90 source
4.4.4.4 source-port 8000 timeout 20
```

Removing UDP echo configuration:

```
switch(config-ipsla-1)# no udp-echo https://device.example.com 8080 history-
interval 160 name-server 10.10.10.2 payload-size 50 source 2.2.2.1 timeout
20
```

Command History

Release	Modification
10.16.1000	Added two new parameters history - interval and timeout . Also, IPv6 addresses can be used.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

udp-jitter-voip

Syntax

```
udp-jitter-voip {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-PORT-  
NUM> [advantage-factor <ADVANTAGE-FACTOR>] [codec-type <CODEC-TYPE>]  
[history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|  
<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [source {<IPV4-  
ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [tos <TYPE-OF-  
SERVICE>]  
no udp-jitter-voip {<DEST-IPV4-ADDR>|<DEST-IPV6-ADDR>|<HOSTNAME>} <DEST-  
PORT-NUM> [advantage-factor <ADVANTAGE-FACTOR>] [codec-type <CODEC-TYPE>]  
[history-interval <HISTORY-INTERVAL>] [name-server {<IPV4-ADDR-DNS-SERVER>|  
<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [source {<IPV4-  
ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [tos <TYPE-OF-  
SERVICE>]
```

Description

Configure UDP jitter VoIP as the IP-SLA test mechanism. Requires destination address/hostname and source address/interface for the IP-SLA of udp-jitter-voip IP-SLA type.

The **no** form of this command removes the UDP jitter VoIP configuration.

Parameter	Description
<code>{<DEST-IPV4-ADDR> <DEST-IPV6-ADDR> <HOSTNAME>}</code>	Selects the destination, either an IPv4 address, an IPv6 address, or hostname, for the IP - SLA.
<code><DEST-PORT-NUM></code>	Selects the port number for the IP - SLA. Range: 1 to 65535.
<code>advantage-factor</code> <code><ADVANTAGE-FACTOR></code>	Selects the value for the advantage factor. Default value is 0. Range: 0 to 20.
<code>codec-type</code> <code><CODEC-TYPE></code>	Selects the codec - type for the Voip IP - SLA test.
<code>history-interval</code> <code><HISTORY-INTERVAL></code>	Configures the history interval for the IP - SLA. Set the history interval to minimum of two times the probe - interval for the SLA. Range: 240 to 7200.
<code>name-server</code> <code>{<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}</code>	Specifies the DNS server for destination hostname resolution.
<code>probe-interval</code> <code><PROBE-INTERVAL></code>	Specifies the probe interval in seconds. Range: 120 to 604800.
<code>source</code> <code>{<IPV4-ADDR> <IPV6-ADDR> <IFNAME>}</code>	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
<code>source-port</code> <code><SOURCE-PORT-NUM></code>	Specifies the value of source port for the IP - SLA probes.
<code>tos</code> <code><TYPE-OF-SERVICE></code>	Specifies the type of service. Range: 0 to 255.

Examples

Configuring udp-jitter-voip with optional parameters, including history interval:

```
switch(config-ipsla-1)# udp-jitter-voip https://device.arubanetworks.com
8080 advantage-factor 10 codec-type g711a history-interval 240 name-server
10.10.10.2 probe-interval 120 source 10.1.1.1 source-port 8888 tos 10
```

Configuring udp-jitter-voip with optional parameters:

```
switch(config-ipsla-1)# udp-jitter-voip 2.2.2.2 8080 advantage-factor 10  
codec-type g711a
```

Disabling udp-jitter-voip:

```
switch(config-ipsla-1)# no udp-jitter-voip https://device.example.com 8080  
advantage-factor 10 codec-type g711a name-server 10.10.10.2 probe-interval  
120 source 10.1.1.1 source-port 8888 tos 10
```

Command History

Release	Modification
10.16.1000	Added new parameters history - interval . Also, IPv6 addresses can be used.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config-ip-sla- <IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

vrf

Syntax

```
vrf <VRF-NAME>  
no vrf [<VRF-NAME>]
```

Description

Configures the VRF on which the SLA will send or receive packets. By default, the default VRF is used.

The **no** form of the command removes VRF from SLA.

Parameter	Description
<code><VRF-NAME></code>	Specifies a VRF name. Length: Default: default. If no VRF name is specified, then the default VRF name will be taken.

Examples

```
switch(config-ip-sla-test)# vrf ipslasrc
switch(config-ip-sla-test)# no vrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-ip-sla- <code><IP-SLA-NAME></code>	Administrators or local user group members with execution rights for this command.

L1-100Mbps downshift

The speed downshift feature allows the user to link-up at sub-optimal speeds when failing to link-up at the highest advertised speed. There are fixed number of link attempts made to establish link at highest advertised speed and when all of them fail and attempt is made to link-up at a lower possible speed.

This feature requires underlying PHY to have support for the same and hence capability is only added to select set of ports. If a link cannot be established at the highest common denominator within a set number of link attempts, the PHY advertises the next highest speed using auto-negotiation.

Subtopics

Limitations with speed downshift

Limitations with speed downshift

- Link up may be delayed as certain number of retries are done to establish the link at highest advertise speeds by both link partners before downshifting.
- Link may be established at sub-optimal speed.

L1-100Mbps downshift commands

Select a command from the list in the left navigation menu.

Subtopics

[downshift enable](#)

[show interface](#)

[show interface downshift-enable](#)

[show running-config interface](#)

downshift enable

Syntax

```
downshift-enable  
no downshift-enable
```

Description

Enables/disables automatic speed downshift on an interface that supports downshift, generally 1GBASE-T ports. When enabled, downshift allows an interface to link at a lower advertised speed when unable to establish a stable link at the maximum speed. Downshifting only applies to physical interfaces that are not members of a LAG and is only available when auto-negotiation is enabled. When only one speed is advertised, downshift will not be triggered.

Examples

```
switch(config-if) # interface 1/1/1  
switch(config-if) # downshift-enable  
Warning: this is a non-standard mode for use only when standards-based
```

```
auto-negotiation is not able to establish a stable link. Enabling this
may cause the port to link at a lower than expected speed and should
not be used on ports that are members of a LAG. Support calls may require
this feature to be disabled
Continue (y/n)?
switch(config-if) #
```

When automatic downshift is enabled:

```
switch(config-if) # show running-config interface
interface 1/1/1
    downshift-enable
```

Disabling automatic speed downshift:

```
switch(config-if) # interface 1/1/1
switch(config-if) # no downshift-enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

show interface

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical]
show interface [<IFNNAME>|<IFRANGE>] [extended [non-zero] | [human-
readable]]
show interface [<IFNNAME>] monitor [human-readable]
```



```
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief]
show interface lag [<LAG-ID>] [extended [non-zero] | [human-readable]]
show interface lag [<LAG-ID>] monitor [human-readable]
```

Description

Shows active configurations and operational status information for interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
brief	Shows brief info in tabular format.
physical	Shows the physical connection info in tabular format.
extended	Shows additional statistics, including the tx filtered and rx filtered counters. - Rx filter packets are protocol packets received when the protocol is disabled on the switch and there is only one port in the VLAN. Protocols include OSPF, PIM, RIP, LACP, and LLDP. - An example of a Tx filtered packet would be a multicast packet being filtered from going out of the ingress port.
human-readable	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G. This is available only in the CLI interface output.
non-zero	Shows only non zero statistics.
LAG	Shows LAG interface information.
monitor	Continuously monitor interface statistics.
LOOPBACK	Shows loopback interface information.
TUNNEL	Shows tunnel interface information.

Parameter	Description
VLAN	Shows VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1 - 256
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0 - 255
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1 - 255
<VLAN-ID>	Specifies the VLAN ID. Range: 1 - 4094
VXLAN	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1

Examples

Showing interface information when it is configured as a route-only port:

```
switch# show interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Link state: up for 2 days (since Sun Jun 21 05:30:22 UTC 2020)
Link transitions: 1
Description: backup data center link
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 1500
Type 1GbT
Full-duplex
qos trust none
Speed 1000 Mb/s
```

Auto-negotiation is on

Flow-control: off

Error-control: off

Energy-Efficient Ethernet is enabled

MDI mode: MDIX

L3 Counters: Rx Enabled, Tx Enabled

Rate collection interval: 300 seconds

Rates	RX	TX	Total (RX+TX)
-----	-----	-----	-----
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00
Statistics	RX	TX	Total
-----	-----	-----	-----
Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
L3 Packets	0	0	0
L3 Bytes	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0
Other	0	0	0

MDI mode: MDIX

VLAN Mode: native-untagged

Native VLAN: 1

Allowed VLAN List: all

Rate collection interval: 300 seconds

Rates	RX	TX	Total (RX+TX)
-----	-----	-----	-----
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00

Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00
Statistics	RX	TX	Total

Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

Showing information when the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

Showing information when the interface is currently linked with energy-efficient-ethernet negotiated:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled and active
```

Showing information when the interface is shut down during a VSX split:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is down
Admin state is up
State information: Disabled by VSX
Link state: down for 3 days (since Tue Mar 16 05:20:47 UTC 2021)
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 04:09:73:62:90:e7
MTU 1500
Type SFP+DAC3
Full-duplex
```

```

qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: 1502-1505
Rate collection interval: 300 seconds

```

Rate	RX	TX	Total
(RX+TX)			

Mbits / sec	0.00	0.00	
0.00			
KPkts / sec	0.00	0.00	
0.00			
Unicast	0.00	0.00	
0.00			
Multicast	0.00	0.00	
0.00			
Broadcast	0.00	0.00	
0.00			
Utilization	0.00	0.00	
0.00			
Statistic	RX	TX	
Total			

Packets	0		
0	0		
Unicast	0		
0	0		
Multicast	0		
0	0		
Broadcast	0		
0	0		
Bytes	0		
0	0		
Jumbos	0		
0	0		
Dropped	0		
0	0		

```

Pause Frames          0
0                      0
Errors                0
0                      0
CRC/FCS              0
n/a                   0
Collision             n/a
0                      0
Runts                0
n/a                   0
Giants               0
n/a                   0

```

Showing information when the interface is configured with EEE and the EEE has auto-negotiated:

```
switch(config-if) # show interface 1/1/1 physical
```

```

-----
Control          EEE      Link  Admin      Speed      Flow-
Port            Type      PoE Power      Port
Config  Status | Config  Status  Config  Status |
Config  Status | Config  (Watts)  State  Information  Description
-----
1/1/1      1GbT      up      up      1G      auto      off
off      on      on      --      10M/100M/1G      --

```

Showing the monitor information:



NOTE

In monitor mode, the CLI refreshes data automatically until it is exited by entering **q**. Pressing **?** opens the help menu to display which options are available in this context.

```

Interface 1/1/1 is up
Rate                      RX                      TX                      Total
(RX+TX)
-----
Mbits / sec              30196.43              30196.43
60392.85
MPkts / sec              58977.39              58977.40
117954.79
Unicast                  0.00                  0.00
0.00
Multicast                58977.39              58977.40

```

```

117954.79
Broadcast                0.00                0.00
0.00
Utilization %           75.49                75.49
150.98
Statistic                RX                TX                Total
(RX+TX)
-----
Packets                  4756527649            4756527865
9513055514
Unicast                  0
0                        0
Multicast                4756527649            4756527865
9513055514
Broadcast                2
0                        2
Bytes                    304417778668          304417795428
608835574096
Jumbos                  0
0                        0
Dropped                  0                19028847730
19028847730
Pause Frames             0
0                        0
Errors                   0
0                        0
CRC/FCS                  0
n/a                      0
help: ?, quit: q
Help for Interface Monitor
h  Toggle human-readable mode
c  Clear interface statistics
Does not apply to rates
Arrows, PgUp, PgDn, Home, End
Navigate interface statistics
Delay: 2
help: ?, quit: q

```

Showing the output for interface 1/1/1 in human-readable format:



NOTE

In human-readable format, the **< 1** symbol for **Utilization** indicates that the amount of packets is between zero and one. This is true in cases where the number of bytes increases but the number of packets and the **Utilization** value is not displayed even in the normal output, where the human-readable parameter is not included in the command.

```
switch(config-if)# show interface 1/1/1 human-readable
Interface 1/1/1 is up
Rate                                     RX                                     TX                                     Total
(RX+TX)
-----
Bits / sec                             3M
3M                                     6M
Pkts / sec                             316                                    316
633
Unicast                               319                                    319
638
Multicast                             0
0                                     0
Broadcast                             0
0                                     0
Utilization %                          < 1                                    < 1
< 1
Statistic                             RX                                     TX
Total
-----
Packets                               577K
577K                                  1M
Unicast                               577K
577K                                  1M
Multicast                             0
51                                    51
Broadcast                             0
15                                    15
Bytes                                 744M
745M                                  1G
Jumbos                                0
0                                     0
Dropped                               0
```


0	0	
Filtered		0
0	0	
Pause Frames		0
0	0	
Errors		0
0	0	
CRC/FCS		0
n/a	0	
Collision		n/a
0	0	
Runts		0
n/a	0	
Giants		0
n/a	0	

Showing information about extended counters:



NOTE

The output of the `show interface extended` command varies depending on the switch model and configuration.

```
switch(config-if)# show interface 1/1/17 extended
```

```
-----
Interface 1/1/17
-----
```

Statistics	Value
Dot1d Tp Port In Frames	547
Dot1d Tp Port Out Frames	608
Dot3 In Pause Frames	0
Dot3 Out Pause Frames	0
Ethernet Stats Broadcast Packets	19
Ethernet Stats Bytes	40162
Ethernet Stats Packets	342
...	

Error-Statistics	Value
Dot1d Base Port MTU Exceeded Discards	0
Dot3 Control In Unknown Opcodes	0
Dot3 Stats Alignment Errors	0
Dot3 Stats FCS Errors	0
Dot3 Stats Frame Too Longs	0

```
Dot3 Stats Internal Mac Transmit Errors      0
Ethernet RX Oversize Packets                 0
...
```

Showing interface link-status:

```
switch# show interface link-status
```

```
-----
Port                Type                Physical      Link          Last
Link State          Transitions    Change
-----
1/1/1               1G-BT          down          0             --
1/1/2               1G-BT          up            1             1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/3               1G-BT          up            1             1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/4               --             down          0             --
1/1/5               --             down          0             --
```

Showing interface loopback 1 link-status:

```
-----
Port                Type                Physical      Link          Last
Link State          Transitions    Change
-----
loopback1           --             up            --            --
```

Showing interface 1/1/2-1/1/3 link-status:

```
-----
Port                Type                Physical      Link          Last
Link State          Transitions    Change
-----
1/1/2               1G-BT          up            1             1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/3               1G-BT          up            1             1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
```

Showing interface link-status:

```
switch# show interface link-status
```

```
-----
Port                Type                Physical      Link          Link Flaps    Last
Link State          Transitions    Ignored        Change
-----
1/1/1               1G-BT          down          0             0             --
1/1/2               1G-BT          up            1             0             1 minute
ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/3               1G-BT          up            1             0             1 minute
```

```
ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/4      --          down      0          0          --
1/1/5      --          down      0          0          --
```

Showing state information when interface is blocked:

```
8360(config-if)# show interface 1/1/1
Interface 1/1/1 is up (Blocked)
Admin state is up
State information: Blocked by UDLD
Link state: up for 1 minute (since Mon Jun 10 09:25:27 UTC 2024)
Link transitions: 1
Description:
Persona:
Hardware: Ethernet, MAC Address: 00:fd:45:67:85:91
MTU 1500
Type 10G-LR / 10G SFP+ LR
Full-duplex
qos trust none
Speed 10000 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rate collection interval: 300 seconds
```

Rate	RX	TX	Total
(RX+TX)			
Mbits / sec	0.00	0.00	
0.00			
KPkts / sec	0.00	0.00	
0.00			
Unicast	0.00	0.00	
0.00			
Multicast	0.00	0.00	
0.00			
Broadcast	0.00	0.00	
0.00			
Utilization %	0.00	0.00	
0.00			
Statistic	RX	TX	
Total			

Packets		15	
15	30		
Unicast		12	
12	24		
Multicast		3	
3	6		
Broadcast		0	
0	0		
Bytes		1350	1350
2700			
Jumbos		0	
0	0		
Dropped		0	
0	0		
Pause Frames		0	
0	0		
Errors		0	
0	0		
CRC/FCS		0	
n/a	0		
Collision		n/a	
0	0		
Runts		0	
n/a	0		
Giants		0	
n/a	0		

Command History

Release	Modification
10.15	Added state information when port goes into down state.
10.11	Added <code>monitor</code> parameter.
10.10	Added <code>human-readable</code> parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface downshift-enable

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] downshift-enable
```

Description

Displays speed downshift information, including the interface speed status and configuration.

Parameter	Description
<i><IFNAME></i>	Specifies a interface name.
<i><IFRANGE></i>	Specifies the port identifier range.

Examples

Showing automatic downshift information:

```
switch(config-if) # show interface downshift-enable
-----
Port          Downshift              Speed
             Enabled | Active      Status   | Config
-----
1/1/1         yes      yes      100M-FDx  auto
1/1/2         yes      no       1G       auto
```

1/1/3	yes	no	100M-FDx	100M-FDx
1/1/4	no	no	--	auto

Showing automatic downshift information on per interface:

```
switch(config-if)# show interface 1/1/2 downshift-enable
```

Port	Downshift		Speed	
	Enabled	Active	Status	Config
1/1/2	yes	no	1G	auto

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.
6400		

show running-config interface

Syntax

```
show running-config interface [<IFNNAME>|<IFRANGE>]
show running-config interface [lag | loopback | tunnel | vlan ] [<ID>]
```

Description

Displays active configurations of various switch interfaces.

Parameter	Description
	Specifies a interface name.

Parameter	Description
<code><IFNAME></code>	
<code><IFRANGE></code>	Specifies the port identifier range.
LAG	Specifies LAG interface information
LOOPBACK	Specifies loopback interface information.
TUNNEL	Specifies tunnel interface information.
VLAN	Specifies VLAN interface information.
<code><LAG-ID></code>	Specifies the LAG number. Range: 1 - 256.
<code><LOOPBACK-ID></code>	Specifies the LOOPBACK number. Range: 0 - 255.
<code><TUNNEL-ID></code>	Specifies the tunnel ID. Range: 1 - 255.
<code><VLAN-ID></code>	Specifies the VLAN ID. Range: 1 - 4094.
VXLAN	Specifies the VXLAN interface information.
<code><VXLAN-ID></code>	Specifies the VXLAN interface identifier. Default: 1.

Examples

Showing 1/1/2 interface configuration:

```
switch(config-if)# show running-config interface 1/1/2
interface 1/1/2
  no shutdown
  description DC-23
  exit
```

Showing loopback interfaces configured:

```
switch(config-if)# show running-config interface loopback
interface loopback 1
  description lb interface 1
  exit
interface loopback 2
  description lb interface 2
  exit
```

Showing loopback interfaces not configured:

```
switch(config-if)# show running-config interface loopback
No loopback interfaces configured.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Mirroring

Mirroring allows you to replicate all traffic arriving and/or leaving the selected system interfaces. This data can be used for collection or analysis.

The traffic replicated using mirroring can be sent to a separate interface on the same switch as the traffic source for analysis or inspection. Such a collection of interfaces and settings is called a mirror session.

A mirror session can be configured with many traffic sources but only a single output, or destination. In the initial configuration, the mirror session is disabled. You have enable the feature to start the replication.



CAUTION

Care must be taken in choosing the number and rates of sources to avoid over-saturating a session destination. A mirror session with multiple 10G sources can overwhelm a single 10G destination and important data may be lost.

This version of AOS-CX support the following mirror capabilities:

- Support for a VLAN as source for a mirror session
- Ability for a given Mirror source in one session to act as source in another Mirror Session
- Support for a Layer 2/bridged Link Aggregation Group (LAG) as Session destination
- Support for a Layer 3/Route-only Link Aggregation Group (LAG) as Session destination



NOTE

The following interface types are not supported as mirror source or destination:

- Management
- Persona
- Loopback
- VxLAN
- Subinterfaces

Hit Count Behavior for Policy Mirroring Actions

When using classifier policies with mirroring actions, it is important to note that packets mirrored via these policy actions will have their hit counts recorded within the Policy hit counts. These packets will not be recorded as output packets in the Mirror Session statistics, except in the case of Mirror-to-CPU sessions. This distinction means that traffic metrics for packets mirrored via Policy Actions should be obtained from Policy hit counts, as these packets will not appear in the Mirror Session output statistics.

Mirroring and CoPP

When configuring a Mirror Session with the destination set to the CPU, it is important to understand how mirrored traffic interacts with Control Plane Policing (CoPP).

Control plane packets that are already destined for the switch (for example, routing protocol packets and management traffic) will be processed by their respective CoPP classes. These packets will still be mirrored to the CPU as part of the Mirror-to-CPU session, but they will not be counted under the Mirror-to-CPU CoPP class. The Mirror-to-CPU CoPP class is specifically designed to manage dataplane traffic that is mirrored to the CPU. Any control plane traffic mirrored to the CPU will bypass this class and be handled by the CoPP

class corresponding to its original purpose (e.g., OSPF, BGP, or ARP). Administrators should account for this distinction when analyzing traffic mirrored to the CPU.

For example, if a mirror session is configured to mirror all traffic from a source interface to the CPU, and that interface receives OSPF packets, the OSPF packets will be mirrored to the CPU. These packets will be processed by the OSPF CoPP class, not the Mirror-to-CPU CoPP class.

Subtopics

Mirror statistics

Classifier policies and mirroring sessions

Mirroring on a VSX configuration

VLAN as a source

Mirroring commands

Mirror statistics

Mirror statistics are reset for a Mirror-to-CPU session when an interface is added or removed from a LAG that is a source interface in the Mirror session and during a failover.

Classifier policies and mirroring sessions

About this task

Network traffic can be mirrored to a destination interface in two ways:

- Using a mirroring session alone.
- Using Classifier Policies with mirror actions in conjunction with a mirroring session.

Basic mirroring sessions provide coarse control over the type of traffic mirrored from a source: all received, all transmitted, or both. However, a traffic class within a Classifier Policy applied to a source can provide much finer grained control of mirrored traffic. For example, a policy can match on many different aspects of the Ethernet or IPv4 or IPv6 header information in each frame or packet received or transmitted on an interface.

The steps to configure a policy and class with a mirror action are the following:

Procedure

1. Configuring a mirroring session with a destination interface.
2. Enabling the mirroring session.
3. Configuring the Classifier Policy, specifying the mirroring session ID in the mirror action.

Results

If the packets being mirrored are received from a VLAN that is not allowed on the mirror destination, the mirrored packets would be dropped at the mirror destination interface. When the mirrored packets are dropped at the destination, the mirror output packet and byte count will increment, however the packets will not be received at the mirror destination.

The mirror destination port among the active mirror sessions must be unique. That is, if an interface is configured as a source or destination in an active mirror session, the same port cannot be used as a destination in another active mirror session.

Scenario 1

Scenario 2

Mirroring on a VSX configuration

On 6400 switch series with a VSX configuration, north to south traffic may enter one switch, then cross the ISL to the other switch to reach its destination. If both 6400 switches have the following mirror configuration and the analyzer is connected to 1/1/10 on both 6400s, then the packet is mirrored by each 6400 switch, and two copies are sent to the analyzer.

```
mirror session 1
source vlan 1 both
source vlan 2 both
...
destination 1/1/10
```

VLAN as a source

About this task

AOS-CX allows configuration of VLAN as a mirroring source. When a VLAN source is configured in the 'rx' direction, all packets are mirrored as they are received in the switch. When a VLAN source is configured in 'tx' direction, all packets are mirrored as they are transmitted out of the switch.

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 1024 source VLANs in each direction of a given mirror session.

Same VLANs can be configured as a mirror source for multiple sessions.



NOTE

When changing a source VLAN in an enabled mirror session (that is, adding, changing direction, or removing), mirrored packets being transmitted out the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways:

Procedure

1. Reenter the `source vlan` command with the new preferred direction.
2. Use the `no` form of the command with a direction (rx or tx) to selectively remove the specified direction. Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

Results

For packets bridged through the switch:

If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in the 'rx' direction, packets are mirrored in the pre-routed form with the destination MAC address as the switch address.
- If the mirror is configured in the 'tx' direction, packets are mirrored in the post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in the 'both' direction, one copy of the packet will be mirrored.

Control plane packets generated by the switch's CPU are processed both in the ingress and the egress packet processing pipeline. The following are the behaviors for mirroring with VLAN as source:

- If the mirror is configured in the 'rx' or 'tx' direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the 'both' direction, two copies of the packets are mirrored to the mirror destination.

Mirroring commands

Select a command from the list in the left navigation menu.

Subtopics

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[clear mirror endpoint](#)
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[copy tcpdump-pcap](#)
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[mirror session](#)
[show mirror](#)
[show mirror endpoint](#)
[shutdown](#)
[source](#)
[source interface](#)
[source vlan](#)

clear mirror

Syntax

```
clear mirror [all | <SESSION-ID>]
```

Description

Clears the mirror statistics for all configured mirror sessions or a specified session

Parameter	Description
all	Specifies all configured sessions.
<SESSION-ID>	Specifies a numeric identifier for the session. Range: 1 to 4

Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

clear mirror endpoint

Syntax

```
clear mirror endpoint [<NAME>]
```

Description

Clears mirror endpoint statistics for all configured mirror endpoints. The optional parameter can be added to clear a specific mirror endpoint.

Parameter	Description
<NAME>	Specifies name of the mirror endpoint instance to be cleared.

Examples

Clearing statistics for all configured mirror endpoints:

```
switch# clear mirror endpoint
```

Clearing mirror statistics for mirror endpoint test:

```
switch# clear mirror endpoint test
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

comment

Syntax

```
comment <COMMENT>  
no comment
```

Description

Specifies a comment for the mirroring session.

When used in mirror endpoint command context, specifies a comment for the mirror endpoint.

The **no** form of this command removes the comment.

Parameter	Description
<COMMENT>	A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3)# comment This Mirror will be removed during next maintenance window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3)# no comment
```

Adding a comment to a mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor endpoint traffic
```

Replacing the existing comment for mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor statistics on each endpoint interfaces
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror- <SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

copy tcpdump-pcap

Syntax

```
copy tcpdump-pcap <FILE-NAME> <REMOTE-URL>
```

Description

Saves packet capture files to external storage.

Parameter	Description
<code><FILE-NAME></code>	Specifies the packet capture file to save.
<code><REMOTE-URL></code>	Specifies the external storage to which the packet capture file will be saved.

Usage

Only four files can be saved at any point on the switch. Packet capture files are not saved after a failover or reboot. View a list of saved files using **diag utilities list-files**.

Examples

Saving my_capture_file.pcap to sftp://root@10.0.0.2/file.pcap:

```
switch# copy tcpdump-pcap my_capture_file.pcap sftp://root@10.0.0.2/  
file.pcap
```

```
root@10.0.0.2's passowrd:
Connected to 10.0.0.2.
sftp > put my_capture_file.pcap file.pcap
Uploading my_capture_file.pcap to /root/file.pcap
my_capture_file.pcap          100%   156   219.8KB/s
00:00
Copied successfully.
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

copy tshark-pcap

Syntax

```
copy tshark-pcap <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Copies the tshark capture data to a file on a TFTP or SFTP server.

Parameter	Description
<code><REMOTE-URL></code>	Specifies the capture file on a remote TFTP or SFTP server. The URL syntax is: {tftp:// sftp://<USER>@} {<IP> <HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
<code>vrf <VRF-NAME></code>	Specifies the name of a VRF. Default: default.

Example

Copying the capture data to a file on SFTP server 10.0.0.2:

```
switch# copy tshark-pcap sftp://root@10.0.0.2/file.pcap
root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp> put packets.pcap file.pcap
Uploading packets.pcap to /root/file.pcap
packets.pcap                               100% 156    219.8KB/s   00:00
Copied successfully.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Manager (#)	Administrators or local user group members with execution rights for this command.

destination cpu

Syntax

```
destination cpu
no destination cpu
```

Description

The command causes the mirror session to transmit mirrored packets to the switch CPU. This destination may be configured for multiple sessions, however only one such configured session may be active at a given time.

The diagnostic utility Tshark may be used to view and capture packets transmitted to the CPU through this route. Ctrl+C must be entered to terminate a Tshark capture session. More details can be found in the **Supportability Guide**.

The **no** form of this command will immediately stops mirroring traffic to the CPU, but will not remove any sources from the mirror configuration.

Examples

Configuring a mirror session with CPU as the destination.

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination cpu
```

Removing the destination entirely.

```
switch(config-mirror-1)# no destination cpu
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

destination interface

Syntax

```
destination interface {<INTERFACE-ID>|<LAG-NAME>}
no destination interface {<INTERFACE-ID>|<LAG-NAME>}
```

Description

Configures the specified interface as the destination of the mirrored traffic.

The **no** form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

Parameter	Description
<code><INTERFACE-ID></code>	Specifies a interface. Format: <code>member/slot/port</code> .
<code><LAG-NAME></code>	Specifies a LAG (link aggregation group) identifier.

Usage

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration. Mirroring traffic to a LAG destination with a device running LACP attached to the destination will cause the LAG link to flap if LACP packets are part of the mirrored traffic source.

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface 1/1/12
```

Switch	Destination interface limit per mirror session (4 possible sessions)
6300	64
6400	64
10040	1

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror- <SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

destination tunnel

Syntax

```
destination tunnel <TUNNEL-IPV4-ADDR> source <SOURCE-IPV4-ADDR>
    dscp <DSCP-VALUE> vrf <VRF-NAME> id <SPAN-ID>
no destination tunnel
destination tunnel <TUNNEL-IPV4> source <SOURCE-IPV4-ADDR>
    dscp <DSCP-VALUE> vrf <VRF-NAME>
no destination tunnel
```

Description

Specifies the tunnel where all mirrored traffic for the session is transmitted. Only one tunnel destination is allowed per session.

You may configure multiple mirror sessions with the same source/destination IP address pair, however, only one of those sessions sharing the same source/destination IP address pair can be enabled at a given time.

ERSPAN is not supported leaving the switch by the OOB port. If VRF management is configured for an ERSPAN session, the session will be in "mirror_err_tunnel_oob_port_not_supported" operation status.

ERSPAN is not supported leaving the switch encapsulated within another tunnel (e.g. GRE IPv4). When the path to the destination IP address will leave via a tunnel, the session will be in "tunnel_route_resolution_not_populated" operation status.



NOTE

The interface/LAG used to transmit ERSPAN packets should not be a source in the same mirror session.

The **no** form of this command will cease the use of the tunnel and disable the session.

Parameter	Description
<code><TUNNEL-IPv4-ADDR></code>	Specifies the tunnel address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
<code><SOURCE-IPv4-ADDR></code>	Specifies the source address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
<code><DSCP-VALUE></code>	Specifies the DSCP value to be carried within the DS field of E RSPAN packet header. Range: 0 to 63. Default: 0.
<code><VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

Creating a Mirror Session and adding tunnel destination, source, dscp, and VRF:

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination tunnel 1.1.1.1 source 2.2.2.2 dscp 10
vrf default
```

Replacing the existing tunnel destination:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp
10 vrf default
```

Replacing the existing destination with a different DSCP value:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp
2 vrf default
```

Removing the destination:

```
switch(config-mirror-1)# no destination tunnel
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

diagnostic

Syntax

```
diagnostic
diag utilities tshark [file]
diag utilities tshark [delete-file]
```

Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.



NOTE

The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the **delete-file** form of this command to delete the most recent capture file.

Since **file** and **delete-file** are optional, the behavior of the base command **diag utilities tshark** does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

Parameter	Description
file	Saves captured packets to a temporary file.
delete-file	Deletes the most recent captured file.

Example

Performing diagnostic:

```
switch# diagnostic
switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

diag utilities tcpdump

Syntax

```
diag utilities tcpdump [command <TEXT> | delete file <FILE-NAME> | list-
files |
    vrf <VRF-NAME> | count <COUNT-NUM> | proto <PROTO-NUM> | host-ip <IP-
ADDR> | source-ip
```

```
<IP-ADDR> | destination-ip <IP-ADDR> | host-port <PORT> | source-port
<PORT> |
destination-port <PORT> | verbosity <LEVEL> | print <DATA> | ethernet-
type <ETH-NUM>]
```

Description

Captures traffic received or transmitted over a network.

Parameter	Description
command <TEXT>	Captures packets based on a specified tcpdump command string.
delete file <FILE-NAME>	Deletes specified tcpdump list files.
list-files	Lists all the tcpdump capture files saved on the device.
vrf <VRF-NAME>	Captures packets on the specified VRF. If no VRF is named, the default is used.
count <COUNT-NUM>	Runs the tcpdump command until the specified number of packets are captured. Range: 1 - 2147483647.
proto <PROTO-NUM>	Captures packets of a particular type based on IP protocol number. Range: 0 - 255.
host-ip <IP-ADDR>	Captures packets matching with the source or destination IP address.
source-ip <IP-ADDR>	Captures packets from the specified IP address.
destination-ip <IP-ADDR>	Captures packets sent to the specified IP address.
host-port <PORT>	Captures packets matching with the source or destination port.
source-port <PORT>	Captures packets from the specified IP port.
destination-port <PORT>	Captures packets sent to the specified IP port.
verbosity <LEVEL>	Captures packets of the specified verbosity. Range: level1 - level4. If no verbosity is specified, the default is level1.

Parameter	Description
<code>print <DATA></code>	Captures the data of each packet. The maximum is 262144 bytes
<code>ethernet-type <ETH-NUM></code>	Captures packets based on the particular ethernet type. Range: 0 - 65535.

Usage

- When using the **command** option, the only traffic captured will be packets that have been mirrored to the CPU.
- When using the **command** option, command line sanitization is performed to prevent options that may cause harm or security issues. The following options are blocked:
 - `-i/--interface`
 - `-Z`
 - `-B/--buffer-size`
 - `-C`
 - `-W`
 - `-Z/--relinquish privileges`
- Non-word operators such as "&" or "|" are not allowed. Use boolean keywords such as "and," "or," and "not."
- When using **command -r** to read a file, do not provide any directory path characters. Use `list-files` command to get the list of file names currently saved on the device, and then use those file names.
- A total of four files can be saved at any given point on the device. Packet capture files are not saved after a failover or reboot, but can be saved to external storage using the **copy tcpdump-pcap** command.

Examples

Inspecting traffic mirrored to the CPU via tcpdump and saving the output to my_capture_file.pcap:

```
switch# diag utilities tcpdump command -c 2 -x -w my_capture_file.pcap
Inspecting traffic mirrored to the CPU via tcpdump until Ctrl-C is entered.
2 packets captured
2 packets received by filter
0 packets dropped by kernel
Ending traffic capture.
```

Listing saved capture files:

```
switch# diag utilities tcpdump list-files
```

```
my_capture_file.pcap
```

Reading my_capture_file.pcap:

```
switch# diag utilities tcpdump command -r my_capture_file.pcap
```

```
reading from file /tmp/tcpdump/my_capture_file1.pcap, link-type EN10MB  
(Ethernet)
```

```
1 11:59:34.047867 IP6 localhost.40318 > localhost.ntp: NTPv2, Reserved,  
length 12
```

```
0x0000: 0000 0304 0006 0000 0000 0000 0000 0000 86dd .....
```

```
0x0010: 600a 7e47 0014 1140 0000 0000 0000 0000 `~G...@.....
```

```
0x0020: 0000 0000 0000 0001 0000 0000 0000 0000 .....
```

```
0x0030: 0000 0000 0000 0001 9d7e 007b 0014 0027 .....~.{...'
```

```
0x0040: 1601 0001 0000 0000 0000 0000 .....  
.....
```

```
2 11:59:34.047915 IP6 localhost.ntp > localhost.40318: NTPv2, Reserved,  
length 12
```

```
0x0000: 0000 0304 0006 0000 0000 0000 0000 0000 86dd .....
```

```
0x0010: 6b8d 23c5 0014 1140 0000 0000 0000 0000 k.#....@.....
```

```
0x0020: 0000 0000 0000 0001 0000 0000 0000 0000 .....
```

```
0x0030: 0000 0000 0000 0001 007b 9d7e 0014 0027 .....{.~....'
```

```
0x0040: d681 0001 c016 0000 0000 0000 .....  
.....
```

Removing my_capture_file.pcap:

```
switch# diag utilities tcpdump delete-file my_capture_file.pcap
```

```
Successfully removed file
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
6300	Manager (#)	Administrators or local user group members with execution rights for this command.
6400		

disable

Syntax

```
disable
```

Description

Disables the mirroring session specified by the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the **show mirror** command for that session ID shows an **Admin Status** of **disable** and an **Operation Status** of **disabled**.

Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

enable

Syntax

```
enable
```

Description

Enables the mirroring session for the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the **show mirror** command for that session ID shows an **Admin Status** of **enable** and an **Operation Status** of **enabled**.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of

rx

or

both

):

- The attempt to enable the mirroring session fails and an error is returned.



NOTE

When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

Example

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Configuring and enabling a mirroring session:

```
switch(config) # mirror session 3
switch(config-mirror-3) # source interface 1/1/2 rx
switch(config-mirror-3) # destination interface 1/1/3
switch(config-mirror-3) # comment Monitor router port ingress-only traffic
switch(config-mirror-3) # enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror- <SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

mirror endpoint

Syntax

```
mirror endpoint <NAME>
no mirror endpoint <NAME>
```

Description

Creates the specified mirror endpoint or enters its context if it already exists. The specifics of a mirror endpoint are created or altered while in the mirror endpoint context and the mirror endpoint is enabled or disabled from this context. It may be possible to support different encapsulations by different ASICs. For example, UDP for PVOS compatibility. Termination of GRE encapsulation is also supported.

The **no** form of this command removes an existing mirror endpoint. An enabled mirror endpoint is automatically disabled first before removal.

Parameter	Description
<code><NAME></code>	Specifies mirror endpoint name.

Examples

Creating a mirror endpoint named test :

```
switch(config) # mirror endpoint test
```

Deleting mirror endpoint named test:

```
switch(config) # no mirror endpoint test
```

Configuring a mirror endpoint named test :

```
6100 (config) # mirror endpoint test
6100 (config-mirror-endpoint-test) #
6100 (config-mirror-endpoint-test) # destination
    interface Specify interfaces to send traffic
6100 (config-mirror-endpoint-test) # destination interface
    IFNAMELIST An interface, a range or a comma seperated list of interfaces
6100 (config-mirror-endpoint-test) # destination interface 1/1/3
    <cr>
6100 (config-mirror-endpoint-test) # destination interface 1/1/3
6100 (config-mirror-endpoint-test) #
6100 (config-mirror-endpoint-test) # source 1.1.1.1 destination 1.1.1.2 id 1
vrf default
6100 (config-mirror-endpoint-test) #
```



NOTE

Only physical ports can be configured as interface for mirror-endpoint destination. LAG port is not supported as interface for mirror-endpoint destination.



NOTE

The maximum allowed number of destination interfaces for both mirror-session and mirror-endpoint is 1.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

mirror session

Syntax

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The **no** form of this command removes an existing mirroring session from the configuration.

Parameter	Description
<SESSION-ID>	Specifies the session identifier. Range: 1 to 4

Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#
switch(config)# mirror session 3
switch(config-mirror-3)#
switch(config)# no mirror session 1
switch(config)#
```



NOTE

When configuring mirroring via the command-line interface, not all configuration errors will result in an immediate error message. After making configuration changes, always check the operation status of your mirror sessions using the **show mirror** command.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show mirror

Syntax

```
show mirror [<SESSION-ID>] [vsx-peer]
```

Description

Shows information about mirroring sessions. If `<SESSION-ID>` is not specified, then the command shows a summary of all configured mirroring sessions. If `<SESSION-ID>` is specified, then the command shows detailed information about the specified mirroring session.

Parameter	Description
<code><SESSION-ID></code>	Specifies the session identifier. Range: 1 to 4
<code>vsx-peer</code>	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

Information in the **Admin Status** column of the command output indicates the configured status. The admin status is one of the following values:

- **enable**: The mirroring session is enabled.
- **disable**: The mirroring session has been configured but not yet enabled, or has been disabled.

Information in the **Operation Status** column indicates the status of the mirroring session. Operation status is one of the following values:

- **dest_doesnt_exist:**The configured destination interface is not found in the system. The mirroring session cannot be enabled.
- **destination_shutdown:** **The mirroring session is enabled, but the destination interface is shut down. No traffic** can be monitored.
- **disabled:** The mirroring session is disabled and is not in an error condition.
- **enabled:** The mirroring session is enabled.
- **external/driver_error:** An internal ASIC hardware error occurred.
- **hit_active_sessions_capacity:** The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.
- **internal_error:** An invalid parameter was passed to the ASIC software layer.
- **no_dest_configured:** The mirroring session does not have a destination interface configured.
- **no_name_configured:** A software error occurred. The mirroring session does not have a session ID in its configuration.
- **null_mirror:** A software error occurred. The session object reference is invalid.
- **out_of_memory:** The system is out of memory, reboot recommended.
- **tunnel_route_resolution_not_populated:** If the destination tunnel IP address is not reachable.
- **unknown_error:** An unexpected error occurred.

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID  Admin Status  Operation Status
---  -
1   enable       enabled
2   disable      disabled
3   disable      disabled
4   enable       internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
Mirror Session: 3
Admin Status: disable
Operation Status: disabled
Comment: Monitor router port ingress-only traffic
```

```
Source: interface 1/1/2 rx
Destination: interface 1/1/3
switch#
```

Show the details of mirror session 1 with an empty LAG as destination:

```
switch: show mirror 1
Admin Status: enable
Operation Status: dest_doesnt_exist
Source: interface 1/1/1 rx
Source: interface tx none
Destination: interface lag1
Output Packets: 0
Output Bytes: 0
```

Show the details of mirror session 1 with a LAG and a LAG member of same LAG as sources. In this scenario, traffic received on all LAG member interfaces is mirrored, is it is not necessary to define 1/1/1 as a source since it is already part of LAG1; this does not impact the mirror output packet or byte counters.

```
switch: show mirror 1
Admin Status: enable
Operation Status: enabled
Source: interface 1/1/1 rx
Source: interface lag1 rx
Destination: interface 1/1/2
Output Packets: 0
Output Bytes: 0
```

Showing the details of mirror **session 1** for a mirror configuration enabled with invalid interface which is not part of a lag:

```
switch# show mirror 1
Mirror Session: 1
Admin Status: enable
Operation Status: disabled
Source: interface 1/1/1 (unknown lag) both
Destination: cpu
Output Packets: 0
Output Bytes: 0
```

Certain configurations can cause this condition while the mirror source configuration remains unchanged:

1. Removing of a port from a LAG (Link Aggregation Group).
2. Port **split** or **unsplit** operations that result in LAG membership removal.

To recover from this issue, unconfigure the affected interface (previously part of a LAG), or restore the LAG membership configuration.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show mirror endpoint

Syntax

```
show mirror endpoint [<NAME>]
```

Description

Shows a list of all configured mirror endpoints, their Admin Status and their Operation Status.

The optional parameter will display the details of the specified mirror endpoint if it exists.

Parameter	Description
<NAME>	Specifies name of the mirror endpoint instance to be displayed.

Examples

Showing a summary of all configured mirror endpoints on the switch:

```
switch# show mirror endpoint
Name      Admin Status  Operation Status
-----
test      enable       enabled
monitor   disable      disabled
```

Showing the details of enabled mirror endpoint test:

```
switch# show mirror endpoint test
Mirror Endpoint: audit
Admin Status: enable
Operation Status: enabled
Comment: Mirror Endpoint Audit
Type: gre
Tunnel: source 1.1.1.1 destination 1.1.1.2 id 1 vrf default
Interface: 1/1/3
Output Packets: 123456789
Output Bytes: 0
```



NOTE

"Output Packets" in "show mirror endpoint [name]" is only supported for statistics.

"Output Bytes" in "show mirror endpoint [name]" is not supported due to ASIC limitation.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

shutdown

Syntax

```
shutdown
no shutdown
```

Description

Enables mirror endpoint from its default disabled state. To verify the mirror endpoint was successfully activated, run the **show mirror endpoint NAME** command and verify that the **Admin Status** and **Operational Status** has changed from disabled to enabled. If the status value remains disabled, consult the system logs to determine the reason for activation failure. To disable the mirror endpoint, first disable the remote mirror session on the switch that's originating the data. Next, use the `shutdown` command to disable the mirror endpoint.

Examples

Enabling a mirror endpoint:

```
switch(config)# mirror endpoint test
switch(config-mirror-endpoint-test)# no shutdown
```

Disabling a mirror endpoint:

```
switch(config)# mirror endpoint test
switch(config-mirror-endpoint-test)# shutdown
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

source

Syntax

```
source <SOURCE-IP> destination <DESTINATION-IP> id <1-4294967295> [vrf
<VRF_NAME>] [type {gre}]
no source
```

Description

Configures tunnel parameters of the mirror endpoint. Configuring a tunnel parameter to a mirror endpoint will replace the existing configuration. By default the VRF is **default**, users can also explicitly provide a custom VRF. The default tunnel type is considered to be GRE and users also have the option to explicitly give type as GRE.

The **no** form removes the tunnel parameters of the mirror endpoint.

Parameter	Description
<SOURCE-IP>	Specifies L3 encapsulated IPv4 source in the form A.B.C.D.
<DESTINATION-IP>	Specifies L3 encapsulated IPv4 destination in the form A.B.C.D.
id	Specifies tunnel identifier from the encapsulated packet.
<VRF_NAME>	Specifies the name of VRF for which the tunnel belongs to.

Examples

Configuring a tunnel parameter to a mirror endpoint:

```
switch(config-mirror-endpoint-test)# source 1.1.1.1 destination 7.7.7.7 id
1 vrf default type gre
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

source interface

Syntax

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]  
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored.

The **no** form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

Parameter	Description
<PORT-NUM>	Specifies a physical port on the switch. Use the format member/slot/port (for example, 1/3/1).
<LAG-NAME>	Specifies the identifier for the LAG (link aggregation group).
<DIRECTION>	Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following: - both: Mirror both transmitted and received packets. - rx: Mirror only received packets. - tx: Mirror only transmitted packets.

Usage

There is a limit of source interfaces in each direction of a given mirror session:

Switch	Source interface limit per mirror session (4 possible sessions)
6300	64
6400	64
10040	128

However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.



NOTE

When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1)# source interface
LAG-NAME      Enter a LAG name. For example, lag10
PORT-NUM      Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config)# mirror session 1
switch(config-mirror-1)# source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config)# mirror session 2
switch(config-mirror-2)# source interface 1/1/3 tx
switch(config-mirror-2)# source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2)# no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3)# source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1)# source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4)# source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4)# no source interface lag1 rx
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

source vlan

Syntax

```
source vlan <VLAN-NUM> {rx | tx | both}
no source vlan <VLAN-NUM> {rx | tx | both}
```

Description

Mirroring with VLAN as a source is supported in the following traffic directions:

- **both** - traffic received and transmitted

- **rx** - only received traffic
- **tx** - only transmitted traffic

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 1024 source VLANs for a given mirror session. There is also a limit of 4096 source VLANs across all mirror sessions.

Same VLAN can be configured as a mirror source for multiple sessions.



NOTE

When changing a source VLAN in an enabled mirror session (i.e. adding, changing direction, or removing) mirrored packets being transmitted out of the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways.

- Reenter the **source vlan <VLAN-NUM> <direction>** command with the new preferred direction.
- Use the **no source vlan <VLAN-NUM> <direction>** form of the command with a direction (**rx** or **tx**) to selectively remove the specified direction.

Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

Mirroring allows configuration of VLAN as a source. When VLAN source is configured in the **rx** direction, all packets are mirrored as they are received in the switch. When VLAN source is configured in **tx** direction, all packets are mirrored as they are transmitted out of the switch.

For packets bridged through the switch:

- If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in **rx** direction, packets are mirrored in the pre-routed form with the Destination MAC address as the switch address.
- If the mirror is configured in **tx** direction, packets are mirrored in post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in **both** direction, one copy of the packet will be mirrored.

Control plane packets generated by the switch's CPU are processed both in the ingress and the egress packet processing pipeline. The following are the behavior for mirroring with VLAN as source:

- If the mirror is configured in the **rx** or **tx** direction, the packets are mirrored to the mirror destination.

- If the mirror is configured in the **both** direction, two copies of the packets are mirrored to the mirror destination.

The **no** form command will cease mirroring traffic from the specified source VLAN and remove the source from the mirror configuration.

Parameter	Description
VLAN-NUM	Selects the VLAN number.
direction	Specifies the direction of mirroring. tx (transmit), rx (receive), or both .

Examples

Creating a mirror session and adding a VLAN as a source of traffic in both directions on that port:

```
switch# configure terminal
switch(config)# mirror session 1
switch(config-mirror-1)# source vlan 10 both
```

Creating a mirror session and adding two VLANs as sources of traffic:

directions:

```
switch# configure terminal
switch(config)# mirror session 2
switch(config-mirror-2)# source vlan 10 tx
switch(config-mirror-2)# source vlan 20 both
```

Configuring the source in session 2 to receive by specifying the source interface configuration:

```
switch(config-mirror-2)# source vlan 10 rx
```

Removing the first source interface in session 2 entirely, and removing the transmit direction from the other so that mirroring only occurs in the receive direction:

```
switch(config-mirror-2)# source vlan 10 rx
switch(config-mirror-2)# source vlan 20 tx
```

Showing maximum of 1024 mirror source VLANs allowed:

```
switch(config-mirror-2)# source vlan 2000 rx
The maximum number of source VLANs per mirror session is 1024 in each
direction
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

Monitoring a device using SNMP

Configuring SNMP: Refer to the SNMP/MIB Guide for information on how to add SNMP so a device can be monitored from a network management system (NMS).

Configuring an SNMP trap receiver: Refer to the SNMP/MIB Guide and specific information about the `show snmp trap` command to enable SNMP traps.

Packet Capture

The Packet Capture is a solution which enables the Central administrator to capture or view packets from clients connected to Halon switches. The solution serves as a tool which an admin can leverage to debug clients connected to managed devices across branches in an enterprise. The switch requires packet exchanges to be copied to the CPU (both ingress and egress packets) per client session, so that they can be streamed to Central where the admin can view the packet exchanges for further analysis.

This feature is supported on the following platforms:

- Rosewood (6200, 6300, 6400, 8360)
- Matrix (4100i, 6000, 6100)

Subtopics

[packet-capture commands](#)

packet-capture commands

Select a command from the list in the left navigation menu.

Subtopics

[copy packet-capture](#)
[packet-capture delete-pcap](#)
[packet-capture enable](#)
[packet-capture](#)
[show packet-capture pcaps](#)
[show packet-capture session](#)

copy packet-capture

Syntax

```
copy packet-capture-pcap <FILE_NAME> <REMOTE_URL>
```

Description

This command copies a capture file created using the [packet-capture](#) to a remote URL.

Parameter	Description
<FILE_NAME>	Specifies the name of the tcpdump capture file to be copied.
<REMOTE_URL>	Specifies a URL to copy the command output. Valid URL syntaxes: - sftp://USER@{IP HOST}[:PORT]/FILE - tftp://{IP HOST}[:PORT][;blocksize=VAL]/FILE

Example

```
8400X# copy packet-capture-pcap my_capture_file.pcap sftp://root@10.0.0.2/
file.pcap
root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp > put my_capture_file.pcap file.pcap
Uploading my_capture_file.pcap to /root/file.pcap
```

```
my_capture_file.pcap 100% 156 219.8KB/s 00:00
Copied successfully.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

packet-capture delete-pcap

Syntax

```
packet-capture delete-pcap <FILE_NAME>
```

Description

This command deletes a capture file created with the [packet-capture](#) command.

Parameter	Description
<FILE_NAME>	Specifies the name of the tcpdump capture file to be deleted.

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

packet-capture enable

Syntax

```
packet-capture <SESSION_NAME> [enable]  
no...
```

Description

Enables the packet capture session.

The **no** form of this command disables the packet capture for the specified session.

Parameter	Description
<SESSION_NAME>	Specifies the session name.
enable	Enables packet capture for the specified session.



NOTE

A session name must be specified first before issuing the enable command or its no form. If the session hasn't been configured as a packet-capture session prior to running these commands, the error message Packet Capture session does not exist is displayed.

Examples

Enable packet capture for session1:

```
switch# packet-capture session1 enable
```

Disable packet capture for session1:

```
switch# no packet-capture session1 enable
```



NOTE

If session1 has been configured as a `packet-capture` session but isn't enabled, the error message Packet Capture session not running is displayed when issuing the no form of the command.

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

packet-capture

Syntax

```
packet-capture <SESSION_NAME> <PORT-ID> ( mac { <SMAC> <DMAC> <ETHERTYPE> }  
| ipv4 {<PROTOCOL>  
<SIP> eq <SPORT> <DIP> eq <DPORT> } | ipv6 {<PROTOCOL> <SIP> eq <SPORT>  
<DIP> eq <DPORT> } )  
file-name <PCAP_NAME> [timeout <TIMEOUT>] [buffer-size <BUFFER_SIZE>] [sw-  
filter <SW_FILTERS>] [vlan <VLAN>]
```

Description

Configures a new packet capture session.

Parameter	Description
<SESSION_NAME>	Configures the session name with a maximum of 32 characters.

Parameter	Description
<PORT-ID>	Configures the port ID. Applies to physical ports only.
<SMAC>	Configures the source MAC address.
<DMAC>	Configures the destination MAC address.
<ETHERTYPE>	Configures either a hexadecimal EtherType value (<H:600–FFF F>) or a supported protocol name (arp, any, appletalk, arp, fcoe, fcoe - init, ip, ipv6, ipx - arpa, ipx - non - arpa, is - is, lldp, mpls - multicast, mpls - unicast, q - in - q, rbridge, trill, wake - on - lan).
<PROTOCOL>	Configures either a numeric protocol value (0–255) or a supported protocol name (ah, any, esp, gre, icmp, igmp, ip, ospf, pim, sctp, tcp, udp).
<SIP>	Configures the source IP address.
<DIP>	Configures the destination IP address.
<SPORT>	Configures the source port.
<DPORT>	Configures the destination port.
<TIMEOUT>	Configures the maximum session time for packet capture in seconds. Default: 300.
<BUFFER_SIZE>	Configures the buffer size in MB. Range: 1 - 100. Default: 100.
<PCAP_NAME>	Configures the packet capture file name. Length: 5 - 128 characters.
<SW_FILTERS>	Configures the software filter with a maximum of 512 characters. Standard tcpdump filters are supported for this parameter.
<VLAN>	Configures the VLAN identifier. Range: 1 - 4094.

Examples

Packet capture using MAC:

```
switch# packet-capture session1 1/1/1 mac 0050.5696.7cef 0050.5696.7cea 0800
```

Help

<600-ffff>	Hexadecimal ethertype value
aarp	AppleTalk Address Resolution Protocol (AARP)
any	Any EtherType
appletalk	AppleTalk/EtherTalk
arp	Address Resolution Protocol (ARP)
fcoe	Fibre Channel over Ethernet
fcoe-init	Fibre Channel over Ethernet Initialization
ip	Internet Protocol version 4
ipv6	Internet Protocol version 6
ipx-arpa	IPX Advanced Research Projects Agency (ARPA)
ipx-non-arpa	IPX non-ARPA
is-is	Intermediate System to Intermediate System
lldp	Link Layer Discovery Protocol
mpls-multicast	MPLS Multicast
mpls-unicast	MPLS Unicast
q-in-q	IEEE 802.1ad encapsulation
rbridge	RBridge Channel Protocol
trill	IETF TRILL protocol
wake-on-lan	Wake on LAN

Packet capture using IPV4:

```
switch# packet-capture session1 1/1/1 ipv4 tcp any eq any any eq any
```

Help

<0-255>	Numeric protocol value
ah	Authentication Header
any	Any Internet Protocol
esp	Encapsulation Security Payload
gre	Generic Routing Encapsulation
icmp	Internet Control Message Protocol
igmp	Internet Group Management Protocol
ip	Any Internet Protocol
ospf	Open Shortest Path First (version 2)
pim	Protocol Independent Multicast
sctp	Stream Control Transmission Protocol
tcp	Transmission Control Protocol
udp	User Datagram Protocol

Packet capture using IPV6:

```
switch# packet-capture session1 1/1/1 ipv6 any 2001:db8:abcd:12::2/  
ffff:ffff::0000 eq any any eq any
```

Help

<0-255>	Numeric protocol value
ah	Authentication Header

any	Any Internet Protocol
esp	Encapsulation Security Payload
gre	Generic Routing Encapsulation
icmpv6	Internet Control Message Protocol Version 6
ipv6	Any Internet Protocol
ospf	Open Shortest Path First (version 3)
pim	Protocol Independent Multicast
sctp	Stream Control Transmission Protocol
tcp	Transmission Control Protocol
udp	User Datagram Protocol

General Limitations

- Only 3 pcap files are allowed at one time. Creating a fourth pcap file will replace the oldest one.
- Packet capture is only supported on physical ports.
- Only one `packet-capture` session can be active a time.
- A `packet-capture` session cannot be active at the same time as a mirror session with destination CPU.
- Packets captured by `packet-capture` sessions will not include the Ethernet frame check sequence in the resulting capture.

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	<code>config</code>	Administrators or local user group members with execution rights for this command.

show packet-capture pcaps

Syntax

```
show packet-capture pcaps
```

Description

This command shows capture files created with the packet-capture command.

Parameter	Description
pcaps	Lists the saved tcpdump capture files.

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

show packet-capture session

Syntax

```
show packet-capture [<SESSION_NAME>]
```

Description

Displays information about packet capture sessions configured on the switch. If a SESSION_NAME is specified, it shows details for that session only.

Parameter	Description
<SESSION_NAME>	Specifies the name of the packet capture session.

Examples

Packet capture using MAC:

```
switch# show packet-capture
Name                : orange0
Admin State         : Disabled
Operation Status    : Inactive
Reason code         : max-timeout-reached
Port                : 1/1/1
Hardware filter     : ipv4
  Ethertype         : -
  Source MAC        : -
  Dest MAC          : -
  Source IP         : -
  Destination IP    : -
  Protocol          : -
  Source port min   : 21
  Source port max   : 21
  Dest port min     : 1001
  Dest port max     : 1001
  Vlan              : -
Software Filter     : -
Buffer size (in MB) : 100
Timeout (in second) : 300
Pcap file name      : session1.pcap
Capture direction   : Ingress and egress
```



NOTE

On 5420, 6200, 6300, 6400, 8100 and 8360 Switch Series, both ingress and egress are supported capture directions.

On 8325, 9300 and 10000 Switch Series, only ingress is a supported capture direction.

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
6300 6400	<code>config</code>	Administrators or local user group members with execution rights for this command.

Power-over-Ethernet

- The Power-over-Ethernet (PoE) subsystem manages power supplied to devices using standard Ethernet data cables. A Power Sourcing Equipment (PSE) supplies DC power as well as Ethernet connectivity to a Powered Device (PD) using a standard Ethernet cable. The maximum current depends on the PD Requested Class.
- A PoE subsystem contains two parts : a PSE and PD. A Power Sourcing Equipment (PSE) is a device that provides power through a standard Ethernet cable. A PoE capable switch functions as PSE. All Aruba PoE switches are considered as PSEs. A PD is a device powered by a PSE. Examples of PD are VoIP phones, Wireless APs, and IP cameras.
- When a PD or any network cable is connected to a PSE port, the PSE applies a detection voltage and measures the resistance value of the PD. If resistance is within IEEE 802.3 standard values (23 - 26k ohm), the connected device is treated as PD and classification begins. For legacy devices to be detected, you must enable prestandard detection on the switch.
- PDs are divided into different types and classes based on PD power requirements. The power supplied by the PSE is higher than the power PD draws to accommodate for the line losses that can result with the use of the standard maximum length cable(100m).
 - Type 1: PSE can supply maximum of 15.4W, and PD can draw a maximum of 13W.
 - Type 2: PSE can supply maximum of 30W, and PD can draw a maximum of 25.5W.
 - Type 3: PSE can supply maximum of 60W, and PD can draw a maximum of 51W.
 - Type 4: PSE can supply maximum of 90W, and PD can draw a maximum of 71W.
- Classes of PD:
 - Class 0: Type1 PD, it can draw a maximum of 13W.
 - Class 1: Type1 PD, it can draw a maximum of 3.84W.
 - Class 2: Type1 PD, it can draw a maximum of 6.49W.

- Class 3: Type1 PD, it can draw a maximum of 13W.
- Class 4: Type2 PD, it can draw a maximum of 25.5W.
- Class 5: Type3 PD, it can draw a maximum of 40W.
- Class 6: Type3 PD, it can draw a maximum of 51W.
- Class 7: Type4 PD, it can draw a maximum of 62W.
- Class 8: Type4 PD, it can draw a maximum of 71.3W.
- IEEE 802.3bt introduced 4-Pair PoE as a means of supplying higher power to PDs that need more than the current 25.5W supplied by IEEE 802.3at. To increase the available power without damaging the Ethernet cable, the standard introduced the ability to use all four pairs within the Ethernet cable instead of the two pairs used by previous standards (802.3at, 802.3af).
- Supported protocols:
 - Compatibility with IEEE 802.3af, 802.3at, 802.3bt and prestandard.
 - Long first class event supported on Type 3-4 PSE.
 - Support for Single Signature (SS) Type 0-6 and Dual Signature (DS) Type 0-4 PDs.
 - Multi-Event classification permits mutual ID of SS Class 0-6 and DS Class 0-4.
 - Support LLDP Data Link Layer (DLL) Type 1-2 extension 12-octet TLV and Type 3-4 extension 29-octet TLV.
 - Default PSE assigned class delivers the maximum PSE capable power at initial power up based on PD requested class.
- Always-on PoE is a feature that provides the ability for a switch to continue to provide power across user initiated reboots through software. Always-on PoE is enabled by default and no additional configuration is needed.



NOTE

PDs only remain powered, no data transfer or PoE power negotiation can occur until the switch has completely booted up and in normal operation. PD faults occurring prior to full switch boot up will result in PoE power removal and restart the detection process only after switch returns to normal operation.

Subtopics

PoE commands

PoE commands

All PoE configuration commands except **quick-poe**, **threshold configuration** and **always-on poe configuration** are entered at the **config-if** context. The PoE threshold command is used at the system level whereas the **always-on poe** and **power-over-ethernet quick-poe** commands are set at the slot level. These commands can only be configured in the global configuration context.

Subtopics

lldp dot3 poe

lldp med poe

power-over-ethernet

power-over-ethernet allocate-by

power-over-ethernet always-on

power-over-ethernet assigned-class

power-over-ethernet power-pairs

power-over-ethernet pre-std-detect

power-over-ethernet priority

power-over-ethernet quick-poe

power-over-ethernet threshold

power-over-ethernet trap

show lldp local

show lldp neighbor

show power-over-ethernet

lldp dot3 poe

Syntax

```
lldp dot3 poe
no lldp dot3 poe
```

Description

Enables 802.3 TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. LLDP dot3 TLV is by default enabled for PoE.

The **no** form of this command disables 802.3 TLV list in LLDP.

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Enabling 802.3 TLV list in LLDP:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 poe
```

Disabling 802.3 TLV list in LLDP:

```
switch(config-if)# no lldp dot3 poe
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

lldp med poe

Syntax

```
lldp med poe [priority-override]
no lldp med poe [priority-override]
```

Description

Enables MED TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. Also enables the lldp-MED TLV priority to override user configured port priority for Power over Ethernet. When both dot3 and MED are enabled, dot 3 will take precedence. MED TLV is by default enabled for PoE. Priority over-ride is by default disabled.

The **no** form of this command disables MED TLV list in LLDP.

Parameter	Description
[priority-override]	System defined name of the interface.

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Enabling and disabling LLDP MED PoE:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp med poe
switch(config-if)# no lldp med poe
```

Enabling and disabling LLDP MED PoE priority override:

```
switch(config-if)# lldp med poe priority-override
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet

Syntax

```
power-over-ethernet
no power-over-ethernet
```

Description

Enables per-interface power distribution. Per-port power is enabled by default with priority low. PoE cannot be disabled for individual ports when Quick PoE is enabled for the entire switch or line module.

The **no** form of this command disables per-interface power distribution.

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Enabling per-interface power distribution:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet
```

Disabling per-interface power distribution:

```
switch(config-if)# no power-over-ethernet
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# no power-over-ethernet
Interface PoE cannot be disabled when Quick PoE is enabled.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet allocate-by

Syntax

```
power-over-ethernet allocate-by {usage | class}
no power-over-ethernet allocate-by {usage | class}
```

Description

Configures the power allocation method. Power allocation method is initially based on usage. PSE Allocated power value will change to LLDP negotiated power if and when LLDP exchange takes place between PSE

and PD. When there is no LLDP negotiation, PSE Allocated Power Value will be the actual instantaneous power draw and reserve power based on actual consumption. In allocate-by class, power allocation is based on PD requested class and PSE allocated power value will be the LLDP negotiated power when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocate Power will be based on PD class. Reserve power is based on PD Class. By default, power allocation is by usage.

The power allocation method can be changed on an interface through port-access (User roles or RADIUS). An allocation method when configured through port-access will replace the user configured method.

The **no** form of this command resets the action to default.

Parameter	Description
usage	Configures the usage - based allocation method.
class	Configures the class - based allocation method.

Usage

If you enable **pd-class-override** for an interface, the **allocate-by** configuration of that interface will be automatically changed to **class**. However, if you change the allocation method to **usage** when **pd-class-override** is still enabled, you will receive an error message stating that "The power allocation method cannot be changed when pd-class-override is enabled."

To remove **pd-class-override**, you can use the **no power-over-ethernet pd-class-override** command . It is important to note that **pd-class-override** requires the allocation method to be set to **class** and is enforced when configured through CLI. However, if you override the allocation method to **usage** via port-access, **pd-class-override** will not be in effect. Therefore, it is recommended that you do not override the allocation method to **usage** through port-access on interfaces configured with **pd-class-override**.

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Configuring the power allocation method:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet allocate-by usage
switch(config-if)# power-over-ethernet allocate-by class
```

Resetting power allocation method:

```
switch(config-if)# no power-over-ethernet allocate-by class
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet always-on

Syntax

```
power-over-ethernet always-on <MODULE-ID>  
no power-over-ethernet always-on <MODULE-ID>
```

Description

Always-on PoE is a feature that provides the ability to the switch to continue to provide power across a soft reboot. It is applicable only to the interfaces which were connected and delivering before the soft reboot. Also, power will not be delivered if power to the switch is interrupted. This command enables or disables the always-on PoE feature at the switch or the slot level. By default, always-on PoE is enabled at the switch or the slot level.

The **no** form of this command disables power distribution on soft reboot.

Parameter	Description
<MODULE-ID>	Module number to apply always - on PoE configuration.

Examples

Enabling per-interface power distribution:

```
switch(config) # power-over-ethernet always-on 1/1
```

Disabling per-interface power distribution:

```
switch(config) # no power-over-ethernet always-on 1/1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

power-over-ethernet assigned-class

Syntax

```
power-over-ethernet assigned-class {3 | 4 | 6}  
no power-over-ethernet assigned-class
```

Description

Limit PoE power based on the assigned class. When an user assigns a maximum class to an interface, the PSE will limit the maximum power delivered to the PD up to a total power draw not exceeding the PSE assigned-class power. Power demotion occurs when a PD requested class is higher than the PSE assigned class, permitting the PD to receive power and operate in a reduced power mode. PoE ports cannot set an assigned class when Quick PoE is enabled on the subsystem. The default assigned class is 4 for 2-pair capable PSE and 6 for 4-pair capable PSE.

The **no** form of this command resets the action to default.

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Setting PoE assigned class:


```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet assigned-class 4
```

Resetting PoE assigned class to default:

```
switch(config-if)# no power-over-ethernet assigned-class 4
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# interface 1/1/1
switch(config)# power-over-ethernet assigned-class 4
Interface assigned class cannot be configured when Quick PoE is enabled.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet power-pairs

Syntax

```
power-over-ethernet power-pairs {alt-a | alt-a-and-alt-b}
no power-over-ethernet power-pairs {alt-a | alt-a-and-alt-b}
```

Description

Configures the four-pair capable switch to operate in a mode, that restricts the power delivery for class 0 to class 4 single signature devices to operate only on ALT-A power pair.



NOTE

When configured, a warning message is displayed. User must accept the warning by entering **Y** to enable the mode.

The **no** form of this command resets the power pairs to default PoE pairs.

Parameter	Description
alt-a	Delivers power only on the ALT - A pair.
alt-a-and-alt-b	Delivers power on the ALT - A and ALT - B pairs. This is the default configuration on all PoE interfaces.

Usage

IEEE 802.3bt devices such as four-pair (class 5 and higher) and dual signature powered devices require power on both pairs. However, there is no such restriction on IEEE 802.3af (class 0 to class 3) and IEEE 802.3at (class 4) powered devices not to draw power on both pairs if the overall consumption does not violate the power class limit. For such powered devices, a **power-pairs** configuration is provided to configure the 4-pair capable switch to restrict power on only one power pair.

Examples

Configuring PoE power pairs:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet power-pairs alt-a
This setting configures the interface to deliver power only on the ALT-A
cable pair when a Class 0-4 device is connected. Devices that require power
on all pairs may not operate correctly.
Continue (y/n)? y
```

Resetting the PoE power pair to default:

```
switch(config-if)# no power-over-ethernet power-pairs alt-a
This setting configures the interface to deliver power on the ALT-A
and ALT-B cable pairs. This is the default and most devices work
properly with this setting, however some older Class 0-4 devices may
not operate correctly.
Continue (y/n)? y
```

Command History

Release	Modification
10.09	Command Introduced

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet pre-std-detect

Syntax

```
power-over-ethernet pre-std-detect  
no power-over-ethernet pre-std-detect
```

Description

Before IEEE 802.3 released the first Power over Ethernet standard (802.3af), vendors had shipped PoE capable switches and PD's. HPE Aruba Networking switches are backward compatible and will support both IEEE standard and pre-standard 802.3af Power over Ethernet PD's concurrently. This CLI allows the user to enable or disable pre-802.3af-standard device detection and powering on the specific port. When pre-std-detect is enabled, power will be delivered on PairA only. Default is disabled.

The **no** form of this command resets the action to default.

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Enabling pre-standard device detection:

```
switch(config)# interface 1/1/1  
switch(config-if)# power-over-ethernet pre-std-detect
```

Disabling pre-standard device detection:

```
switch(config-if)# no power-over-ethernet pre-std-detect
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet priority

Syntax

```
power-over-ethernet priority {critical | high | low}  
no power-over-ethernet priority {critical | high | low}
```

Description

Sets PoE priority for an interface. Specifying critical, high, or low indicates the priority of the interface in the event of power over-subscription. Within the same priority level, higher power-priority line-module ports have higher precedence. With same PoE priority and same line-module priority, lower numbered line-module ports have higher precedence. Per-interface PoE priority is low by default.

The **no** form of this command resets the priority to default PoE priority "low".

Examples

Configuring PoE priority:

```
switch(config)# interface 1/1/1  
switch(config-if)# power-over-ethernet priority critical  
switch(config-if)# power-over-ethernet priority high
```

Resetting the PoE priority to default:

```
switch(config-if)# no power-over-ethernet priority high
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet quick-poe

Syntax

```
power-over-ethernet quick-poe <MODULE-ID>  
no power-over-ethernet quick-poe <MODULE-ID>
```

Description

Quick PoE is a feature that provides the ability for the switch to provide power to the connected powered device as soon as switch goes through cold reboot. When quick PoE is enabled on the subsystem PoE port disablement and PD demotion is not allowed. also quick PoE enablement is not allowed if any of the port is disabled on the subsystem. User should not over-subscribe the PoE power when quick PoE is enabled. Quick PoE saved configuration will work irrespective of the configuration change at reboot.

Enables quick PoE feature on the switch or the subsystem level. By default, quick-PoE is disabled for the subsystem.

The **no** form of this command disables quick PoE.

Parameter	Description
<MODULE-ID>	Specifies module number for quick PoE configuration .

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Enabling and disabling quick PoE:

```
switch(config)# power-over-ethernet quick-poe 1/2
switch(config)# no power-over-ethernet quick-poe 1/2
switch(config-if)# power-over-ethernet quick-poe 1/1
PoE must be enabled on all interfaces before enabling Quick PoE
switch(config-if)# power-over-ethernet quick-poe 1/3
All interfaces must use the default assigned class before enabling Quick PoE
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet threshold

Syntax

```
power-over-ethernet threshold <PERCENTAGE>
no power-over-ethernet threshold <PERCENTAGE>
```

Description

Sets the threshold at which the system will send an excess power consumption notification trap. Default value is 80 percentage.

The **no** form of this command resets the action to default.

Parameter	Description
<PERCENTAGE>	Excess power consumption trap threshold. Range 1 - 99.

Examples

Setting the power-over-ethernet threshold:

```
switch(config)# power-over-ethernet threshold 75
```

Resetting the power-over-ethernet threshold to default:

```
switch(config-if)# no power-over-ethernet threshold 75
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

power-over-ethernet trap

Syntax

```
power-over-ethernet trap
no power-over-ethernet trap
```

Description

This command enables/disables the SNMP trap generation for PoE related events at system level. PoE trap generation is enabled by default.

The **no** form of this command resets the priority to default PoE priority "low".

Examples

Enabling SNMP trap generation for PoE:

```
switch(config)# power-over-ethernet trap
```

Disabling SNMP trap generation for PoE:

```
switch(config-if)# no power-over-ethernet trap
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	config-if	Administrators or local user group members with execution rights for this command.

show lldp local

Syntax

```
show lldp local-device [<INTERFACE-ID>]
```

Description

Displays information advertised by the switch if the LLDP feature is enabled by user.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Showing LLDP local device:

```
switch# show lldp local-device 1/1/10
Local Port Data
=====
Port-ID          : 1/1/10
Port-Desc        : "1/1/10"
Port VLAN ID     : 0
PoE Plus Information
PoE Device Type  : Type 2 PSE
Power Source     : Primary
Power Priority    : low
PSE Allocated Power: 25.0 W
PD Requested Power : 25.0 W
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor

Syntax

```
show lldp neighbor [<INTERFACE-ID>]
```

Description

Displays detailed information about a particular neighbor connected to a particular interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Showing LLDP neighbor information when there is only one neighbor:

```
switch# show lldp neighbor-info 1/1/10
Port                               : 1/1/10
Neighbor Entries                   : 1
Neighbor Entries Deleted           : 0
Neighbor Entries Dropped           : 0
Neighbor Entries Aged-Out          : 0
Neighbor Chassis-Name              : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description       : ArubaOS (MODEL: 325), Version HPE_ANW IAP
Neighbor Chassis-ID                : 84:d4:7e:ce:5d:68
Neighbor Management-Address        : 169.254.41.250
Chassis Capabilities Available     : Bridge, WLAN
Chassis Capabilities Enabled       :
Neighbor Port-ID                   : 84:d4:7e:ce:5d:68
Neighbor Port-Desc                  : eth0
TTL                                : 120
Neighbor Port VLAN ID              :
Neighbor PoEplus information       : DOT3
Neighbor Device Type               : TYPE2 PD
Neighbor Power Priority             : Unkown
Neighbor Power Source              : Primary
Neighbor Power Requested           : 25.0 W
Neighbor Power Allocated           : 0.0 W
Neighbor Power Supported           : No
```

Neighbor Power Enabled	: No
Neighbor Power Class	: 5
Neighbor Power Paircontrol	: No
Neighbor Power Pairs	: SIGNAL

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show power-over-ethernet

Syntax

HPE Aruba Networking 6300 Switch Series:

```
show power-over-ethernet [member <MEMBER-ID>] [brief]
```

HPE Aruba Networking 6400 Switch Series:

```
show power-over-ethernet [<MODULE-ID>] [brief]
```

HPE Aruba Networking 6300, 6400 Switch Series:

```
show power-over-ethernet [<IFRANGE>] [brief]
```

Description

Displays the status information of the full system. Displays the brief status of all port or given port if parameter brief is used. Displays the detailed status of given port.

Parameter	Description
-----------	-------------

	Displays detailed status for the given module.
--	--

Parameter	Description
<code><MODULE-ID></code>	
<code><IFRANGE></code>	Port identifier range.
<code><IFNAME></code>	Display the detailed status of given port.
<code>brief</code>	Display the brief status of all ports or the given port.

Examples

Showing sample output for show power-over-ethernet on standalone box with VSF capability:

```
switch# show power-over-ethernet
System Power Status for member 1
  Configured Power Status      : No redundancy
  Operational Power Status     : No redundancy
  Total Available Power        : 740 W
  Total Failover Pwr Avl       : 0 W
  Total Redundancy Power       : 0 W
  Total Power Drawn            : 0 W +/- 6W
  Total Power Reserved         : 0 W
  Total Remaining Power        : 740 W
  Trap Threshold               : 80 %
  Trap Enabled                 : Yes
  Always-on PoE Enabled        : 1/1
  Quick PoE Enabled            : None
Internal Power
  Total Power
  PS      (Watts)      Status
  ----  -
  1       0            Absent
  2      740           Ok
System Power Status for member 2
  Configured Power Status      : No redundancy
  Operational Power Status     : No redundancy
  Total Available Power        : 600 W
  Total Failover Pwr Avl       : 0 W
```

```

Total Redundancy Power      :    0 W
Total Power Drawn           :    0 W +/-  6W
Total Power Reserved        :    0 W
Total Remaining Power       : 600 W
Trap Threshold              : 80 %
Trap Enabled                : Yes
Always-on PoE Enabled       : None
Quick PoE Enabled           : None

```

Internal Power

	Total Power	
PS	(Watts)	Status
-----	-----	-----
1	0	Absent
2	600	Ok

Showing sample output for power-over-ethernet member:

```

switch# show power-over-ethernet member 1
System Power Status for member 1
Configured Power Status      : No redundancy
Operational Power Status     : No redundancy
Total Available Power        : 740 W
Total Failover Pwr Avl       :    0 W
Total Redundancy Power       :    0 W
Total Power Drawn            :    0 W +/-  6W
Total Power Reserved         :    0 W
Total Remaining Power        : 740 W
Trap Threshold               : 80 %
Trap Enabled                 : No
Always-on PoE Enabled        : 1/1
Quick PoE Enabled            : 1/1

```

Internal Power

	Total Power	
PS	(Watts)	Status
-----	-----	-----
1	0	Absent
2	740	Ok

Showing sample output for power-over-ethernet brief in a VSF stack:

```

switch# show power-over-ethernet brief
Status and Configuration Information for PoE
Member 1 Power Status
  Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
  Always-on PoE Enabled: 1/1
  Quick PoE Enabled: None

```

PoE	Pwr	Power	Pre-std	Alloc	PSE	Pwr	PD	Pwr	PoE	Port	PD	Cls
-----	-----	-------	---------	-------	-----	-----	----	-----	-----	------	----	-----

```

Type
Port      En  Priority Detect  Act   Rsrvd  Draw  Status      Sign
-----  ---  -
1/1/1     Yes Low      Off    Class  0.0 W  0.0 W Denied      None  4  2
1/1/2     Yes Critical Off    Usage  1.6 W  1.5 W Delivering* Single 0  1
1/1/3     Yes High    Off    Class  54.0 W 25.5 W Delivering*^ Dual  1/3 3
1/1/4     No  Low      On     Usage  0.0 W  0.0 W Disabled   None  N/A
N/A

```

Member 2 Power Status

Available: 600 W Reserved: 0.00 W Remaining: 600 W

Always-on PoE Enabled: None

Quick PoE Enabled: None

```

PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port  PD  Cls
Type
Port      En  Priority Detect  Act   Rsrvd  Draw  Status      Sign
-----  ---  -
2/1/1     Yes Low      Off    Class  0.0 W  0.0 W Searching   None  N/A
N/A
2/1/2     Yes Critical Off    Usage  0.0 W  0.0 W Searching   None  N/A
N/A
2/1/3     Yes High    Off    Class  0.0 W  0.0 W Searching   None  N/A
N/A
2/1/4     No  Low      On     Usage  0.0 W  0.0 W Disabled   None  N/A
N/A

```

*This port may go down in the event of a PSU failure.

^This port is power demoted due to user config or power availability.

Showing sample output for power-over-ethernet brief for a Chassis system:

```
switch# show power-over-ethernet brief
```

Status and Configuration Information for PoE

Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1,1/3,1/4,1/7

Quick PoE Enabled: None

```

PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port  PD  Cls
Type
Port      En  Priority Detect  Act   Rsrvd  Draw  Status      Sign
-----  ---  -
1/1/1     Yes Low      Off    Class  0.0 W  0.0 W Denied      None  4  2
1/1/2     Yes Critical Off    Usage  1.6 W  1.5 W Delivering* Single 0  1
1/1/3     Yes High    Off    Class  54.0 W 25.5 W Delivering^ Dual  1/3 3

```

```
1/1/4    No    Low      On      Usage  0.0 W   0.0 W Disabled  None  N/A
N/A
```

*This port may go down in the event of a PSU failure.

^This port is power demoted due to user config or power availability.

Showing sample output for power-over-ethernet brief per-port:

```
switch# show power-over-ethernet 1/1/1 brief
```

Status and Configuration Information for port 1/1/1

Member 1Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1

PoE Type	Port	En	Priority	Pre-std Detect	Alloc Act	PSE Rsrvd	Pwr Draw	PD Status	Pwr PoE Sign	Port	PD	Cls
-----	----	---	-----	-----	-----	-----	-----	-----	-----	-----	-----	---
----	1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2	

Showing sample output for power-over-ethernet brief for interface range:

For 6300 Switch series:

```
switch# show power-over-ethernet 1/1/1-1/1/2 brief
```

Status and Configuration Information for port 1/1/1-1/1/2

Member 1Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1

Quick PoE Enabled: None

PoE Type	Port	En	Priority	Pre-std Detect	Alloc Act	PSE Rsrvd	Pwr Draw	PD Status	Pwr PoE Sign	Port	PD	Cls
-----	----	---	-----	-----	-----	-----	-----	-----	-----	-----	-----	---
----	1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2	
----	1/1/2	Yes	Critical	Off	Usage	1.6 W	1.5 W	Delivering*	Single	0	1	

For 6400 Switch series:

```
switch# show power-over-ethernet 1/1/1-1/1/2 brief
```

Status and Configuration Information for port 1/1/1-1/1/2

Power Status

Available: 360 W Reserved: 0.00 W Remaining: 360.00 W

Always-on PoE Enabled: 1/1

Quick PoE Enabled: None

PoE Type	Port	En	Priority	Pre-std Detect	Alloc Act	PSE Rsrvd	Pwr Draw	PD Status	Pwr PoE Sign	Port	PD	Cls
-----	----	---	-----	-----	-----	-----	-----	-----	-----	-----	-----	---

```

-----
1/1/1      Yes Low      Off      Usage  0.0 W   0.0 W Searching  N/A   N/A
N/A
1/1/2      Yes Low      Off      Usage  0.6 W   0.0 W Searching  N/A   N/A
N/A

```

Showing sample output for power-over-ethernet for a missing line card:

```
switch# show power-over-ethernet 1/3 brief
```

```
Module 1/3 is not physically present.
```

Showing sample output for power-over-ethernet brief for a missing member:

```
switch# show power-over-ethernet member 3 brief
```

```
Member 3 is not physically present.
```

Showing sample output for power-over-ethernet port when physical interface is not present:

```
switch# show power-over-ethernet 2/1/1
```

```
Interface 2/1/1 is not present.
```

Showing power-over-ethernet port with dual signature PD connected:

```
switch# show power-over-ethernet 1/1/1
```

```
Status and Configuration Information for port 1/1/1*
```

```
Power Enable           : Yes           PD signature           : Dual
```

```
PoE PairA Status       : Delivering    PoE PairB Status       :
```

```
Delivering
```

```
Alloc-by Configured    : Class          Alloc-by Actual         : Class
```

```
User Profile Priority   : High           Port Config Priority     : Low
```

```
Port Priority           : High           Pre-std Detect          :
```

```
Disabled
```

```
PD Type                 : Type3          User Assigned Class     : Class6
```

```
PairA Requested Class   : Class1         PairB Requested Class   : Class4
```

```
PairA Assigned Class    : Class1         PairB Assigned Class    : Class4
```

```
Fault Status PairA      : None           Fault Status PairB      : None
```

```
PD Class Override       : Disabled        Power Pairs Configured  : alt-a
```

```
Power Pairs Applied     : alt-a-
```

```
and-alt-b
```

```
PoE Counter Information
```

```
Over Current Cnt PairA : 0              MPS Absent Cnt PairA   : 0
```

```
Power Denied Cnt PairA : 0              Short Cnt PairA        : 0
```

```
Over Current Cnt PairB : 0              MPS Absent Cnt PairB   : 0
```

```
Power Denied Cnt PairB : 0              Short Cnt PairB        : 0
```

```
Power Information
```

```
PSE Voltage             : 56.3 V          PSE Reserved power     : 34.0 W
```

```
PD Current Draw          : 4.1 A           PD Power Draw           : 24.6 W
```

```
PD Average Power Draw    : 24.0 W          PD Peak Power Draw      : 25.1 W
```

```
LLDP Information
```

```
MED Override            : Enabled
```



```

MED Priority                : High
PSE TLV Configured         : dot3, med
PSE TLV Sent Type          : dot3-ext
PD TLV Sent Type           : med, dot3-ext
DS PSE Allocated Power Value Alt A : 2.5 W
DS PD Requested Power Value Mode A : 2.5 W
DS PSE Allocated Power Value Alt B : 25.0 W
DS PD Requested Power Value Mode B : 25.0 W

```

Showing power-over-ethernet port with single signature PD connected:

```
switch# show power-over-ethernet 1/1/1
```

Status and Configuration Information for port 1/1/9*

```

Power Enable                : Yes          PD signature                : None
PoE Port Status             : Delivering   PD Type                      : Type3
Alloc-by Configured         : Usage        Alloc-by Actual              : Usage
User Profile Priority        : High         Port Config Priority         : Low
Port Priority                : High         Pre-std Detect               : Disabled
PD Requested Class          : Class1       PSE Assigned Class          : Class1
Fault Status                : None         User set Assigned Class     : Class6
PD Class Override           : Disabled     Power Pairs Configured      : alt-a-
and-alt-b                  Power Pairs Applied           : alt-a-
and-alt-b

```

PoE Counter Information

```

Over Current Cnt           : 0            MPS Absent Cnt              : 0
Power Denied Cnt           : 0            Short Cnt                   : 0

```

Power Information

```

PSE Voltage                 : 56.3 V      PSE Reserved power         : 8.6 W
PD Current Draw              : 1.1 A      PD Power Draw              : 8.6 W
PD Average Power Draw        : 8.0 W      PD Peak Power Draw         : 9.1 W

```

LLDP Information

```

LLDP Detect                 : Disabled
PSE TLV Configured          : N/A
PSE TLV Sent Type           : N/A
PD TLV Sent Type            : N/A
PSE Allocated Power Value   : 0.0 W
PD Requested Power Value    : 0.0 W

```

Showing power-over-ethernet for a port range:

```
switch# show power-over-ethernet 1/1/3-1/1/4
```

Status and Configuration Information for port 1/1/3

```

Power Enable                : Yes          PD signature                : None
PoE Port Status             : Delivering   PD Type                      : Type3
Alloc-by Config             : Usage        Alloc-by Actual              : Usage
User Profile Priority        : High         Port Config Priority         : Low

```

```

Port Priority           : High           Pre-std Detect         : Disabled
PD Requested Class     : Class1         PSE Assigned Class     : Class1
Fault Status           : None           User set Assigned Class : Class6
PD Class Override      : Disabled       Power Pairs Configured : alt-a-
and-alt-b
Power Pairs Applied    : alt-a-
and-alt-b

PoE Counter Information
Over Current Cnt       : 0             MPS Absent Cnt        : 0
Power Denied Cnt       : 0             Short Cnt              : 0

Power Information
PSE Voltage            : 56.3 V         PSE Reserved power    : 8.6 W
PD Current Draw        : 1.1 A         PD Power Draw         : 8.6 W
PD Average Power Draw  : 8.0 W         PD Peak Power Draw    : 9.1 W

LLDP Information
LLDP Detect            : Disabled
PSE TLV Configured     : N/A
PSE TLV Sent Type      : N/A
PD TLV Sent Type       : N/A
PSE Allocated Power Value : 0.0 W
PD Requested Power Value : 0.0 W

Status and Configuration Information for port 1/1/4*
Power Enable           : Yes           PD signature           : None
PoE Port Status        : Delivering    PD Type                 : Type3
Alloc-by Config        : Usage         Alloc-by Actual         : Usage
User Profile Priority   : High          Port Config Priority     : Low
Port Priority           : High          Pre-std Detect          : Disabled
PD Requested Class     : Class1         PSE Assigned Class     : Class1
Fault Status           : None           User set Assigned Class : Class6
PD Class Override      : Disabled       Power Pairs Configured : alt-a
Power Pairs Applied    : alt-a

PoE Counter Information
Over Current Cnt       : 0             MPS Absent Cnt        : 0
Power Denied Cnt       : 0             Short Cnt              : 0

Power Information
PSE Voltage            : 56.3 V         PSE Reserved power    : 4.3 W
PD Current Draw        : 1.1 A         PD Power Draw         : 4.3 W
PD Average Power Draw  : 4.0 W         PD Peak Power Draw    : 4.3 W

LLDP Information
LLDP Detect            : Disabled
PSE TLV Configured     : N/A
PSE TLV Sent Type      : N/A
PD TLV Sent Type       : N/A

```

PSE Allocated Power Value : 0.0 W

PD Requested Power Value : 0.0 W

Command History

Release	Modification
10.09	Added power - pairs configuration in the show power - over - ethernet <IFRA NGE> output.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6300 6400	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Aruba AirWave

You can manage and monitor the AOS-CX switch through Aruba AirWave. The following benefits and functions include:

- Configuration (partial configuration)
- Device topology
- Immediate and historical trend reports
- Monitoring of the device and user connected to the network.
- Network discovery
- Syslogs and trap receiver

For information about which versions of Aruba AirWave support AOS-CX, see the AOS-CX Release Notes.

Subtopics

SNMP support and AirWave

Supported features with AirWave and the AOS-CX switch

Configuring the AOS-CX switch to be monitored by AirWave

SNMP support and AirWave

For AirWave to discover and monitor the switch, you must:

- Enable the SNMP services on the switch.
- Configure the SNMP agent to use the SNMP version supported by the management station.

SNMP on the switch

The switch provides SNMP services through the management channel and the data interfaces. Functionality, such as device discovery from NMS, syslog and trap forwarding, can be any channel configured by you.

Although the SNMP server can be enabled on both VRFs (`mgmt` and `default`), only one instance of SNMP can be running. The highest priority is on the `default` VRF.

For example, assume that SNMP is first enabled on the `mgmt` VRF (`snmp-server vrf mgmt`). Then, SNMP is enabled on the `default` VRF (`snmp-server vrf default`) without disabling SNMP on the `mgmt` (using an equivalent `no` form of the command). The `show running-config` command displays both `snmp-server vrf` commands; however, the SNMP instance is running only on the `default` VRF (highest priority).

```
switch# config
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.01.
led locator on
!
!
!
snmp-server vrf default
snmp-server vrf mgmt
!
...
```

Supported features with AirWave and the AOS-CX switch

AirWave supports the following features with the AOS-CX switch:

Device management	Device discovery using SNMPv2C and SNMPv3
	Device dashboards
Monitoring management	Device health attributes (device status/reachability)
	Interface and VLAN management
	Initiates an SSH connection from Aruba AirWave to AOS-CX so that the device outputs from the AOS-CX CLI can be displayed in the Aruba AirWave user interface.
	Firmware versions
	Displays neighbor devices connected to AOS-CX switches
Configuration management	Device topology
	Partial configuration
Alarm management	Alarm triggers (device and interface up/down, new device discoveries, custom event triggers)
	Syslogs and traps
Report management	Device inventory, interface utilization, and device reachability reports
	Summary report of device model, firmware, and boot loader version

Configuring the AOS-CX switch to be monitored by AirWave

Prerequisites

Aruba AirWave is active on the network.

Procedure

1. Enable SNMP on the switch by entering the `snmp-server vrf` command.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
```

2. Configure the SNMPv2C community to public by entering the `snmp-server community public` command. In this instance, `public` is a read-only community string.

```
switch(config)# snmp-server community public
```

3. The community-string is used by SNMPv1 and SNMPv2C for unencrypted authentication. SNMPv3 lets you encrypt the authentication mechanism. To enable SNMPv3, enter the `snmpv3 user` and `snmpv3 context` commands.

```
switch(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
switch(config)# snmpv3 context Admin
```

For discovering devices in AirWave through the SNMPv3 community, the SNMPv3 context name is not mandatory. Devices can still be discovered in Aruba AirWave without the SNMPv3 context name.

4. Enter the `logging` command for enabling syslog forwarding to a remote syslog server, such as AirWave:

```
switch(config)# logging 10.0.10.2 severity debug
```

5. SNMP traps enable an agent to notify the management station of significant events by way of an unsolicited SNMP message. Enable SNMP traps by entering the `snmp-server host` command:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c vrf default
```

SNMP traps cannot be forwarded from AOS-CX 10.00 switches that have the VRF configured as mgmt. Later versions of AOS-CX support SNMP trap forwarding even when the VRF is configured as default or mgmt.

6. For information on how to add a device for monitoring in the Aruba AirWave user interface, see the documentation for Aruba AirWave.

AirWave commands

Select a command from the list in the left navigation menu.

Subtopics

Remote syslog commands

Select a command from the list in the left navigation menu.

Subtopics

[clear accounting-logs](#)

[diag persistent-storage](#)

[logging](#)

[logging accounting-format-native](#)

[logging filter](#)

[logging facility](#)

[logging persistent-storage](#)

clear accounting-logs

Syntax

```
clear accounting-logs
```

Description

Use this command to clear accounting logs. Once issued, only logs generated after this command is run will be displayed in the output of the **show accounting log** commands.



NOTE

This command will not clear logs when the [logging accounting-format-native](#) feature is configured. To clear accounting logs on switches with this feature enabled, users should first revert the native accounting format back to the default AOS-CX format by executing the **no logging accounting-format-native** command.

Example

```
switch(config) # clear accounting-logs
```

The following example shows that accounting logs cannot be cleared using the clear accounting-logs command if the **logging accounting-native-format** command has been enabled, and that disabling this

option with the **no logging accounting-format-native** command again allows the accounting logs to be cleared.

```
switch# logging audit-format-native
switch# clear accounting-logs
Warning: Clear accounting-logs is not supported for 'audit-format-native'.
switch# no logging audit-format-native
switch# clear accounting-logs
switch# show accounting log last 5
-----
Command logs from current boot
-----
No command logs has been logged in the system
```

Command History

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

diag persistent-storage

Syntax

```
diag persistent-storage enable [{severity <LEVEL> | timeout <TIMEOUT> |
filter <FILTER_NAME> | storage-writes <STORAGE_WRITES> | usb <STORAGE_URL>}]
diag persistent-storage disable
```


Description

Enables or disables storage of logs in storage. Only logs of the specified severity and above will be preserved in the storage.



NOTE

To switch the persistent storage setting from one option (e.g., USB) to another (e.g., storage-writes), you must first disable persistent storage by running the command `diag persistent-storage disable`. That the storage-writes option applies only to local storage and does not work with USB.

Parameter	Description
<code>enable</code>	Enable persistent logging.
<code>disable</code>	Disable persistent logging.
<code>severity <LEVEL></code>	Specifies the severity of the syslog messages: • alert: Preserves syslog messages with the severity of alert (6) and emergency (7) • crit: Preserves syslog messages with the severity of critical (5) and above. Default. • debug: Preserves syslog messages with the severity of debug (0) and above. • emer: Preserves syslog messages with the severity of emergency (7) only. • err: Preserves syslog messages with the severity of err (4) and above. • info: Preserves syslog messages with the severity of info (1) and above. • notice: Preserves syslog messages with the severity of notice (2) and above. • warn: Preserves syslog messages with the severity of warning (3) and above.
<code>filter <FILTER_NAME></code>	Preserve logs for specified filter name.
<code>timeout <TIMEOUT></code>	Preserve logs for specified duration. Range: 1 to 60 minutes. Default: 20 minutes.
<code>storage-writes <STORAGE_WRITES></code>	Limit the amount of storage writes.
<code>usb <STORAGE_URL></code>	Preserve logs in the USB storage.

Examples

Enabling storage of logs in storage with severity **err** and timeout 30 minutes:

```
switch(config)#diag persistent-storage enable severity err timeout 30  
Logs will be written to storage. Prolonged usage of persistent logging can  
affect lifetime of storage. Hence enable persistent logging only for short  
duration and also for the right severity.  
Do you want to continue (y/n)? y
```

Enabling storage of logs in storage with severity **err** and **timeout** 30 minutes and 512 storage-writes:

```
switch(config)#diag persistent-storage enable severity err timeout 30  
storage-writes 512  
Logs will be written to storage. Prolonged usage of persistent logging can  
affect lifetime of storage. Hence enable persistent logging only for short  
duration and also for the right severity.  
Do you want to continue (y/n)? y
```

Disabling log storage:

```
switch(config)# diag persistent-storage disable
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

logging

Syntax

```
logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME>} {udp [<PORT-NUM>]} |  
{tcp [<PORT-NUM>]} || {tls [<PORT-NUM>]}
```

```

auth-mode {certificate|subject-name}
disable
filter <FILTER-NAME>
include-auditable-events
legacy-tls-renegotiation]
rate-limit-burst <BURST>
rate-limit-interval <INTERVAL>] ]
severity <LEVEL>]
vrf <VRF-NAME>]
no logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME> }

```

Description

Enables syslog forwarding to a remote syslog server.

The **no** form of this command disables syslog forwarding to a remote syslog server.

Starting with AOS-CX 10.11, payload information is present in accounting logs.

The maximum REST payload that can be sent to RADIUS/TACACS server is 1024 characters, and the maximum of REST payload that can be sent to syslog server is 3500 characters. If this limit is reached, the log will display three dots (...) to indicate that the log exceeded the character limit and is incomplete.

Parameter	Description
{<IPV4-ADDR> <IPV6-ADDR> <HOSTNAME>}	Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.
[udp [<PORT-NUM>] tcp [<PORT-NUM> tls [<PORT-NUM>]]	Specifies the UDP port, TCP port, or TLS port of the remote syslog server to receive the forwarded syslog messages.
udp [<PORT-NUM>]	Range: 1 to 65535. Default: 514
tcp [<PORT-NUM>]	Range: 1 to 65535. Default: 1470
tls [<PORT-NUM>]	Range: 1 to 65535. Default: 6514
auth-mode	Specifies the TLS authentication mode used to validate the certificate. • certificate: Validates the peer using trust anchor certificate based authentication. Default. • subject - name: Validates the peer using trust anchor certificates as well as subject - name based authentication.
disable	Disable remote syslog configuration. This does not delete the configuration, just disables/pauses the forwarding of syslog messages to the remote server. The config/forwarding can be reenabled.

Parameter	Description
	ed (un - paused) again using the no logging <hostname> disable command.
<code>filter <FILTER-NAME></code>	Specifies the name of the filter to be applied on the syslog messages.
<code>include-auditable-events</code>	Specifies that auditable messages are also logged to the remote syslog server.
<code>legacy-tls-renegotiation</code>	Enables the TLS connection with a remote syslog server supporting legacy renegotiation.
<code>rate-limit-burst <BURST></code>	Specifies the rate limit for the messages sent to the remote syslog server.
<code>rate-limit-interval <INTERVAL></code>	Specifies the rate limit interval in seconds. Default: 30 Seconds
<code>severity <LEVEL></code>	Specifies the severity of the syslog messages: • alert: Forwards syslog messages with the severity of alert (6) and emergency (7). • crit: Forwards syslog messages with the severity of critical (5) and above. • debug: Forwards syslog messages with the severity of debug (0) and above. • emerg: Forwards syslog messages with the severity of emergency (7) only. • err: Forwards syslog messages with the severity of err (4) and above. • info: Forwards syslog messages with the severity of info (1) and above. Default. • notice: Forwards syslog messages with the severity of notice (2) and above. • warning: Forwards syslog messages with the severity of warning (3) and above.
<code>vrf <VRF-NAME></code>	Specifies the VRF used to connect to the syslog server. Optional. Default: <code>default</code>

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of **err (4)** and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF **lab_vrf**:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)# logging example.com tls auth-mode subject-name
```

Applying log filtering for syslog server forwarding:

```
switch(config)# logging 10.0.10.6 severity info filter filter_lldp_logs vrf mgmt
```

Applying log filtering and enabling the rate limit for syslog server forwarding over TCP port:

```
switch(config)# logging 10.0.10.2 tcp 3440 severity err vrf mgmt include-auditable-events filter filter_lldp_logs rate-limit-burst 3 rate-limit-interval 35
```

Command History

Release	Modification
10.12.1000	The disable parameter is introduced
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logging accounting-format-native

Syntax

```
logging accounting-format-native
[no] logging accounting-format-native
```

Description

Change the accounting log message format to native Linux format. (Default: AOS-CX format)

The 'no' form of this command will change the accounting log message format to AOS-CX format.

Usage

This option enables the switch to show all types of accounting records to the user. When configured, the same format will be used while sending messages to syslog servers. When upgrading to this version of AOS-CX from AOS-CX 10.10 or earlier versions, if native accounting logs are preferred, then best practices is to issue this command as a part of the upgrade. If the switch upgrades from AOS-CX 10.10 or earlier without configuring this setting, by default, the accounting log message format will be AOS-CX Format.

Example

This example changes the accounting log message format to native Linux format.

```
switch(config)# logging accounting-format-native
```

The following example returns the accounting log message format to the default AOS-CX format.

```
switch(config)# no logging accounting-format-native
```

Command History

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logging filter

Syntax

```
logging filter <FILTER-NAME>
    [{enable | disable}]
    [<SEQUENCE-ID> {permit | deny} [event-id <EVENT-ID-RANGE>] [includes
<REGEX>] [severity <COMPARISON-OPERATOR>
    <LEVEL>]
    no <SEQUENCE-ID>
    resequence <OLD-SEQUENCE-ID>
    <NEW-SEQUENCE-ID>
no logging filter <FILTER-NAME>
```

Description

Creates a filter to restrict what event or debug logs are logged. A filter can be used to either permit or deny:

- The event logs from being generated on the switch, or
- The event or debug logs generated on the switch from being forwarded to a syslog server.

A filter is identified by a filter name and can have up to 20 rules or entries, each with a different sequence number, matching criteria, and corresponding action (deny or permit). When a filter is applied on a log, the log is matched against the criteria mentioned in the rules or entries in ascending numerical order of their sequence numbers until a matching entry is found. Once a matching entry is found, its corresponding action is applied on the log. If no matching rule is found, the default action (permit) is applied.

The **no** form of this command removes the filter.

Parameter	Description
<FILTER-NAME>	Specifies the unique name to identify the filter.
enable	Filter event logs generated on the switch.
<SEQUENCE-ID>	Specifies the filter criteria sequence number. Default: Increments by 10 from the largest sequence - id currently used in this filter.
deny	Prevents the matching log from being logged.
permit	Allows the matching log.

Parameter	Description
<code><event-id></code>	Matches logs by event ID. Specify an event ID or a range of event IDs. It supports a maximum of 100 event IDs.
<code>includes <REGEX></code>	Matches the log message against a regular expression string.
<code>severity</code>	Matches the logs by severity level. The following options are used to compare the severity: • eq: Match events of severity equal to the specified. • ge: Match events of severity greater than or equal to the specified. • gt: Match events of severity greater than the specified. • le: Match events of severity lesser than or equal to the specified. • lt: Match events of severity lesser than the specified. The following are the severity levels: • alert: Logs with the severity alert (6). • crit: Logs with the severity critical (5). • debug: Logs with the severity debug (0). • emerg: Logs with the severity emergency (7). • err: Logs with the severity err (4). • info: Logs with the severity info (1). • notice: Logs with the severity notice (2). • warning: Logs with the severity warning (3).

Usage

Filtering event logs on the switch: To permit or deny event logs from being generated on the switch. In this case, the matching event logs are filtered at generation. The denied event logs are neither logged to the switch events nor forwarded to any remote syslog servers. Multiple filters can be configured, but only one filter can be applied to filter the events on the switch. Such a filter can be chosen by adding the **enable** command under its configuration. Configuring the **enable** command under a new filter automatically removes it from the filter where it was previously used.

For example:

```
logging filter low_severity_logs
enable
10 deny severity lt info
```


This configuration denies the event logs which have a severity less than info.



NOTE

If a filter contains **enable** command, it is not recommended to configure this filter in the **logging** command used for remote syslog server configuration. This is because, any event logs denied by the filter are already not available for forwarding to a remote server.

A filter with **enable** command will not affect debug logs. Consider the configuration in the following example of a filter with **enable** command and two rules applied **10 permit severity ge info** and **20 deny**. This implies permit only those event logs which have severity greater than or equal to **info**.

Example:

```
logging filter low_severity_logs
enable
10 permit severity ge info
20 deny
```

Filtering event or debug logs when forwarding to a remote syslog server: The filter name must be configured in the logging command that is used to configure remote syslog server. The logs will be generated on the switch and the filter only decides whether to deny or permit the syslog forwarding for the matching log. For example: **logging 10.0.10.6 filter filter_ldap_logs**



NOTE

The filter affects debug logs only when the command **debug destination syslog** is configured on the switch.



NOTE

The severity mentioned in the remote syslog server configuration using logging command under configuration context has more precedence than the severity mentioned in a filter entry. If a log with warning severity is permitted by a filter, but the remote syslog configuration has severity **err** mentioned in it, the log will not be forwarded to the remote syslog server (since warning(3) is lesser than err(4)). On the other hand, if a log with **err** severity is permitted by a filter and the remote syslog configuration has severity warning mentioned in it, the log will be forwarded to the remote syslog server.

Examples

Configuring a new logging filter:

```
switch(config)# logging filter example_filter
```

To deny logs having event ID 1301 and a range of event IDs from 1305 to 1309:

```
switch(config-logging-filter)# 20 deny event-id 1301,1305-1309
```

To permit logs having event ID 1300:

```
switch(config-logging-filter)# 30 permit event-id 1300
```

To permit logs with severity greater than or equal to `err`:

```
switch(config-logging-filter)# 30 permit severity ge err
```

To deny logs with severity greater than info:

```
switch(config-logging-filter)# 30 deny severity gt info
```

To deny logs with event ID 1024 and a message matching the regular expression LLDP:

```
switch(config-logging-filter)# 40 deny event-id 1024 includes LLDP
```

Denying all logs:

```
switch(config-logging-filter)# 40 deny
```

Changing the sequence ID of an existing rule:

```
switch(config-logging-filter)# resequence 20 70
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code> and <code>config logging-filter</code>	Administrators or local user group members with execution rights for this command.

logging facility

Syntax

```
logging facility {local0 | local1 | local2 | local3 | local4 | local5 |  
local6 | local7}  
no logging facility
```

Description

Sets the logging facility to be used for remote syslog messages. Default: `local7`

The **no** form of this command disables the logging facility to be used for remote syslog messages.

Parameter	Description
<code>{local0 local1 local2 local3 local4 local5 local6 local7}</code>	Selects the logging facility to be used for remote syslog messages. Required. Specifies the severity of the syslog messages: • local0 • local1 • local2 • local3 • local4 • local5 • local6 • local7

Examples

Sets the local5 logging facility to be used for remote syslog messages:

```
switch(config)# logging facility local5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logging persistent-storage

Syntax

```
logging persistent-storage [severity {alert|crit|debug|emerg|err|info|notice|warning}]  
no logging persistent-storage
```

Description

Enables or disables storage of logs in storage. Only logs of the specified severity and above will be preserved in the storage.

The **no** form of this command disables storage of logs in storage.



NOTE

This command is deprecated. Please use "diag persistent-storage" command to preserve logs in storage.

Parameter	Description
severity <LEVEL>	Specifies the severity of the syslog messages: • alert: Preserves syslog messages with the severity of alert (6) and emergency (7) • crit: Preserves syslog messages with the severity of critical (5) and above. Default. • debug: Preserves syslog messages with the severity of debug (0) and above. • emerg: Preserves syslog messages with the severity of emergency (7) only. • err: Preserves syslog messages with the severity of err (4) and above.

Parameter	Description
	• info: Preserves syslog messages with the severity of info (1) and above. • notice: Preserves syslog messages with the severity of notice (2) and above. • warning: Preserves syslog messages with the severity of warning (3) and above.

Usage

These logs can be copied out by using the **copy support-files all** or **copy support-files previous-boot**.

Examples

Enabling storage of logs in storage with severity **info**:

```
switch(config)#logging persistent-storage severity info
Logs will be written to storage and made available across reboot.
Do you want to continue (y/n)?
```

Disabling storage of logs in storage:

```
switch(config)# no logging persistent-storage
```

Command History

Release	Modification
10.17	Command deprecated.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

SNMP commands

Subtopics

- [event-trap-enable](#)
- [lldp trap enable](#)
- [mac-notify traps](#)
- [rmon alarm {enable | disable} {index | all}](#)
- [rmon alarm](#)
- [show configuration-changes trap](#)
- [show mac-notify port](#)
- [show mac-notify](#)
- [show rmon alarm](#)
- [show snmp agent-port](#)
- [show snmp community](#)
- [show snmpv3 context](#)
- [show snmpv3 engine-id](#)
- [show snmpv3 security-level](#)
- [show snmp system](#)
- [show snmp trap](#)
- [show snmpv3 users](#)
- [show snmp views](#)
- [show snmp vrf](#)
- [snmpv3 context](#)
- [snmpv3 engine-id](#)
- [snmpv3 security-level](#)
- [snmp-server agent-port](#)
- [snmp-server community view](#)
- [snmp-server community](#)
- [snmp-server historical-counters-monitor](#)
- [snmp-server host](#)
- [snmp-server system-contact](#)
- [snmp-server system-description](#)
- [snmp-server system-location](#)
- [snmp-server trap-source interface vrf](#)
- [snmp-server vrf](#)
- [snmpv3 user view](#)
- [snmpv3 user](#)
- [snmp-server response-source](#)
- [snmp-server snmpv3-only](#)
- [snmp-server trap](#)
- [snmp-server trap aaa-server-reachability-status](#)
- [snmp-server trap configuration-changes](#)
- [snmp-server trap mac-notify](#)
- [snmp-server trap module](#)
- [snmp-server trap port-security](#)
- [snmp-server trap snmp](#)

event-trap-enable

Syntax

```
event-trap-enable
no event-trap-enable
```

Description

Enables the notification of events to be sent as traps to the SNMP management stations. It is enabled by default.

The **no** form of this command disables the event traps.

Examples

Enabling the event traps:

```
switch(config) # event-trap-enable
```

Disabling the event traps:

```
switch(config) # no event-trap-enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp trap enable

Syntax

```
lldp trap enable
no lldp trap enable
```

Description

Enables sending SNMP traps for LLDP related events from a particular interface. LLDP trap generation is enabled by default on all the interfaces and has to be disabled for interfaces on which traps are not required to be generated.

The **no** form of this command disables the LLDP trap generation.



NOTE

LLDP trap generation is disabled by default at the global level and must be enabled before any LLDP traps are sent.

Examples

Enabling LLDP trap generation on global level:

```
switch(config) # lldp trap enable
```

Enabling LLDP trap generation on interface level:

```
switch(config-if) # lldp trap enable
```

Disabling LLDP trap generation on global level:

```
switch(config) # no lldp trap enable
```

Disabling LLDP trap generation on interface level:

```
switch(config-if) # no lldp trap enable
```

Displaying LLDP global configuration:

```
switch# show lldp configuration
LLDP Global Configuration
=====
LLDP Enabled                : No
LLDP Transmit Interval      : 30
LLDP Hold Time Multiplier   : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval  : 2
LLDP Trap Enabled           : No
```



```

TLVs Advertised
=====
Management Address
Port Description
Port VLAN-ID
System Description
System Name
LLDP Port Configuration
=====
PORT                TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
-----
1/1/1                Yes                 Yes                 Yes
1/1/2                Yes                 Yes                 Yes
1/1/3                Yes                 Yes                 Yes
1/1/4                Yes                 Yes                 Yes
1/1/5                Yes                 Yes                 Yes
1/1/6                Yes                 Yes                 Yes
.....
.....
mgmt                 Yes                 Yes                 Yes

```

Displaying LLDP Configuration for the interface:

```

switch# show lldp configuration 1/1/1
LLDP Global Configuration
=====
LLDP Enabled          : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled     : No
LLDP Port Configuration
=====
PORT                TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
-----
1/1/1                Yes                 Yes                 Yes

```

Displaying LLDP Configuration for the management interface:

```

switch# show lldp configuration mgmt
LLDP Global Configuration
=====
LLDP Enabled          : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2

```

```

LLDP Reinit Timer Interval    : 2
LLDP Trap Enabled             : Yes
LLDP Port Configuration
=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
mgmt          Yes             Yes             Yes

```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code> and <code>config -if</code>	Administrators or local user group members with execution rights for this command.

mac-notify traps

Syntax

```

mac-notify traps {aged | learned | moved | removed}
no mac-notify traps {aged | learned | moved | removed}

```

Description

Configures a Layer 2 or VXLAN interface to generate SNMP trap notifications for up to four different types of MAC address related events on the trunk or access in physical or lag interfaces.

MAC notification trap addition to or removal from an interface can be in any combination, quantity, or order. The addition of existing configured traps or removal of non-configured traps will be accepted and ignored.



NOTE

The **mac-notify** feature must be enabled globally for any interface configurations to generate SNMP traps. Enabling **mac-notify traps** may impact the system performance on networks with a large number of mac-notify events.

The **no** form of this command removes the traps from the interface.

Parameter	Description
aged	Notifies when a MAC address aged out on the interface.
learned	Notifies when a MAC address is learned on the interface.
moved	Notifies when a MAC address moved from the interface.
removed	Notifies when a MAC address is removed from the interface.



NOTE

MAC notification cannot be configured on a Layer 3 (routing) interface. A Layer 2 interface that is changed to a Layer 3 interface through the `routing` command will discard any existing MAC notification configurations.

When MACs are learned on VXLAN tunnels or **port-access port-security** enabled ports, the move scenario is handled by the EVPN/port-access feature respectively. It performs the move by deleting the MAC from the old port and installing it on the new port. In this scenario, MAC trap notifications, if enabled, will reflect that by producing **removed** and **learned** notifications.

Usage

- MAC notify trap will not generate for static MACs.
- **vsx-sync** is not supported. You must enable the MAC notify traps explicitly on secondary to ensure the traps are generated.
- For EVPN MAC move between the following interfaces, the respective event types are produced (not always removed or learned)
 - Port to port: **moved**
 - Port to tunnel: **removed/learned**
 - Tunnel to port: **removed/learned**
 - Tunnel to Tunnel: **moved**

Examples



NOTE

MAC notification types and the associated events only apply to Layer 2 and VXLAN interfaces, hence routing might need to be disabled on the relevant interfaces.

Enable MAC notification traps within the SNMP module at a global level:

```
switch(config)# snmp-server trap
  aaa-server-reachability-status  Enable SNMP trap for AAA server
reachability status
  configuration-changes           Enable configuration changes traps
  cpu-utilization                 Enable high CPU utilization traps
  link-status                     Enable link status traps for all physical
interfaces
  mac-notify                      Enable MAC table change notification traps
  memory-utilization             Enable high memory utilization traps
  module                         Enable module event traps
  port-security                   Enable port-security violation traps.
                                (Default: enable)
  rmon-events                     Enable RMON event traps
  snmp                           Enable snmp traps
```

For more information, see [snmp-server trap mac-notify](#).

Enabling the traps on an L2 interface:

```
switch(config)# interface 1/1/1
switch(config-if)# mac-notify traps learned
1/1/1 is not an L2 interface or tunnel
switch(config-if)# no routing
switch(config-if)# mac-notify traps learned removed
switch(config-if)# mac-notify traps moved
switch(config-if)# mac-notify traps aged
switch(config)# interface vxlan 1
switch(config-vxlan-if)# mac-notify traps learned removed
switch(config)# interface lag101
switch(config-if)# mac-notify traps removed
```

Disabling the `learned` and `removed` traps from the interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no mac-notify traps learned removed
switch(config)# interface vxlan 1
```

```
switch(config-vxlan-if)# no mac-notify traps learned removed
```

Enable sending SNMP notifications for MAC table changes:

```
switch(config-vxlan-if)# mac-notify traps
aged          Notify when a MAC address aged out on the interface
learned       Notify when a MAC address was learned on the interface
moved        Notify when a MAC address moved from the interface
removed       Notify when a MAC address was removed from the interface
switch(config-vxlan-if)# mac-notify traps learned aged removed moved
```

Command History

Release	Modification
10.13.1000	Support for SNMP MAC notify traps on VXLAN tunnels.
10.10	Support for port access features with mac - notify added.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command. Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

```
rmon alarm {enable | disable} {index | all}
```

Syntax

```
rmon alarm {enable | disable} {index <INDEX> | all}
no rmon alarm [enable | disable] [index <INDEX> | all]
```

Description

Enables and disables the RMON alarm and its index. RMON alarm is enabled by default.

Parameter	Description
<code>enable</code>	Enables the RMON alarm index
<code>disable</code>	Disables the RMON alarm index.
<code>index <INDEX></code>	Specifies the RMON alarm index. Range: 1 to 20.
<code>all</code>	Specifies all the RMON alarms.

Examples

Enabling or disabling all the RMON alarm:

```
switch(config)# rmon alarm enable all
switch(config)# rmon alarm disable all
```

Enabling or disabling RMON alarm by index:

```
switch(config)# rmon alarm enable index 1
switch(config)# rmon alarm disable index 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

rmon alarm

Syntax

```
rmon alarm index <INDEX> snmp-oid <SNMP-OID> rising-threshold <RISING-THRESHOLD>
    falling-threshold <FALLING-THRESHOLD> [sample-interval <SAMPLE-INTERVAL>] [sample-type <ABSOLUTE|DELTA>]
no rmon alarm [index <INDEX>]
```

Description

Stores configuration entries in an alarm table that defines the sample interval, sample-type, and threshold parameters for an SNMP MIB object. Only the SNMP MIB objects that resolve to an ASN.1 primitive type of INTEGER (INTEGER, Integer32, Counter32, Counter64, Gauge32, or TimeTicks) will be monitored.

The **no** form of this command removes all RMON alarms and allows you to specify an index to remove a particular RMON alarm.

Parameter	Description
index <INDEX>	Specifies the RMON alarm index. Range: 1 to 20.
snmp-oid <SNMP-OID>	Specifies the SNMP MIB object to be monitored by RMON.
rising-threshold <RISING-THRESHOLD>	Specifies the upper threshold value for the RMON alarm.
falling-threshold <FALLING-THRESHOLD>	Specifies the falling threshold value for the RMON alarm. The falling threshold must be less than the rising threshold.
sample-interval <SAMPLE-INTERVAL>	Sample interval in seconds. Default: 30.
sample-type <ABSOLUTE DELTA>	Specifies the method of sampling of the SNMP MIB object. Default: Absolute.

Examples

Configuring RMON for the MIB object **ifOutErrors.15** with an index **1**, rising threshold of **2147483647** and falling threshold of **-2134** using **absolute** sampling for a sample interval of **100** seconds:

```
switch(config)# rmon alarm index 1 snmp-oid ifOutErrors.15 rising-threshold
2147483647
                falling-threshold -2134 sample-type absolute sample-interval 100
```

Removing RMON alarm with the index 5:

```
switch(config)# no rmon alarm index 5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show configuration-changes trap

Syntax

```
show configuration-changes trap
```

Description

Shows the SNMP configuration changes trap settings.

Example

Showing the SNMP configuration changes trap:

```
switch# show configuration-changes trap
SNMP Configuration changes trap : Enabled
...
```


Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show mac-notify port

Syntax

```
show mac-notify [port <PORTS>]
```

Description

Displays the MAC notification configuration on a range of ports.

Parameter	Description
[port <PORTS>]	Specifies a port, range of ports, or list of ports.

Examples

Showing the MAC notification configuration on a range of ports:

```
switch(config)# show mac-notify port 1/1/1,1/1/3,1/1/5,lag101-lag104
MAC notification global Setting: Enabled
Port          Enabled Traps
-----
1/1/1         aged learned moved
1/1/3         --
1/1/5         moved
lag101        removed
lag102        --
```

```
lag103      --
lag104      aged learned moved removed
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show mac-notify

Syntax

```
show mac-notify
```

Description

Displays whether the MAC notification feature in the SNMP module is enabled or not. It also displays the trap notification types configured on the Layer 2 ports in the system.

Examples

Showing the MAC notification configuration on all configured ports in the system:

```
switch# show mac-notify
MAC notification global setting : Enabled
Port          Enabled Traps
-----
1/1/1         aged learned moved
1/1/5         moved
lag101        removed
lag104        aged learned moved removed
...
...
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show rmon alarm

Syntax

```
show rmon alarm [index <INDEX>]
```

Description

Displays the RMON alarm configurations.

Parameter	Description
index <INDEX>	Specifies the RMON alarm index. Range: 1 to 20.

Examples

Showing all RMON alarm configurations:

```
switch# show rmon alarm
Index           : 1
Enabled         : true
Status          : valid
MIB object      : ifOutErrors.15
Sample type     : delta
Sampling interval : 6535 seconds
Rising threshold : 100
```

```
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11
Index : 3
Enabled : true
Status : invalid
MIB object : IF-MIB::ifDescr.19
Sample type : absolute
Sampling interval : 10000 seconds
Rising threshold : 4000
Falling threshold : 10
Last sampled value : 0
```

Showing RMON alarm with alarm index 1:

```
switch# show rmon alarm index 1
Index : 1
Enabled : true
Status : valid
MIB object : ifOutErrors.15
Sample type : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-06-21 05:58:11
```

Showing disabled RMON alarm information:

```
switch# show rmon alarm
Index : 1
Enabled : false
Status : valid
MIB object : ifOutErrors.15
Sample type : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11
Index : 3
Enabled : false
Status : invalid
MIB object : IF-MIB::ifDescr.19
Sample type : absolute
Sampling interval : 10000 seconds
Rising threshold : 4000
```

```
Falling threshold : 10
Last sampled value : 0
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
-----------	-----------------	-----------

All platforms	config	Administrators or local user group members with execution rights for this command.
---------------	--------	--

show snmp agent-port

Syntax

```
show snmp agent-port
```

Description

Displays SNMP agent UDP port number.

Example

Displaying SNMP agent UDP port number:

```
switch# show snmp agent-port
SNMP agent port : 161
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp community

Syntax

```
show snmp community
```

Description

Displays a list of all configured SNMPv1/v2c communities.

Usage

When a user creates a custom community before enabling an SNMP agent, AOS-CX automatically removes the default **public** community from the system.

Example

Displaying a list of all configured SNMPv1/v2c communities:

```
switch#show snmp community
```

```
SNMP-COMMUNITIES
```

```
-----  
Community      Access-level  ACL Name    ACL Type    View  
-----  
private        ro           my_acl      ipv4         view1  
private        ro           my_acl      ipv6         none  
private2       rw           new_Acl     ipv6         view2  
private3       rw           none        none         none
```

When the switch is configured to use SNMPv3 only, the output of the **show snmp community** command displays the message **SNMP v1/v2c is disabled while snmpv3-only mode is configured**:

```
switch# show snmp community
```

```
-----  
-----  
Community      Access-level  ACL Name    ACL  
Type    View
```


SNMP v1/v2c is disabled while snmpv3-only mode is configured

Command History

Release	Modification
10.14	The output of this command now displays an error message when the switch is in SNMPv3 - only mode.
10.10	Output has been updated with SNMP view details. A View column is added to the command output.
10.08	Added ACL Type column to the command output.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show snmpv3 context

Syntax

```
show snmpv3 context
```

Description

Displays all configured SNMP contexts.

Examples

Displaying all configured SNMP contexts:

```
switch# show snmpv3 context
-----
name                                vrf                                community
-----
```

contextA	default	private
contextB	vrf_A	public

```
switch# show snmpv3 context
```

```
-----
Name          vrf          Community      type[Instance_id]
-----
A             default      public         vrf
switch#
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmpv3 engine-id

Syntax

```
show snmpv3 engine-id
```

Description

Displays the configured SNMPv3 snmp engine-id.

If the SNMPv3 engine-id is not configured, by default a unique engine-id is created by the switch using a combination of the enterprise OID value and the switch's mac address.

Example

Displaying the configured SNMPv3 engine-id:


```
switch# show snmpv3 engine-id
SNMP engine-id : 80:00:B8:5C:08:00:09:1d:de:a5
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmpv3 security-level

Syntax

```
show snmpv3 security-level
```

Description

Displays the configured SNMPv3 security level.

Examples

Displaying the configured SNMPv3 security level:

```
switch# show snmpv3 security-level
SNMPv3 security-level : auth
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp system

Syntax

```
show snmp system
```

Description

Displays SNMP description, location, and contact information.

Example

Displaying SNMP description, location, and contact information:

```
switch# show snmp system
SNMP system information
-----
System description : Aggregation router
System location   : Main lab
System contact    : John Smith, Lab Admin
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp trap

Syntax

```
show snmp trap
```

Description

Displays all configured SNMP traps/informs receivers.

Example

Displaying all configured SNMP trap and informs receivers:

```
switch# show snmp trap
HOST          PORT  TYPE  VER  COMMUNITY/USER NAME  VRF      NOTIFICATION
TYPES
-----
-----
10.10.10.10    162   trap   v1   public              default   bgp
10.10.10.10    162   inform v2c  public              default   bgp, ospf,
fan, mstp
10.10.10.10    162   inform v3   name              default
```

Command History

Release	Modification
10.14	Updated the example output.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmpv3 users

Syntax

```
show snmpv3 users
```

Description

Displays all configured SNMPv3 users.

For more details on the user enabled status, see [snmpv3 security-level](#).

Example

Displaying all configured SNMPv3 users:

```
switch# show snmpv3 users
```

User	AuthMode	PrivMode	Status	Context	Access-level	View
name	md5	none	Enabled	context2 context1 context3	ro	view1
name2	none	none	Disabled	none	ro	view2
name3	none	none	Disabled	none	ro	none

Command History

Release	Modification
10.10	Output has been updated with SNMP view details. A View column is added to the command output.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp views

Syntax

```
show snmp views
```

Description

Displays the list of all the configured SNMP views.

Usage

The following table contains the status and its description of the configured SNMP views:

Status	Description
pending_validation	Default value that indicates SNMP view is yet to be validated.
operational	OID and mask validated.
invalid	Invalid OID/mask.
failed	Validation failed for reasons other than OID/mask.

Examples

Displaying the list of all the configured SNMP views:

```
switch# show snmp views
-----
SNMP MIB Views
-----
View      : new
OID Tree: sysUpTime.0
Mask      : ff
```

```
Type      : included
Status    : pending_validation
View      : admin
OID Tree  : ifIndex.1
Mask      : ff:a0
Type      : included
Status    : operational
View      : user
OID Tree  : sysb
Mask      : none
Type      : excluded
Status    : invalid
View      : admin
OID Tree  : .1.3.6.1.2.1.1
Mask      : none
Type      : excluded
Status    : operational
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp vrf

Syntax

```
show snmp vrf
```

Description

Displays the VRF on which the SNMP agent service is running.

Example

Displaying SNMP services enabled on VRF:

```
switch#show snmp vrf
SNMP enabled VRF
-----
mgmt
default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

snmpv3 context

Syntax

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]
no snmpv3 context <NAME> [vrf <VRF-NAME>] [community <STRING>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The **no** form of this command removes the specified SNMP context.

Parameter	Description
	Specifies the name of the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark (?).

Parameter	Description
<code><NAME></code>	
<code>vrf <VRF-NAME></code>	Specifies the VRF associated with the context. Default: default.
<code>community <STRING></code>	Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Creating an SNMPv3 context named **newContext** on VRF **myVrf** and with community string **private**.

```
switch(config)# snmpv3 context newContext vrf myVrf community private
```

Removing the SNMPv3 context named **newContext** on VRF **myVrf**:

```
switch(config)# no snmpv3 context newContext vrf myVrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 engine-id

Syntax

```
snmpv3 engine-id <ENGINE-ID>
no snmpv3 engine-id <ENGINE-ID>
```

Description

Configures the SNMPv3 SNMP engine-id allowing an administrator to configure a unique SNMP engine-id for the switch. This engine-id is used by the NMS management tool to identify and distinguish multiple switches on the same network.

The **no** form of this command restores the default engine-id, created by the switch using a combination of the enterprise OID value and the switch's mac address.

Parameter	Description
<code><ENGINE-ID></code>	SNMPv3 SNMP engine - id in colon separated hexadecimal notation.

Examples

Configuring the SNMPv3 engine-id:

```
switch(config)#
switch(config)# snmpv3 engine-id
WORD SNMPv3 snmp engine-id in colon separated hexadecimal notation
switch(config)# snmpv3 engine-id 01:23:45:67:89:ab:cd:ef:01:23:45:67
```

Restoring the default SNMPv3 engine-id:

```
switch(config)# no snmpv3 engine-id
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 security-level

Syntax

```
snmpv3 security-level {auth | auth-privacy}
no snmpv3 security-level {auth | auth-privacy}
```

Description

Configures the SNMPv3 security level. The security level determines which SNMPv3 users defined by the command `snmpv3 user` are able to connect.

The **no** form of this command changes the security level as follows:

- **no snmpv3 security-level auth:** Sets the security level to **auth-privacy**.
- **no snmpv3 security-level auth-privacy:** Sets the security level to no authentication or privacy, allowing any SNMP user to connect.

Parameter	Description
auth	SNMPv3 users that support authentication, or authentication and privacy are allowed.
auth-privacy	Only SNMPv3 users with both authentication and privacy are allowed. This is the highest level of SNMPv3 security. Default.

Examples

Setting the SNMPv3 security level to authentication and privacy:

```
switch(config)# snmpv3 security-level auth-privacy
```

Setting the SNMPv3 security level to authentication only:

```
switch(config)# snmpv3 security-level auth
```

Setting the SNMPv3 security level to no authentication and no privacy:

```
switch(config)# no snmpv3 security-level auth-privacy
```

Restoring the default SNMPv3 security level to authentication and privacy:

```
switch(config)# no snmpv3 security-level auth
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server agent-port

Syntax

```
snmp-server agent-port <PORT>
no snmp-server agent-port [<PORT>]
```

Description

Sets the UDP port number that the SNMP master agent uses to communicate. UDP port 161 is the default port.

The **no** form of this command sets the SNMP master agent port to the default value.

Parameter	Description
<PORT>	Specifies the UDP port number that the SNMP master agent will use. Range: 1 to 65535. Default: 161.

Examples

Setting the SNMP master agent port to **2000**:

```
switch(config)# snmp-server agent-port 2000
```

Resetting the SNMP master agent port to the default value:

```
switch(config-schedule)# no snmp-server agent-port 2000
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
-----------	-----------------	-----------

All platforms	config	Administrators or local user group members with execution rights for this command.
---------------	--------	--

snmp-server community view

Syntax

```
snmp-server community <STRING> [view <VIEW-NAME>]  
no snmp-server community <STRING> [view <VIEW-NAME>]
```

Description

Associates an SNMP MIB view with the SNMP community.

The **no** form of this command removes the associated SNMP MIB view from the SNMP community.

Parameter	Description
-----------	-------------

<STRING>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.
----------	--

<VIEW-NAME>	Specifies the view name for the SNMP MIB view. Accepts a maximum of 32 characters.
-------------	--

Examples

Configuring the SNMPv1/SNMPv2c community:

```
switch(config)# snmp-server community my_community
switch(config-community)#
```

Adding SNMP MIB view to the SNMP community:

```
switch(config-community)# view name1
```

Removing SNMP MIB view from the SNMP community:

```
switch(config-community)# no view name1
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config config-community	Administrators or local user group members with execution rights for this command.

snmp-server community

Syntax

```
snmp-server community <STRING>
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is like a password that controls read/write access to the SNMP agent. A network management program must supply this name when attempting

to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The **no** form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value **public** is automatically defined.

Parameter	Description
<code><STRING></code>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Subcommands

```
access-level {ro | rw}
no access-level {ro | rw}
```

This subcommand changes the access level of the SNMP community. The default access level is read-only (**ro**).

The **no** form of this subcommand changes the access level of the community to default.

Parameter	Description
<code>ro</code>	Specifies Read - Only access with the SNMP community.
<code>rw</code>	Specifies Read - Write access with the SNMP community.

```
access-list {ip | ipv6} <ACL-NAME>
no access-list {ip | ipv6} <ACL-NAME>
```

This subcommand associates an ACL with the SNMP community. If an ACL is not associated with the SNMP community, the default access is allowed for all the hosts.

The **no** form of this subcommand removes association of the ACL with the SNMP community.

Parameter	Description
<code>ip</code>	Specifies the IPv4 ACL type.
<code>ipv6</code>	Specifies the IPv6 ACL type.
	Specifies the ACL name. It supports a maximum of 64 characters.

Parameter	Description
<ACL-NAME>	

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

Configuring the access level for the SMNP community to read-only:

```
switch(config-community)# access-level ro
```

Changing the access level of the SNMP community to default:

```
switch(config-community)# no access-level rw
```

Associating an IPv4 ACL named **my_acl** with the SMNP community:

```
switch(config-community)# access-list ip my_acl
```

Removing the associated IPv4 ACL named **my_acl** from the SNMP community:

```
switch(config-community)# no access-list ip my_acl
```

Configuration supported for SNMP ACL:

```
access-list ip ipv4_acl
  10 permit any 4.4.4.4 4.4.4.1
  20 permit any 3.3.3.3 3.3.3.1
access-list ipv6 ipv6_acl
  10 permit any 2001::2 2001::1
  20 permit any 3001::2 3001::1
snmp-server vrf default
snmp-server community my_comm_1
  access-list ip ipv4_acl
  access-list ipv6 ipv6_acl
```

Configuration not supported for SNMP ACL:

```
access-list ip ipv4_acl
  10 deny any 6.6.6.6 6.6.6.1
access-list ipv6 ipv6_acl
  10 deny any 6001::6 6000::1
```

```
snmp-server vrf default
snmp-server community my_comm_1
  access-list ip ipv4_acl
  access-list ipv6 ipv6_acl
```



NOTE

hitcounts for SNMP ACL will not be incremented.

Example: `show access-list hitcounts ip all` will not show the hit count of SNMP ACL.

Command History

Release	Modification
10.14	Replaced the ipv4 parameter with the ip parameter. The ipv4 parameter is deprecated.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config config- community	Administrators or local user group members with execution rights for this command.

snmp-server historical-counters-monitor

Syntax

```
snmp-server historical-counters-monitor
no snmp-server historical-counters-monitor
```

Description

Enables the Remote Network Monitoring agent (**rmond**) to start collecting historical interface statistics. The **no** form of this command stops the historical interface statistics collection.

Example

Enabling the `rmond` agent to start historical interface statistics collection:

```
switch(config)# snmp-server historical-counters-monitor
```

Disabling the `rmond` agent to stop historical interface statistics collection:

```
switch(config)# no snmp-server historical-counters-monitor
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (<code>></code>) or Manager (<code>#</code>)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (<code>></code>) only.

snmp-server host

Syntax

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION>
[community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
```

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform
version v3] user <NAME> [port <UDP-PORT>] [<VRF-NAME>] [notification-type
<NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform
version v3] user <NAME> [port <UDP-PORT>] [<VRF-NAME>] [notification-type
<NOTIFICATION-TYPE>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The **no** form of this command removes the specified trap/inform receiver.



NOTE

Avoid configuring the same receiver for both SNMP traps and informs on the same UDP port for v1, v2, and v3.

Parameter	Description
<IPv4-ADDR>	Specifies the IP address of a trap receiver in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100 .
<IPv6-ADDR>	Specifies the IP address of a trap receiver in IPv6 format (x:x::x:x).
trap version <VERSION>	Specifies the trap notification type for SNMPv1, v2c or v3. Available options are: v1 , v2c or v3 .
inform version v2c	Specifies the inform notification type for SNMPv2c.
trap version v3	Specifies the trap notification type for SNMPv3.
user <NAME>	Specifies the SNMPv3 user name to be used in the SNMP trap notifications.
community <STRING>	Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: public .
	Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.

Parameter	Description
<code><UDP-PORT></code>	
<code><VRF-NAME></code>	Specifies the VRF on which the SNMP agent listens for incoming requests.
<code><notification-type></code>	Specifies the type of notification to be sent to the trap receiver. If no type is specified, all notifications are sent. The supported notification types are: • aaa - server • alarm • bgp • card • config • entity • fan • interface • lldp • loop - protect • mac - notify • mstp • mvrp • ospf • ospfv3 • port - security • power • power - ethernet • rmon • rpvst • stp • temperature • vrrp • vsf • vsx

Examples

```
switch(config) # snmp-server host 10.10.10.10 trap version v1
switch(config) # no snmp-server host 10.10.10.10 trap version v1
```

```

switch(config)# snmp-server host a:b::c:d trap version v1
switch(config)# no snmp-server host a:b::c:d trap version v1

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public

switch(config)# snmp-server host a:b::c:d trap version v2c community public
switch(config)# no snmp-server host a:b::c:d trap version v2c community
public

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d trap version v2c community public
port 5000
switch(config)# no snmp-server host a:b::c:d trap version v2c community
public port 5000

switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public

switch(config)# snmp-server host a:b::c:d inform version v2c community
public
switch(config)# no snmp-server host a:b::c:d inform version v2c community
public

switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public port 5000

```

```

switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public port 5000 vrf default

switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d inform version v2c community
public port 5000
switch(config)# no snmp-server host a:b::c:d inform version v2c community
public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
port 2000

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin port
2000
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin
port 2000

```

SNMP trap notification type examples:

```

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public notification-type bgp fan interface power entity
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public notification-type bgp
switch(config)# snmp-server host a:b::c:d inform version v3 user Admin
notification-type bgp fan interface power-ethernet
switch(config)# no snmp-server host a:b::c:d inform version v3 user Admin
notification-type bgp interface
switch(config)# snmp-server host a:b::c:d inform version v3 user Admin
notification-type ?
    aaa-server      Sends AAA notifications.
    alarm           Sends Alarm notifications.
    bgp             Sends Border Gateway Protocol (BGP) state change

```

```

notifications.
  card          Sends Card notifications.
  config        Sends Configuration change notifications.
  entity        Sends Entity notifications.
  fan           Sends Fan notifications.
  interface     Sends Interface notifications.
  lldp          Sends Link Layer Discovery Protocol (LLDP) notifications.
  loop-protect  Sends Loop Protect notifications.
  mac-notify    Sends MAC Notify notifications.
  mstp          Sends Multiple Spanning Tree Protocol (MSTP)
notifications.
  mvrp          Sends Multiple VLAN Registration Protocol (MVRP)
notifications.
  ospf          Sends Open Shortest Path First (OSPFv2) notifications.
  ospfv3        Sends Open Shortest Path First version 3 (OSPFv3)
notifications.
  port-security Sends Port Security notifications.
  power         Sends Power notifications.
  power-ethernet Sends Power over Ethernet (PoE) notifications.
  rmon          Sends Remote Network Monitoring (RMON) notifications.
  rpvst         Sends Rapid Per VLAN Spanning Tree (RPVST) notifications.
  snmp          Sends Sends Simple Network Management Protocol (SNMP)
notifications.
  stp           Sends Spanning Tree Protocol (STP) notifications.
  temperature   Sends Temperature notifications.
  vrrp          Sends Virtual Router Redundancy Protocol (VRRP)
notifications.
  vsf           Sends Virtual Switching Framework (VSF) notifications.
  vsx           Sends Virtual System Extension (VSX) notifications.

```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

snmp-server system-contact

Syntax

```
snmp-server system-contact <INFO>
no snmp-server system-contact [<INFO>]
```

Description

Sets SNMP contact information.

The **no** form of this command removes the SNMP contact information.

Parameter	Description
<INFO>	Specifies SNMP contact information. Range: 1 to 128 printable ASCII characters, except for question mark (?).

Examples

Defines SNMP contact information to be **John Smith, Lab Admin**:

```
switch(config)# snmp-server system-contact John Smith, Lab Admin
```

Removes SNMP contact information:

```
switch(config)# no snmp-server system-contact
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server system-description

Syntax

```
snmp-server system-description <DESCRIPTION>
no snmp-server system-description
```

Description

Sets the SNMP system description.

The **no** form of this command removes the SNMP system description.

Parameter	Description
<DESCRIPTION>	Specifies the SNMP system description. Typical content to include would be the full name and version of the following: • Hardware type of the system • Software operating system • Networking software Range: 1 to 64 printable ASCII characters, except for the question mark (?).

Examples

Defines the SNMP system description to be **mainSwitch**:

```
switch(config)# snmp-server system-description mainSwitch
```

Removes the SNMP system description:

```
switch(config)# no snmp-server system-description mainSwitch
```


Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server system-location

Syntax

```
snmp-server system-location <INFO>  
no snmp-server system-location
```

Description

Sets the SNMP location information.

The **no** form of this command removes the SNMP location information.

Parameter	Description
<INFO>	Specifies the SNMP location information. Range: 1 to 128 printable ASCII characters, except for the question mark (?).

Examples

Defines the SNMP location information to be **Main Lab**:

```
switch(config)# snmp-server system-location Main Lab
```

Removes the SNMP location information:

```
switch(config)# no snmp-server system-location
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
-----------	-----------------	-----------

All platforms	config	Administrators or local user group members with execution rights for this command.
---------------	--------	--

snmp-server trap-source interface vrf

Syntax

```
snmp-server trap-source {interface <IF-NAME> | <IPv4-Address> | <IPv6-Address>} [vrf <VRF-NAME>]  
no snmp-server trap-source {interface <IF-NAME> | <IPv4-Address> | <IPv6-Address>} [vrf <VRF-NAME>]
```

Description

Configures SNMP trap source interface or IP address for a VRF.

The **no** form of this command removes the SNMP **trap-source** configuration for a VRF.

Parameter	Description
-----------	-------------

<IF-NAME>	Specifies the source interface name. Interface name can be physical interface, loopback interface, LAG interface, or VLAN interface.
-----------	--

	Specifies the IPv4 address of source interface for the SNMP trap.
--	---

Parameter	Description
<code><IPv4-Address></code>	
<code><IPv6-Address></code>	Specifies the IPv6 address of source interface for the SNMP trap.
<code><VRF-NAME></code>	Specifies the name of a VRF associated to the source interface for the SNMP trap.

Examples

Configuring SNMP trap source interface for a VRF.

```
switch(config) # snmp-server trap-source interface 1/1/12 vrf sample
switch(config) # snmp-server trap-source interface loopback10 vrf sample
switch(config) # snmp-server trap-source interface vlan23 vrf sample
```

Configuring SNMP trap source IP address for a VRF.

```
switch(config) # snmp-server trap-source 10.0.0.1 vrf red
switch(config) # snmp-server trap-source 1001::1 vrf red
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server vrf

Syntax

```
snmp-server vrf <VRF-NAME>
no snmp-server vrf <VRF-NAME>
```

Description

Configures a VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF. 4100i, 6000, and 6100 only support default VRF. **The SNMP agent can listen on multiple VRFs.**

The **no** form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Parameter	Description
<VRF-NAME>	Specifies the name of a VRF.

Examples

Configuring the SNMP agent to listen on VRF **default**.

```
switch(config) # snmp-server vrf default
```

Configuring the SNMP agent to listen on VRF **mgmt**.

```
switch(config) # snmp-server vrf mgmt
```

Configuring the SNMP agent to listen on user-defined VRF **myvrf**.

```
switch(config) # snmp-server vrf myvrf
```

Stopping the SNMP agent from listening on VRF **default**.

```
switch(config) # no snmp-server vrf default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user view

Syntax

```
snmpv3 user <USER-NAME> view <VIEW-NAME>
no snmpv3 user <USER-NAME> view <VIEW-NAME>
```

Description

Associates a user with an existing SNMP MIB view.

The **no** form of this command removes the associated user from the specified SNMP MIB view.

Parameter	Description
<code><USER-NAME></code>	Specifies the user name for the SNMP MIB view. Accepts a maximum of 32 characters.
<code><VIEWNAME></code>	Specifies the view name for the SNMP MIB view. Accepts a maximum of 32 characters.

Examples

Adding a user in the existing SNMP MIB view:

```
switch(config)# snmpv3 user nw-admin view my-nw-view
```

Removing the user from the SNMP MIB view:

```
switch(config)# no snmpv3 user nw-admin view my-nw-view
```

Attaching unconfigured or unknown SNMP view to an SNMPv3 user:

```
switch(config)# snmpv3 user nw-admin view myView
View myView is not configured.
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user

Syntax

```
snmpv3 user <NAME>
    [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
    [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
    [access-level ro|rw]
no snmpv3 user <NAME>
    [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
    [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
    [access-level ro|rw]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context. The SNMPv3 security level (set with command **snmpv3 security-level**) determines which users are allowed to authenticate.

The **no** form of this command removes the specified SNMPv3 user.



NOTE

When updating the authentication protocols and privacy protocols for the existing SNMPv3 users, you must also update the access level. Otherwise, the access level will be set to read-only.

Parameter	Description
<code><NAME></code>	Specifies the SNMPv3 username. Range 1 to 32 printable ASCII characters, excluding space and question mark (?).
<code>access - level</code>	Configures the access level for the SNMPv3 user: • ro: Allow read - only access for the SNMPv3 user • rw: Allow read - write access for the SNMPv3 user
<code>auth <AUTH-PROTO></code>	Sets the authentication protocol used to validate user logins. Supported protocols are md5 , sha , sha224 , sha256 , sha384 , and sha512 .
<code>auth-pass [{plaintext ciphertext} <AUTH-PASS>]</code>	Specifies the SNMPv3 user authentication password. Range for plaintext is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for ciphertext is 1 to 256 printable ASCII characters. Ciphertext is used when copying user configuration settings between switches.  NOTE Authentication passwords that include special characters must be enclosed in single quotation marks ('). For example, 'auth - pwd20246!@#'.
<code>priv <PRIV-PROTO></code>	Sets the SNMPv3 privacy protocol (encryption method). Supported privacy protocols are aes , aes192 , aes256 , and des .
<code>priv-pass [{plaintext ciphertext} <PRIV-PASS>]</code>	Specifies the SNMPv3 user privacy encryption password. Range for plaintext is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for ciphertext is 1 to 256 printable ASCII characters. Ciphertext is used when copying user configuration settings between switches.  NOTE

Parameter

Description



Authentication passwords that include special characters must be enclosed in single quotation marks ('). For example, '**priv - pwd20246!@#**'.



NOTE

When the authentication password is not provided on the command line, plaintext authentication password prompting occurs upon pressing Enter, followed by privacy encryption protocol prompting, and finally plaintext encryption password prompting. The entered password characters are masked with asterisks.



NOTE

When the authentication type and password plus the privacy protocol (encryption method) are provided on the command line but the encryption password is not provided, plaintext encryption password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Defining SNMPv3 user **Admin1** using **sha** authentication and **des** privacy encryption with provided plaintext passwords:

```
switch(config)# snmpv3 user Admin1 auth sha auth-pass plaintext F82#450h
                  priv des priv-pass plaintext F82#4eva
```

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with provided authentication password and privacy encryption type but prompted encryption password:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass plaintext F82#450h
                  priv aes priv-pass
Enter the privacy encryption key: *****
Re-Enter the privacy encryption key: *****
```

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with plaintext password prompting and privacy encryption selection:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass
Enter the authentication password: *****
Re-Enter the authentication password: *****
Configure the privacy protocol (y/n)? y
Enter the privacy protocol (aes/des)? aes
```



```
Enter the privacy encryption key: *****
Re-Enter the privacy encryption key: *****
```

Removing SNMPv3 user **Admin1**:

```
switch(config)# no snmpv3 user Admin1
```

Creating an SNMP user on switch 1 and then creating the same user on switch 2 by copying from the switch 1 configuration:

On switch 1, configure a user named **Admin3**, and then use the **show running-config** command to display switch configuration. Save a copy of the full **snmpv3 user** command (shown by **show running-config**). This saved command is used on switch 2.

```
switch1(config)# snmpv3 user Admin3 auth sha auth-pass plaintext F82#450h
                    priv des priv-pass plaintext F82#4eva
switch1(config)# exit
switch1# show running-config
Current configuration:
!
!Version AOS-CX xx.xx.xx.xxxxxx
!
snmpv3 user Admin3 auth sha auth-pass ciphertext AQBaf2d...FJVcZ3o=
priv des priv-pass ciphertext AQBah2p...2jfTFwQ=
ssh server vrf mgmt
!
interface mgmt
no shutdown
ip dhcp
vlan 1
```

On switch 2, execute the **snmpv3 user** command that you saved from switch 1 (as shown by **show running-config**). This creates the user on switch 2 with the same configuration.

```
switch2(config)# snmpv3 user Admin3 auth sha auth-pass ciphertext
AQBaf2d...FJVcZ3o=
                    priv des priv-pass ciphertext AQBah2p...2jfTFwQ=
```

The following command sets a read-write access level for an SNMPv3 user with the user name **user1**.

```
switch(config)# snmpv3 user user1 auth md5 auth-pass plaintext abc1234
access-level rw
```

Command History

Release	Modification
10.13	Following authentication protocols are supported: sha224 , sha256 , sha384 , and sha512 . Following privacy protocols are supported: aes192 and aes256 .
10.09	The access - level parameter was introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server response-source

Syntax

```
snmp-server response-source {interface <IF-NAME> | <IPv4-ADDRESS> | <IPv6-ADDRESS>} [vrf <VRF-NAME>]  
no snmp-server response-source {interface <IF-NAME> | <IPv4-ADDRESS> | <IPv6-ADDRESS>} [vrf <VRF-NAME>]
```

Description

Configures the source interface or IP address for sending SNMP responses. Each SNMP can independently have its own unique response source IP address.

The **no** form of this command removes the source interface name or IP address for sending SNMP responses.



NOTE

- It is recommended to use the loopback interface or ip address of the loopback interface as the response source. If a device does not support a loopback interface, then configure SVI interface or SVI IP address as the response source.
- The active gateway IP address cannot be configured as the response source.
- It is recommended to limit the maximum number of response source to five.
- The interface used for the response source should be in the up state. If the interface is down, the default source IP will be used.
- The use of **udp6** is mandatory for IPv6 SNMP operations. For example, you can use the following syntax: **snmpwalk -v2c -c public -m ALL udp6: [2100::2] .1.3.6.1.2.1.1.**

Parameter	Description
<code>interface <IF-NAME></code>	Specifies the source interface name. The interface can be a physical interface, loopback interface, or VLAN interface.
<code><IPv4-ADDRESS></code>	Specifies the IPv4 address of the source interface for the SNMP response.
<code><IPv6-ADDRESS></code>	Specifies the IPv6 address of the source interface for the SNMP response.
<code>vrf <VRF-NAME></code>	Specifies the VRF name associated to the source interface for the SNMP response.

Examples

Configuring a response source for the interface **1/1/12**:

```
switch(config) # snmp-server response-source interface 1/1/12 vrf red
```

Configuring a response source for interface **loopback10**:

```
switch(config) # snmp-server response-source interface loopback10 vrf red
```

Configuring a response source for the IPv4 address **10.0.0.1**:

```
switch(config) # snmp-server response-source 10.0.0.1 vrf sample
```

Configuring a response source for the IPv6 address **2001::1**:

```
switch(config) # snmp-server response-source 2001::1 vrf default
```

Command History

Release	Modification
10.13	Added support for IPv6 address.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server snmpv3-only

Syntax

```
snmp-server snmpv3-only  
no snmp-server snmpv3-only
```

Description

Accepts SNMPv3 messages only, SNMPv1 and SNMPv2c will be disabled. By default SNMPv1, SNMPv2c and SNMPv3 will all be enabled.

The **no** form of this command restores the default setting and reenables SNMPv1 and SNMPv2c .

Examples

Configuring SNMPv3 messages only, and disabling SNMPv1 and SNMPv2c:

```
switch(config)# snmp-server snmpv3-only
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap

Syntax

```
snmp-server trap {cpu-utilization | memory-utilization | rmon-events}  
no snmp-server trap {cpu-utilization | memory-utilization | rmon-events}
```

Description

Enables the SNMP traps. The SNMP traps are enabled by default.

The **no** form of this command disables the SNMP traps.

Parameter	Description
cpu-utilization	Enables the CPU utilization traps.
memory-utilization	Enables the memory utilization traps.
rmon-events	Enables the RMON event traps.

Examples

Enabling the SNMP traps:

```
switch(config)# snmp-server trap cpu-utilization
switch(config)# snmp-server trap memory-utilization
switch(config)# snmp-server trap rmon-events
```

Disabling the SNMP traps:

```
switch(config)# no snmp-server trap cpu-utilization
switch(config)# no snmp-server trap memory-utilization
switch(config)# no snmp-server trap rmon-events
```

Displaying the SNMP trap configuration:

```
switch(config)# show running-config all | inc snmp
snmp-server trap rmon-events
snmp-server trap cpu-utilization
snmp-server trap memory-utilization
```

Displaying CPU and Memory usage:

```
switch(config)# show system
Hostname           : XXXX
System Description : XX.10.07.0001CI
System Contact     :
System Location    :
Vendor             : Aruba
Product Name       : JLXXXX XXXX Base Chassis/3xFT/18xFans/Cbl Mgr/X462
Bundle
Chassis Serial Nbr : SG6Z009068
Base MAC Address   : f40343-806400
AOS-CX Version     : XX.10.07.0001CI
Time Zone          : UTC
Up Time            : 8 minutes
CPU Util (%)       : 1
Memory Usage (%)   : 10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap aaa-server-reachability-status

Syntax

```
snmp-server trap aaa-server-reachability-status
no snmp-server trap aaa-server-reachability-status
```

Description

Enables the SNMP trap for AAA server status. When enabled, traps are sent whenever AAA server (RADIUS, TACACS) status changes from reachable to unreachable and vice versa.

The **no** form of this command disables sending SNMP trap for AAA server status.

Examples

Enabling the SNMP trap for AAA server status:

```
switch(config)# snmp-server trap aaa-server-reachability-status
```

Disabling the SNMP trap for AAA server status:

```
switch(config)# no snmp-server trap aaa-server-reachability-status
```

Command History

Release	Modification
10.09	Command introduced on 6200, 6300 and 6400 switches.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap configuration-changes

Syntax

```
snmp-server trap configuration-changes
no snmp-server trap configuration-changes
```

Description

Enables sending SNMP traps whenever the configuration changes. Configuration trap generation is disabled by default.

The **no** form of this command disables sending SNMP traps for configuration changes.

Parameter	Description
configuration-changes	Specifies SNMP traps for configuration changes.

Examples

Enabling the SNMP traps for configuration changes:

```
switch(config)# snmp-server trap configuration-changes
```

Disabling the SNMP traps for configuration changes:

```
switch(config)# no snmp-server trap configuration-changes
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap mac-notify

Syntax

```
snmp-server trap mac-notify
no snmp-server trap mac-notify
```

Description

Enables the MAC notification traps within the SNMP module at a global level. When enabled, traps are sent for interfaces that are configured for MAC notification events.

The **no** form of this command disables sending MAC notification traps at a global level. When disabled, existing **mac-notify** interface configuration is preserved but MAC notification events on configured interfaces will not cause SNMP traps to be transmitted.

Examples

Enabling the SNMP MAC notification feature in the system globally:

```
switch(config) # snmp-server trap mac-notify
```

Disabling the SNMP MAC notification feature in the system globally:

```
switch(config) # no snmp-server trap mac-notify
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap module

Syntax

```
snmp-server trap module
no snmp-server trap module
```

Description

Enables SNMP trap generation for modules. Module trap generation is enabled by default. Generates the module event traps whenever a modular line or fabric card changes state, which includes inserted, removed, ready, and down, as well as when a modular card is unrecognized.

The **no** form of this command disables the SNMP trap generation for module events.

Parameter	Description
module	Specifies SNMP traps for module events.

Examples

Enabling the SNMP traps for modules:

```
switch(config)# snmp-server trap module
```

Disabling the SNMP traps for modules:

```
switch(config)# no snmp-server trap module
switch(config)# show running-config
no snmp-server trap module
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
6400	config	Administrators or local user group members with execution rights for this command.

snmp-server trap port-security

Syntax

```
snmp-server trap port-security  
no snmp-server trap port-security
```

Description

Enables SNMP port-security violation traps on the system. Port-security violation traps are enabled by default.

The **no** form of this command disables the SNMP port-security violation traps on the system.

Parameter	Description
port-security	Specifies SNMP traps for port - security.

Examples

Enabling the SNMP port-security violation traps on the system:

```
switch(config) # snmp-server trap port-security
```

Disabling the SNMP port-security violation traps on the system:

```
switch(config) # no snmp-server trap port-security
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
6300 6400	config	Administrators or local user group members with execution rights for this command.

snmp-server trap snmp

Syntax

```
snmp-server trap snmp {authentication | coldstart | warmstart} [vrf  
<VRF_NAME>]  
no snmp-server trap snmp {authentication | coldstart | warmstart} [vrf  
<VRF_NAME>]
```

Description

Enables SNMPv2 MIB traps. The SNMPv2 traps are disabled by default.

The **no** form of this command disables the SNMPv2 MIB traps.

SNMPv2 MIB supports the following traps:

- authentication:** Authentication trap is sent when the SNMP server receives a protocol message that is not properly authenticated.
- coldstart:** A coldstart trap is sent when the switch reboots.
- warmstart:** A warmstart trap is sent when there is a user intervention to enable or disable the SNMP service on the switch.



NOTE

SNMPv2 Authentication traps do not support source IP configuration.

Parameter	Description
authentication	Enables the authentication traps.
coldstart	Enables the coldstart traps.
warmstart	Enables the warmstart traps.
<VRF_NAME>	Specifies the VRF name. Enables the SNMPv2 traps for a VRF.

Examples

Enabling all SNMPv2 traps:

```
switch(config) # snmp-server trap snmp
```

Enabling only SNMPv2 authentication traps:

```
switch(config) # snmp-server trap snmp authentication
```

Disabling all SNMP traps:

```
switch(config) # no snmp-server trap snmp
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap vsx

Syntax

```
snmp-server trap vsx
no snmp-server trap vsx
```

Description

Enables sending the SNMP traps for VSX related events. VSX trap generation is disabled by default.

The **no** form of this command disables sending the SNMP traps for VSX related events.

The trap support is available for the following VSX events:

- ISL up and down
- KA up and down
- MCLAG up and down

Parameter

Description

vsx

Specifies SNMP traps for VSX events.

Examples

Enabling the VSX traps:

```
switch(config)# snmp-server trap vsx
switch(config)# show vsx configuration trap
SNMP traps : Enabled
```

Disabling the VSX traps:

```
switch(config)# no snmp-server trap vsx
switch(config)# show vsx configuration trap
SNMP traps : Disabled
```

Command History

Release

Modification

10.07 or earlier

- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server view

Syntax

```
snmp-server view <VIEWNAME>
                <OID_TREE> [<MASK>] <included/excluded>
no snmp-server view <VIEWNAME>
                <OID_TREE> [<MASK>] <included/excluded>
```

Description

Configures an SNMP MIB view.

The **no** form of this command removes the specified SNMP MIB view.

Parameter	Description
<VIEWNAME>	Specifies the name of the SNMP MIB view. Supports up to a maximum of 32 characters.
<OID_TREE>	Specifies the OID tree to be included or excluded in SNMP MIB view.
<MASK>	Specifies the OID mask value. The values must be in hexadecimal character separated with : (colon).
	Specifies the OID tree that is included in or excluded from the SNMP MIB view.

Parameter	Description
<code><included/excluded></code>	

Usage

You can configure a maximum of 50 SNMP MIB views. The following VTY message is displayed when the configuration exceeds the maximum SNMP MIB views:

```
switch(config)# snmp-server view name51 1.3.6.1.2.1.1 fe:00 included
Configuration failed: Maximum allowed views are configured.
```

Examples

Configuring the SNMP MIB views:

```
switch(config)# snmp-server view name1 .1.3.6.1.2.1.2.2.1.1.1 FF:A0 included
switch(config)# snmp-server view name2 IF-MIB::ifindex included
switch(config)# snmp-server view name4 1.3.6.1.2.1.1 fe:00 included
```

Removing an SNMP MIB view:

```
switch(config)# no snmp-server view name4 1.3.6.1.2.1.1 fe:00 included
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

Support and Other Resources

Access HPE Aruba Networking support and updates, and view warranty and regulatory information

Subtopics

[Accessing HPE Aruba Networking Support](#)

[Accessing Updates](#)

[Warranty Information](#)

[Regulatory Information](#)

[Documentation Feedback](#)

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	<u>https://www.hpe.com/us/en/networking/hpe - aruba - networking - support - services.html</u>
AOS - CX Switch Software Documentation Portal	<u>https://arubanetworking.hpe.com/techdocs/ArubaDocPortal/content/new-portal/aoscx.html</u>
HPE Aruba Networking Support Portal	<u>https://networkingsupport.hpe.com/home</u>
North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	<u>https://www.hpe.com/psnow/doc/a50011948enw</u>

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS - CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release : https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End - of - Life information	https://networkingsupport.hpe.com/end-of-life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.

See Also

Subtopics

[queue-monitor](#)

queue-monitor

Syntax

```
queue-monitor  
no queue-monitor
```

Description

This command enables the queue monitoring feature, allowing the switch to collect queue statistics. then display this information as a queue statistics history. These aggregate statistics can be used to monitor network health, troubleshoot network issues, and identify normal network behavior patterns and performance anomalies. This feature is disabled by default, but can be enabled on a maximum of 52 interfaces per switch.

Examples

Enabling the queue monitoring feature on interface 1/1/1:

```
(config)# interface 1/1/1
(config-if)# queue-monitor
```

Disabling the queue monitoring feature on interface 1/1/1:

```
(config)# interface 1/1/1
(config-if)# no queue-monitor
```

Related Commands

Command	Description
show queue - monitor status	This command is used to view the global state of the queue monitor feature. Output includes the memory consumed by the feature, the statistics being monitored for collection, and the interfaces currently enabled for monitoring.

Command History

Release	Modification
10.17	Command introduced on 5420, 6200, 6300 and 6400 Switch series.

Command Information

Platforms	Command context	Authority
6300M/F (except for JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, J	config-if	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
L668A, JL669 A, JL670A, JL 762A, JL761A , JL760A) 6400v2		